

CSE 311:

Data Communication

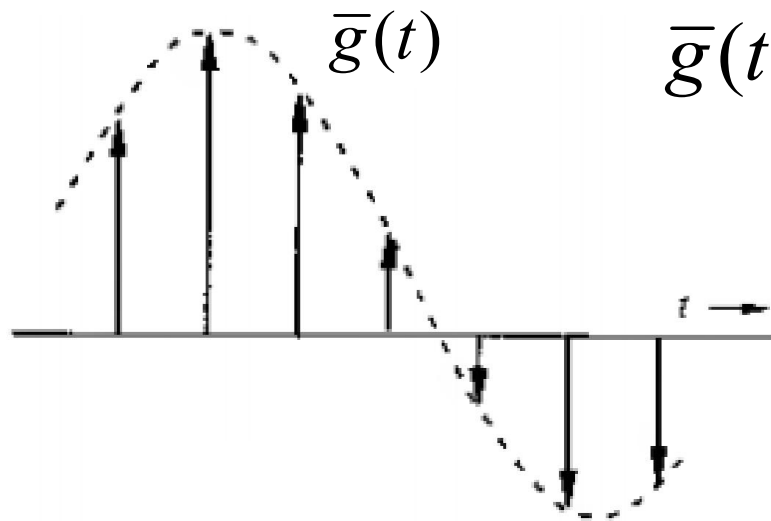
Instructor:
Dr. Md. Monirul Islam

Sampling and Analog-to-Digital Conversion

Sampling Theorem

Review

Uniform sampling $g(t)$ at a rate $f_s = 1/T_s$



$$\bar{g}(t) = g(t)\delta_{T_s}(t) = \frac{1}{T_s} \sum_n g(t) e^{jn2\pi f_s t}$$

$$= \frac{1}{T_s} \sum_n g(t) e^{j2\pi n f_s t}$$

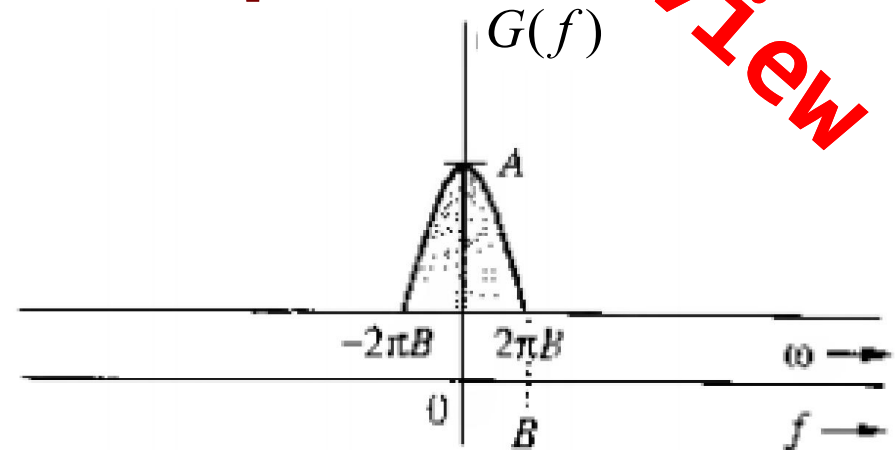
$\bar{g}(t)$ is multiplication of $g(t)$ and impulse train $\delta_{T_s}(t)$ and can be written as

$$\bar{g}(t) = g(t)\delta_{T_s}(t) = \sum_n g(nT_s)\delta(t - nT_s)$$

Signal Reconstruction from Uniform Samples

Review

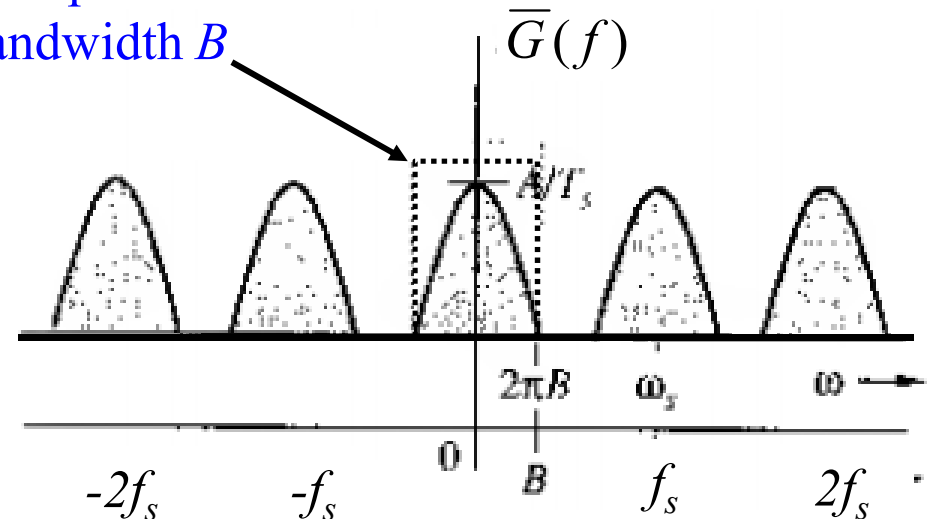
- We proved theoretically exact reconstruction is possible provide
 - the signal is sampled at rate $f_s > 2B$
 - The sampled signal is passed through an ILPF with bandwidth B



$$\bar{G}(f) = \frac{1}{T_s} \sum_n G(f - nf_s)$$

$$H(f) = T_s \Pi\left(\frac{f}{2B}\right)$$

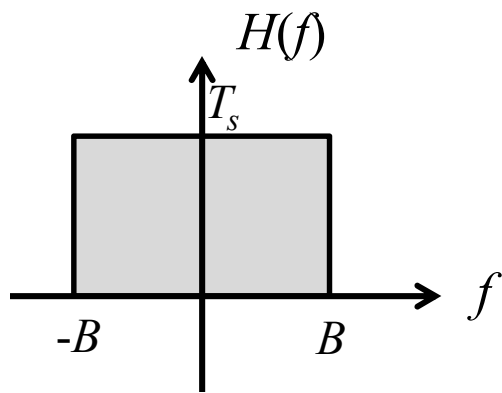
Low pass filter of bandwidth B



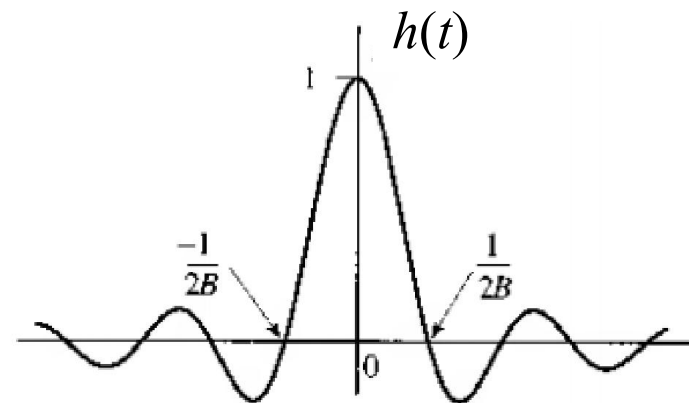
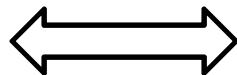
Signal Reconstruction from Uniform Samples

Review

- To recover $g(t)$, the transfer function will be $H(f) = T_s \Pi(\frac{f}{2B})$



$$H(f) = T_s \Pi\left(\frac{f}{2B}\right)$$



$$\begin{aligned} h(t) &= 2BT_s \text{sinc}(2\pi Bt) \\ &= \text{sinc}(2\pi Bt) \end{aligned}$$

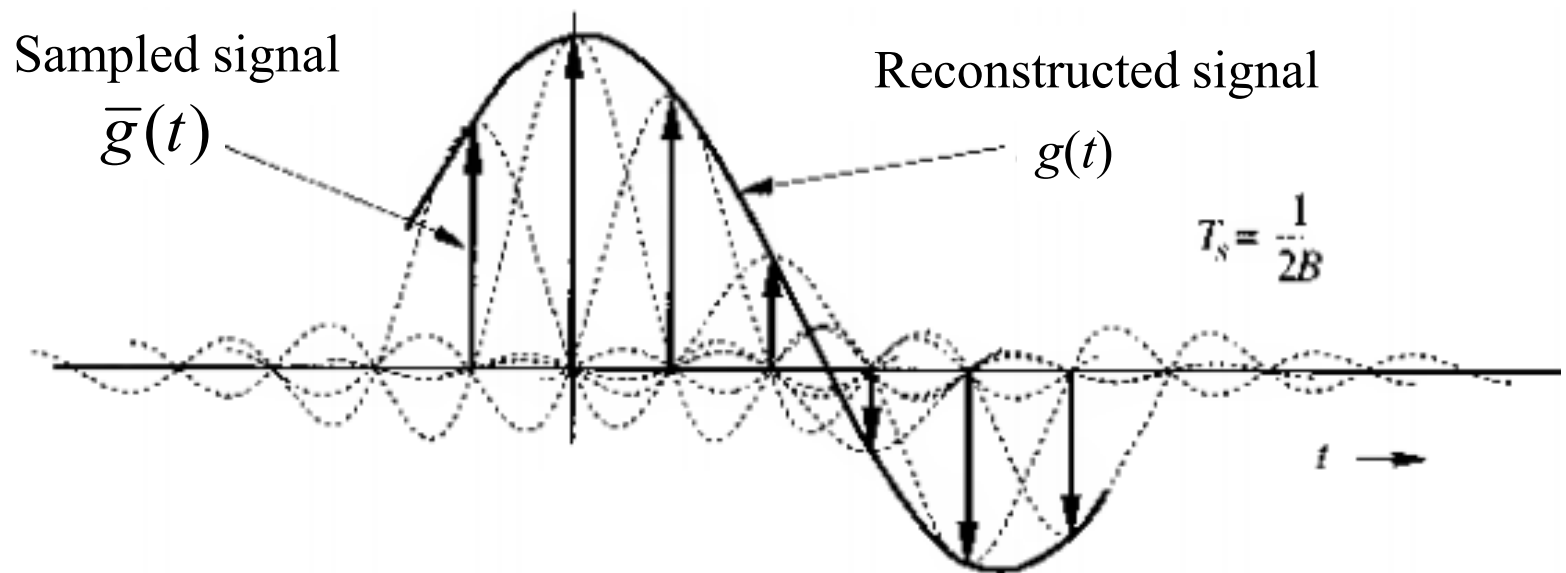
Assuming Nyquist sampling interval $T_s = \frac{1}{2B}$

Signal Reconstruction from Uniform Samples

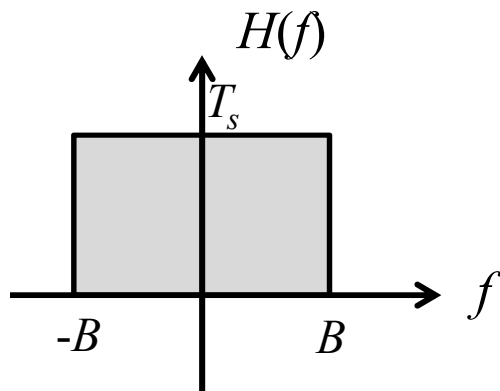
Review

Significance of Interpolation formula

$$g(t) = \sum_n g(nT_s) \text{sinc}(2\pi Bt - n\pi)$$



Practical Signal Reconstruction (Interpolation)



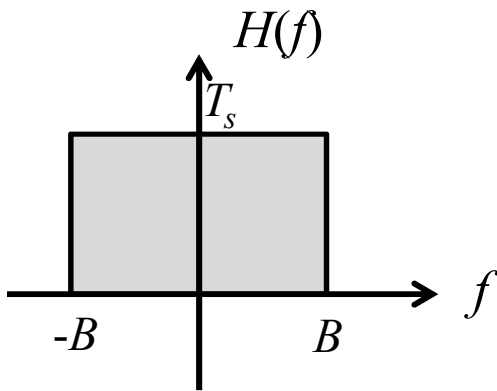
ILPF



$$g(t) = \sum_n g(nT_s) \text{sinc}(2\pi Bt - n\pi)$$

Interpolation formula

Practical Signal Reconstruction (Interpolation)

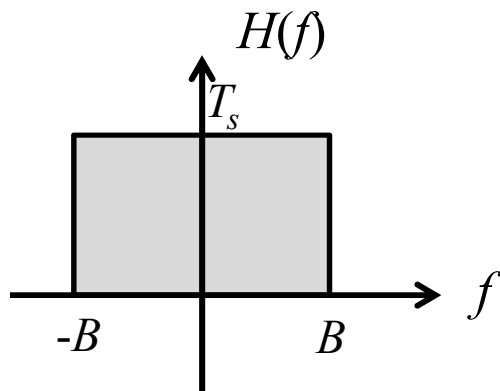


$$g(t) = \sum_n g(nT_s) \text{sinc} 2\pi B(t - n\pi)$$

Interpolation formula

ILPF:
Noncausal
and
Unrealizable

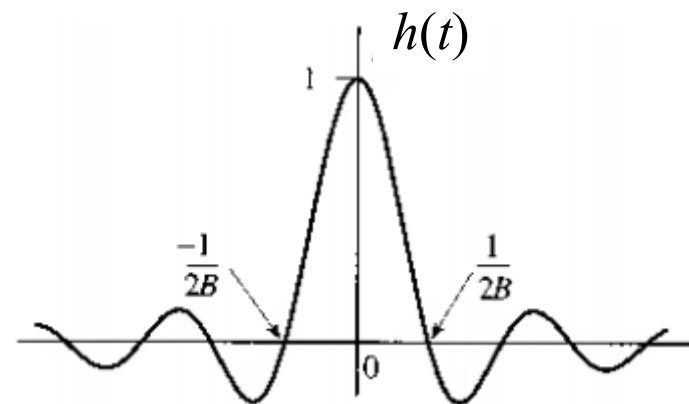
Practical Signal Reconstruction (Interpolation)



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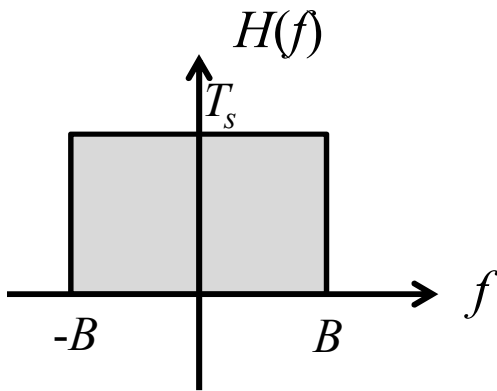
Interpolation formula

ILPF:
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Also seen form long nature of sinc
reconstruction pulse

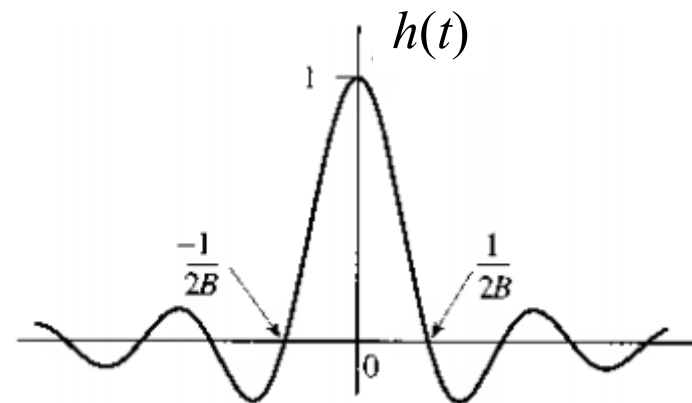
Practical Signal Reconstruction (Interpolation)



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Interpolation formula

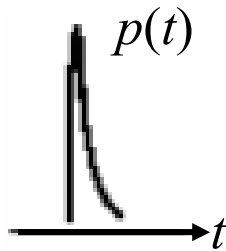
Sharp transition of
ILPF at $\pm B$ causes
this problem



Also seen from **long nature** of **sinc**
reconstruction pulse

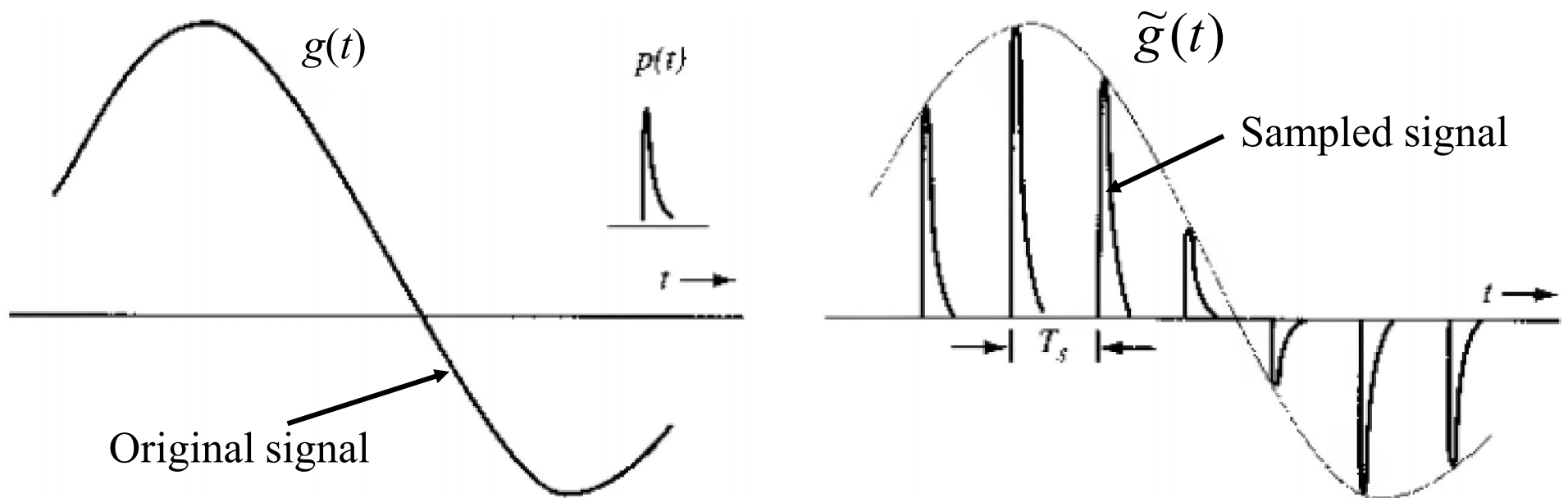
Practical Signal Reconstruction (Interpolation)

Alternate solution is to use
realizable pulse for interpolation



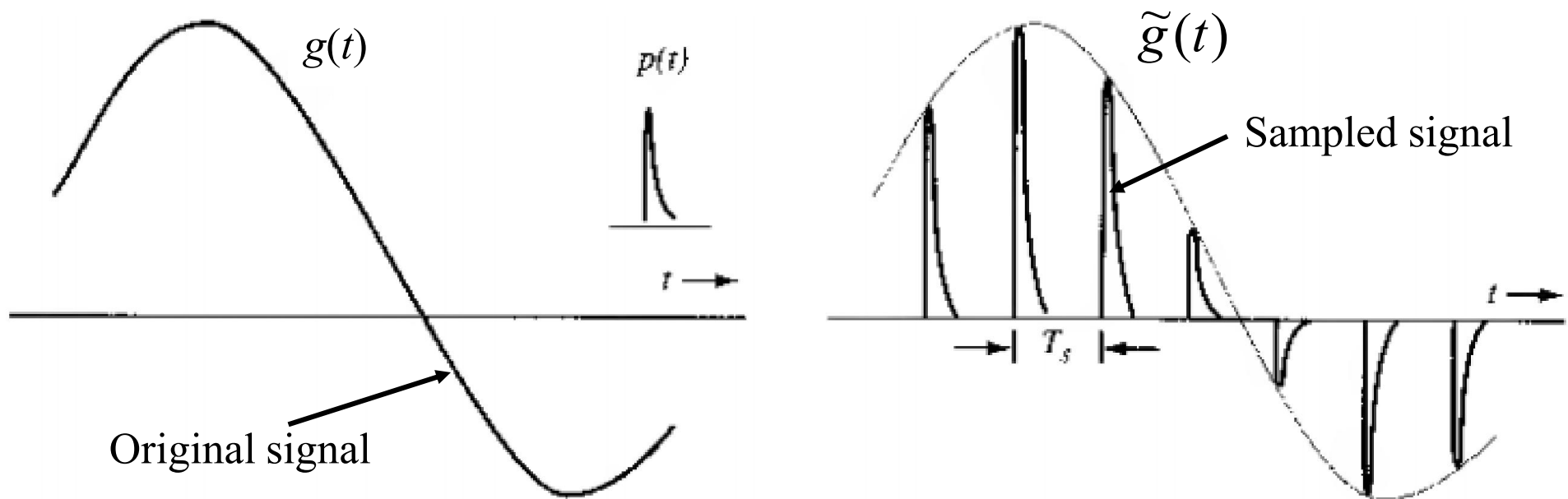
Practical Signal Reconstruction (Interpolation)

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Practical Signal Reconstruction (Interpolation)

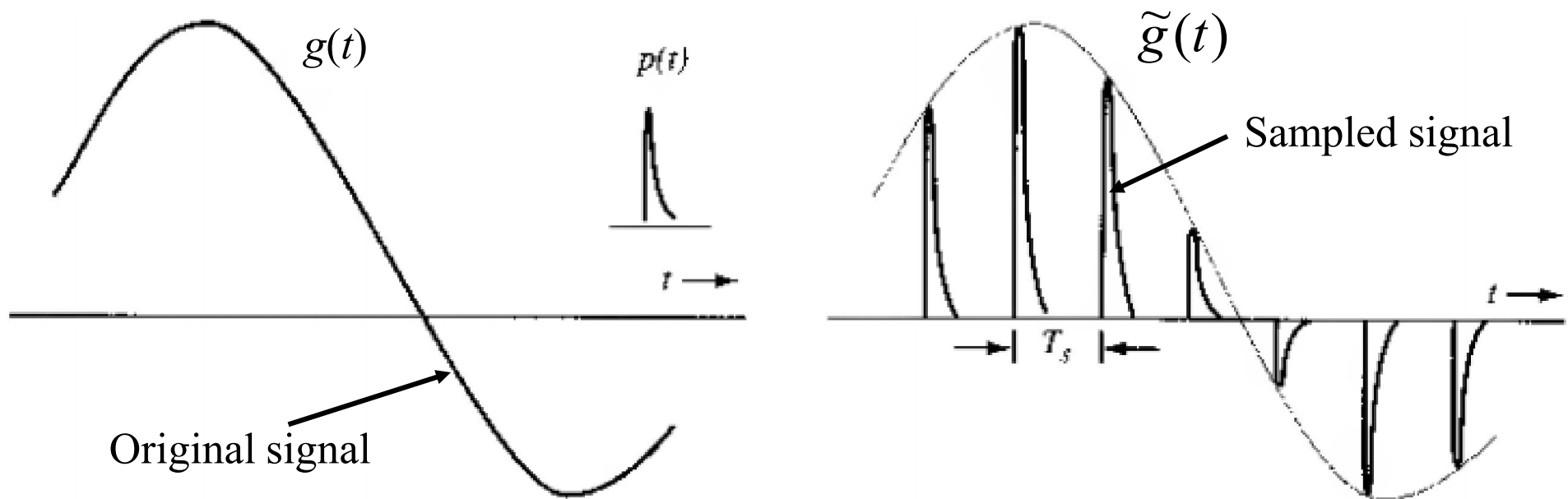
Alternate solution is to use realizable pulse for interpolation



$$\tilde{g}(t) = \sum_n g(nT_s) p(t - nT_s)$$

Practical Signal Reconstruction (Interpolation)

Alternate solution is to use realizable pulse for interpolation



$$\tilde{g}(t) = \sum_n g(nT_s) p(t - nT_s) = \bar{g}(t) * p(t)$$

Practical Signal Reconstruction (Interpolation)

New interpolation formula:

$$\tilde{g}(t) = \sum_n g(nT_s) p(t - nT_s) = \bar{g}(t) * p(t)$$

Practical Signal Reconstruction (Interpolation)

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In frequency domain, it becomes

$$\tilde{G}(f) = P(f) \frac{1}{T_s} \sum_n G(f - nf_s)$$

Practical Signal Reconstruction (Interpolation)

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Remember, $\bar{G}(f) = \frac{1}{T_s} \sum_n G(f - nf_s)$

Practical Signal Reconstruction (Interpolation)

New interpolation formula:

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In frequency domain, it becomes

$$\tilde{G}(f) = P(f) \frac{1}{T_s} \sum_n G(f - nf_s)$$

The reconstructed signal $\tilde{g}(t)$ also contains multiples replicas of $G(f)$ at every f_s distance apart and filtered by $P(f)$

Practical Signal Reconstruction (Interpolation)

New interpolation formula:

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In frequency domain, it becomes

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To fully recover $g(t)$ from $\tilde{g}(t)$, we need another filter or Equalizer $E(f)$

Practical Signal Reconstruction (Interpolation)

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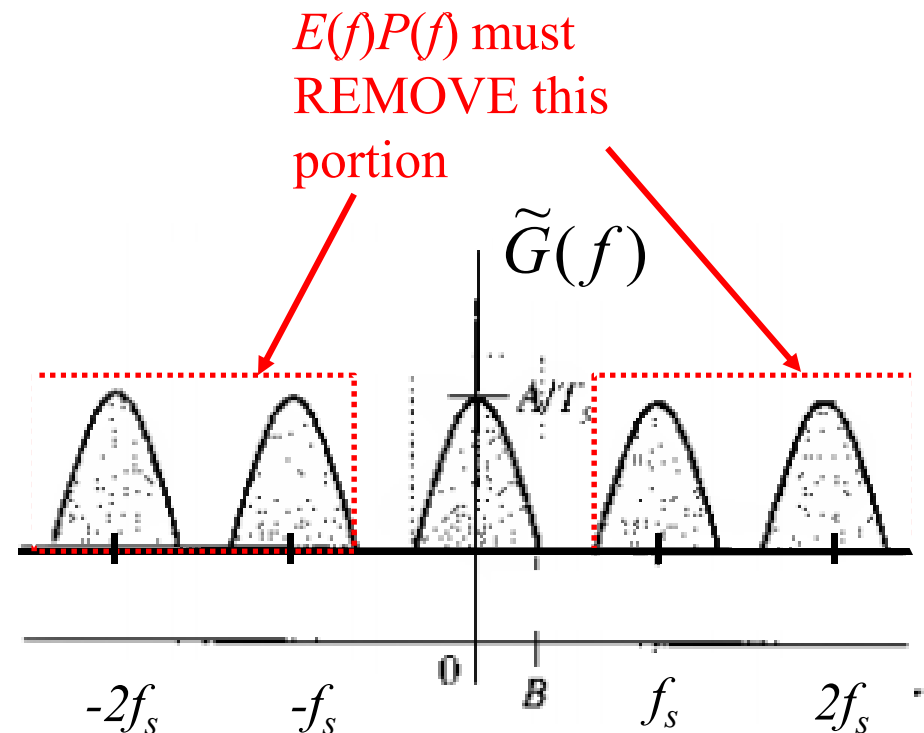
That means,

$$\begin{aligned} G(f) &= E(f) \tilde{G}(f) \\ &= E(f) P(f) \frac{1}{T_s} \sum_n G(f - nf_s) \end{aligned}$$

Practical Signal Reconstruction (Interpolation)

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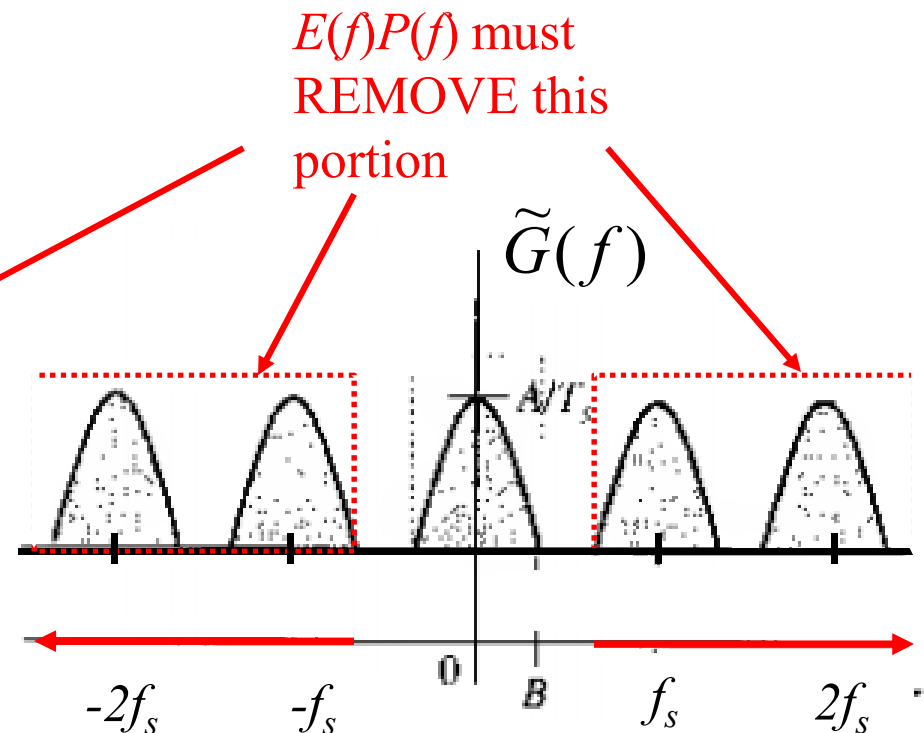


Practical Signal Reconstruction (Interpolation)

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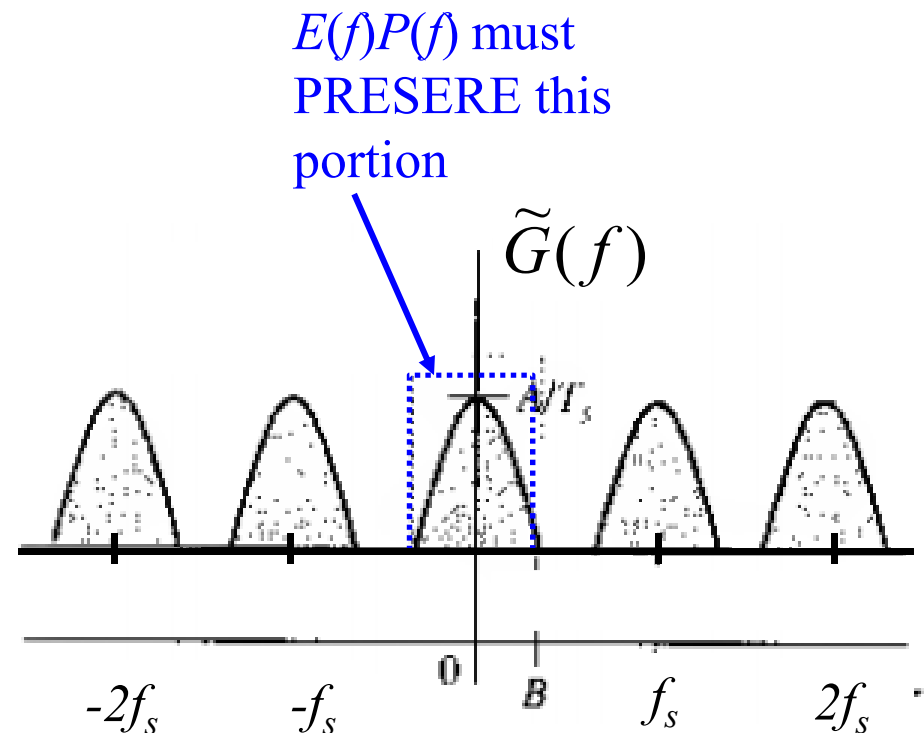
$$E(f)P(f) = 0 \quad |f| > f_s - B$$



Practical Signal Reconstruction (Interpolation)

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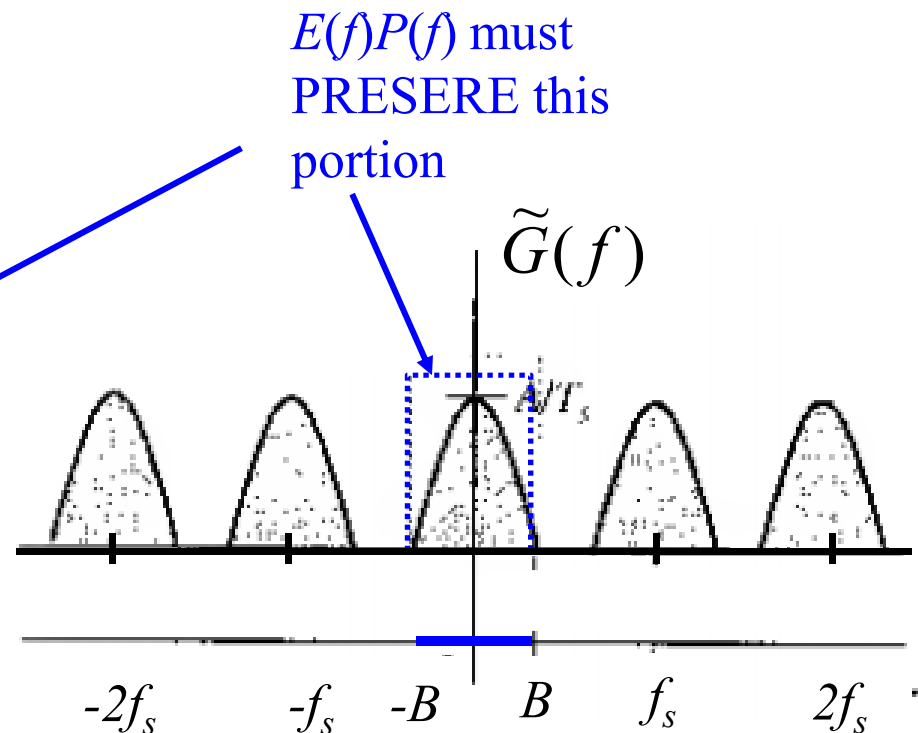


Practical Signal Reconstruction (Interpolation)

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$$E(f)P(f) = T_s \quad |f| < B$$

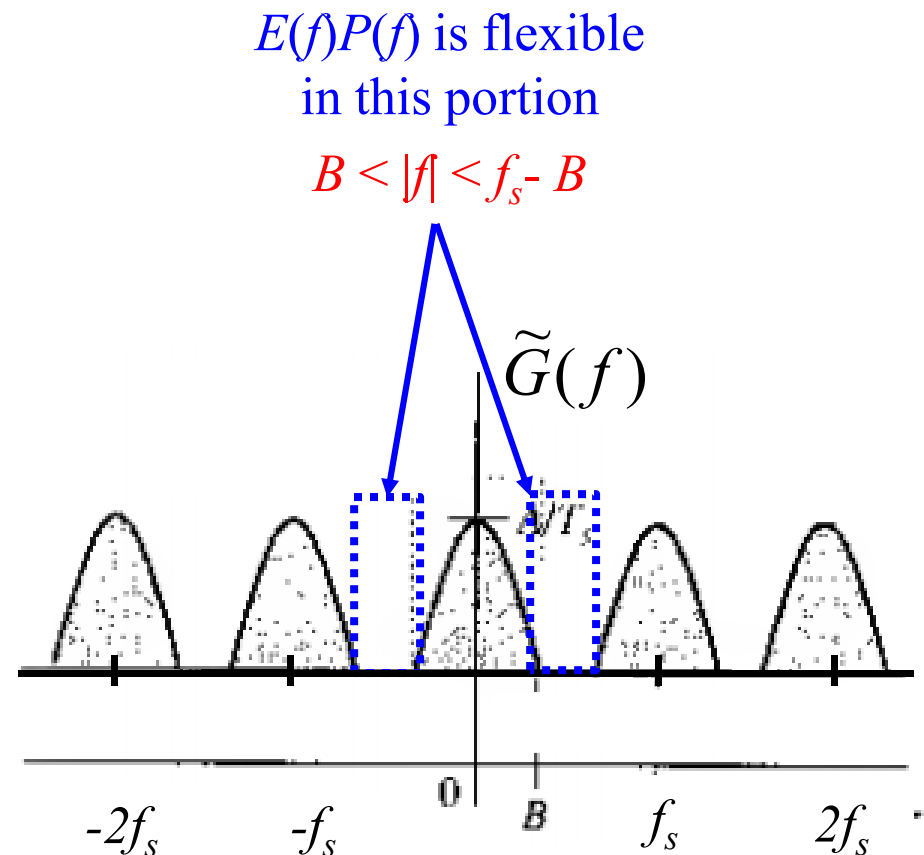


Practical Signal Reconstruction (Interpolation)

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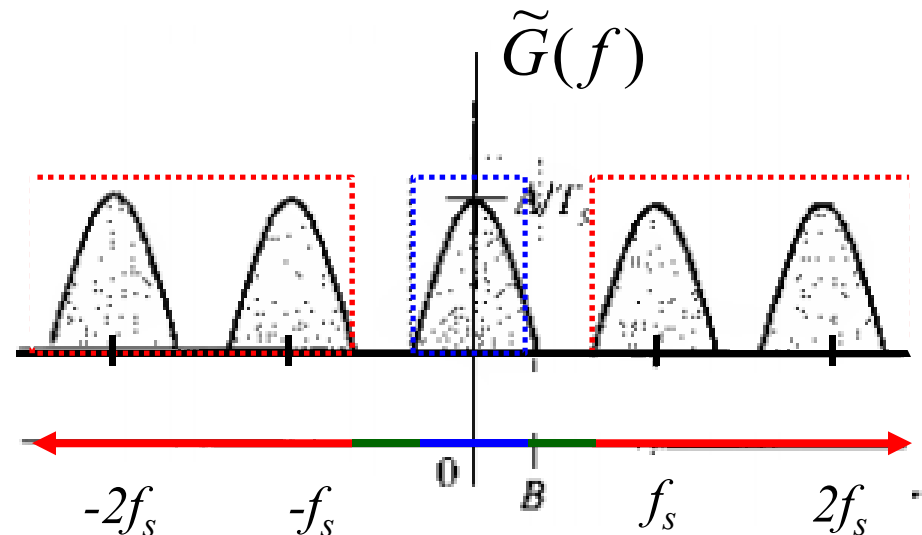
Practical Signal Reconstruction (Interpolation)

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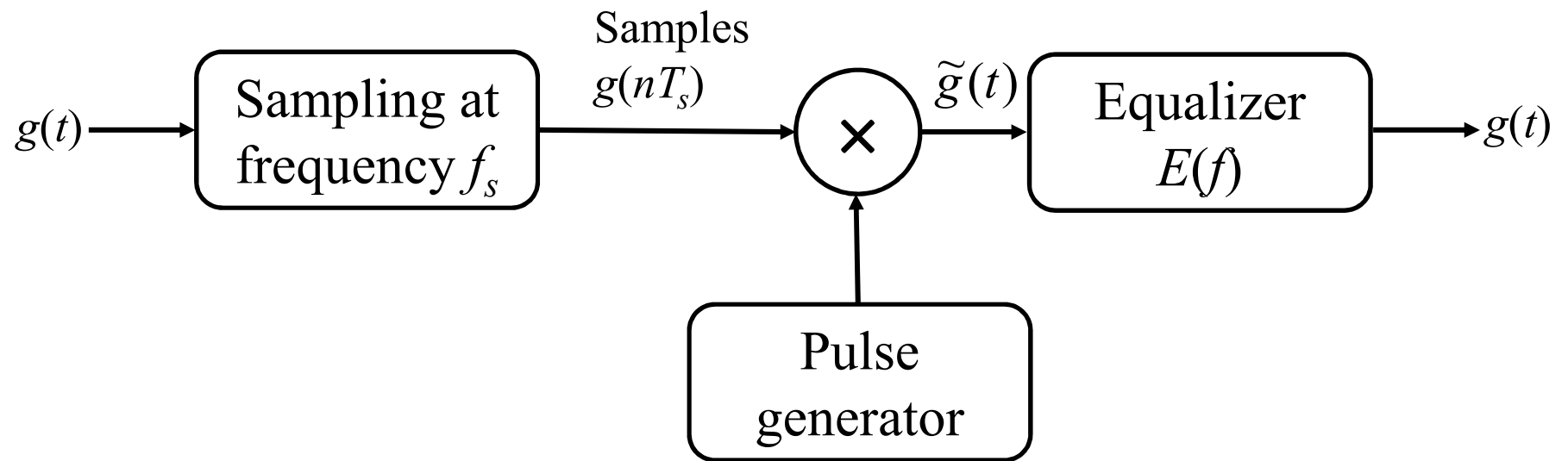
$$\begin{aligned} G(f) &= E(f) \tilde{G}(f) \\ &= E(f) P(f) \frac{1}{T_s} \sum_n G(f - nf_s) \end{aligned}$$

3 conditions together

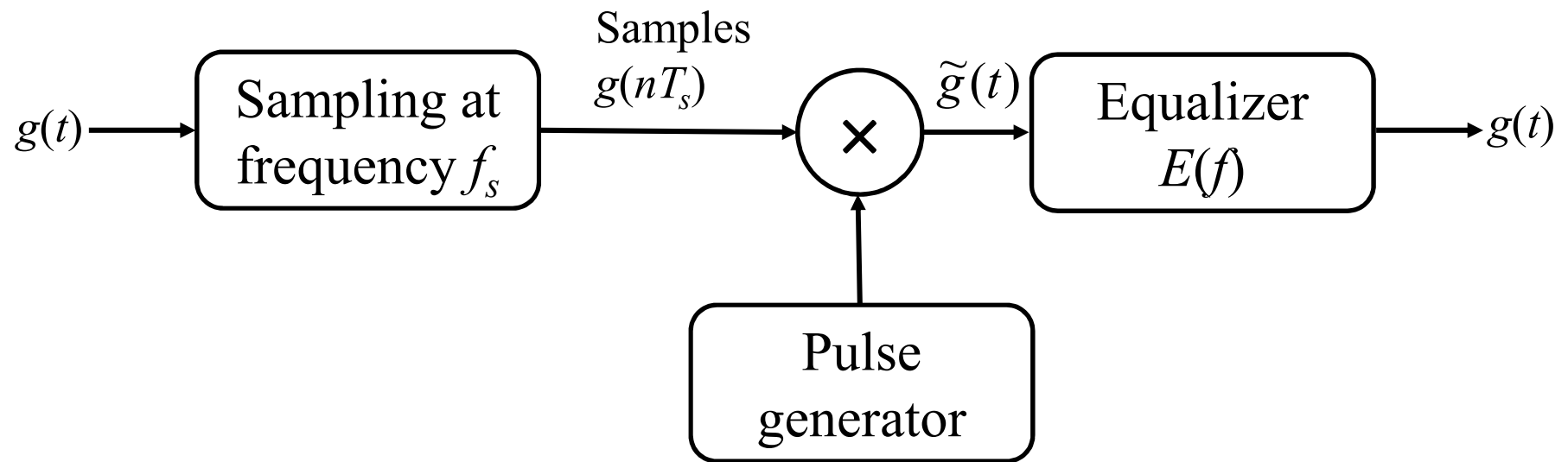
$$E(f) = \begin{cases} T_s / P(f) & |f| \leq B \\ \text{Flexible} & B < |f| \leq f_s - B \\ 0 & |f| \geq f_s - B \end{cases}$$



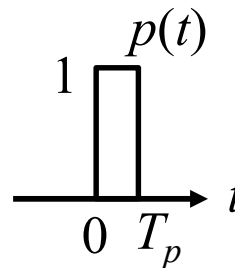
Practical Signal Reconstruction (Interpolation)



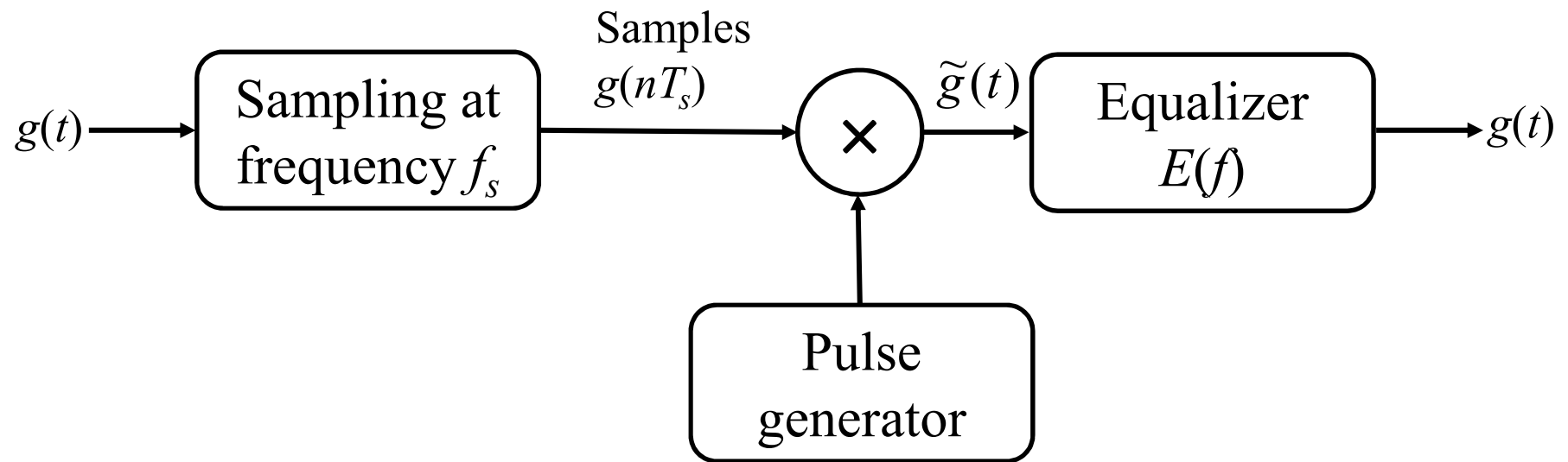
Practical Signal Reconstruction (Interpolation)



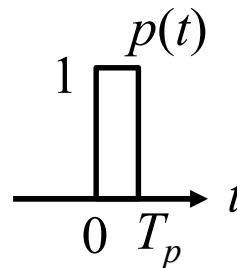
Let this pulse generator generates this short pulse



Practical Signal Reconstruction (Interpolation)

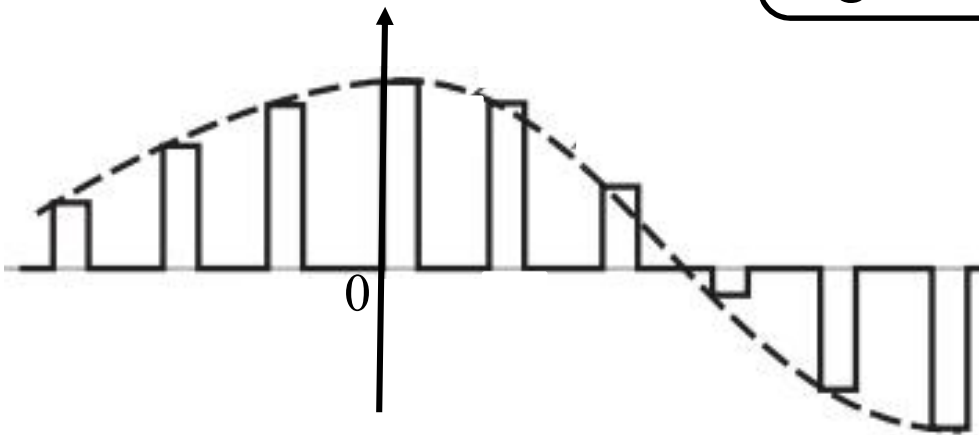
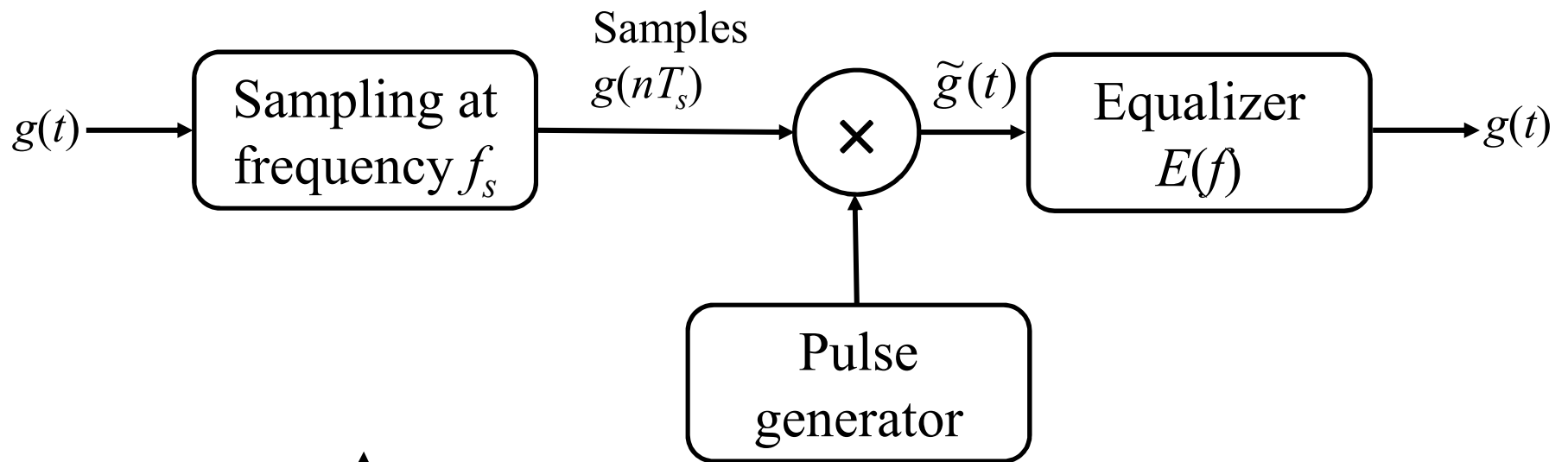


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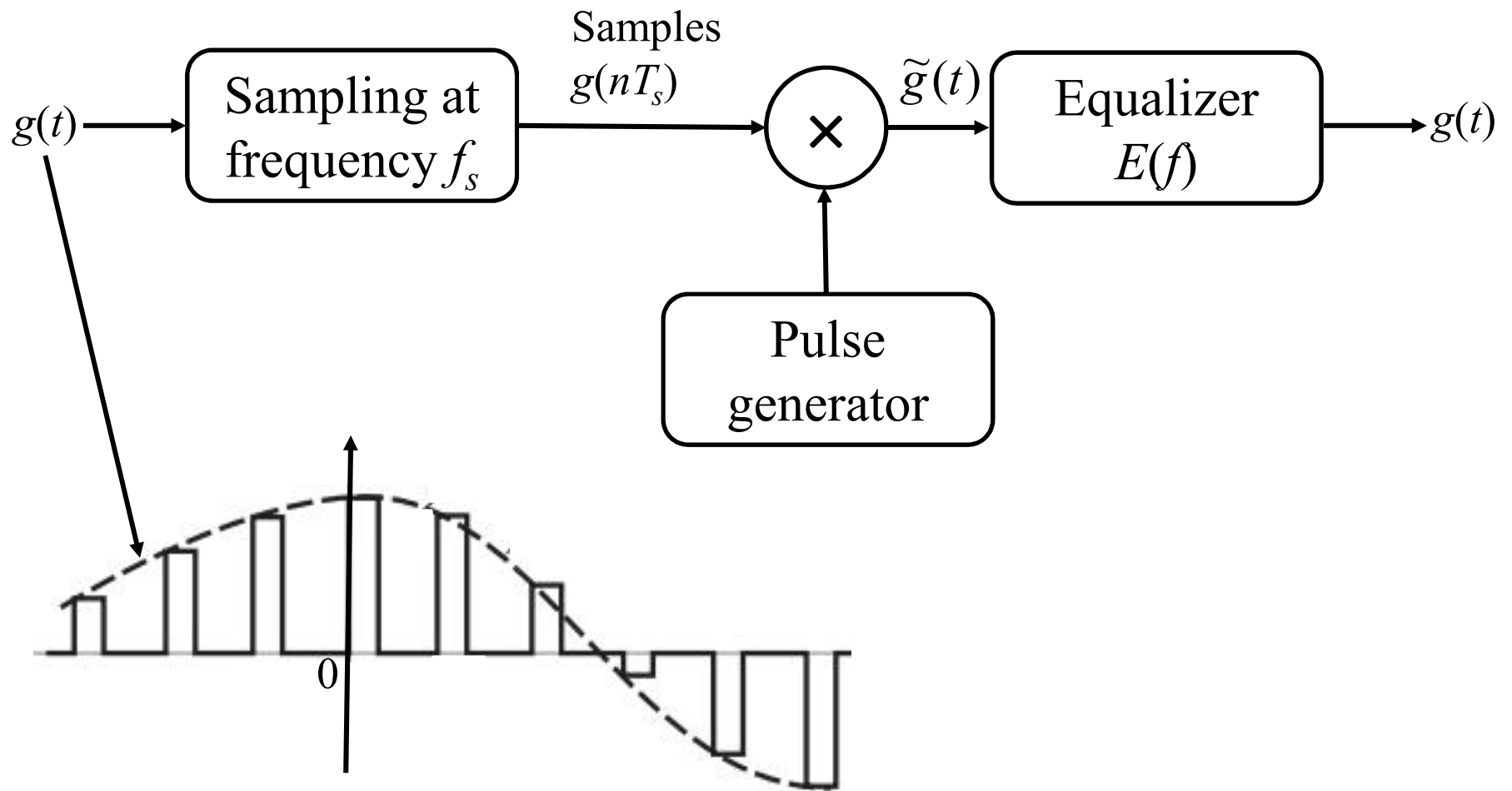


$$p(t) = \Pi\left(\frac{t - 0.5T_p}{T_p}\right)$$

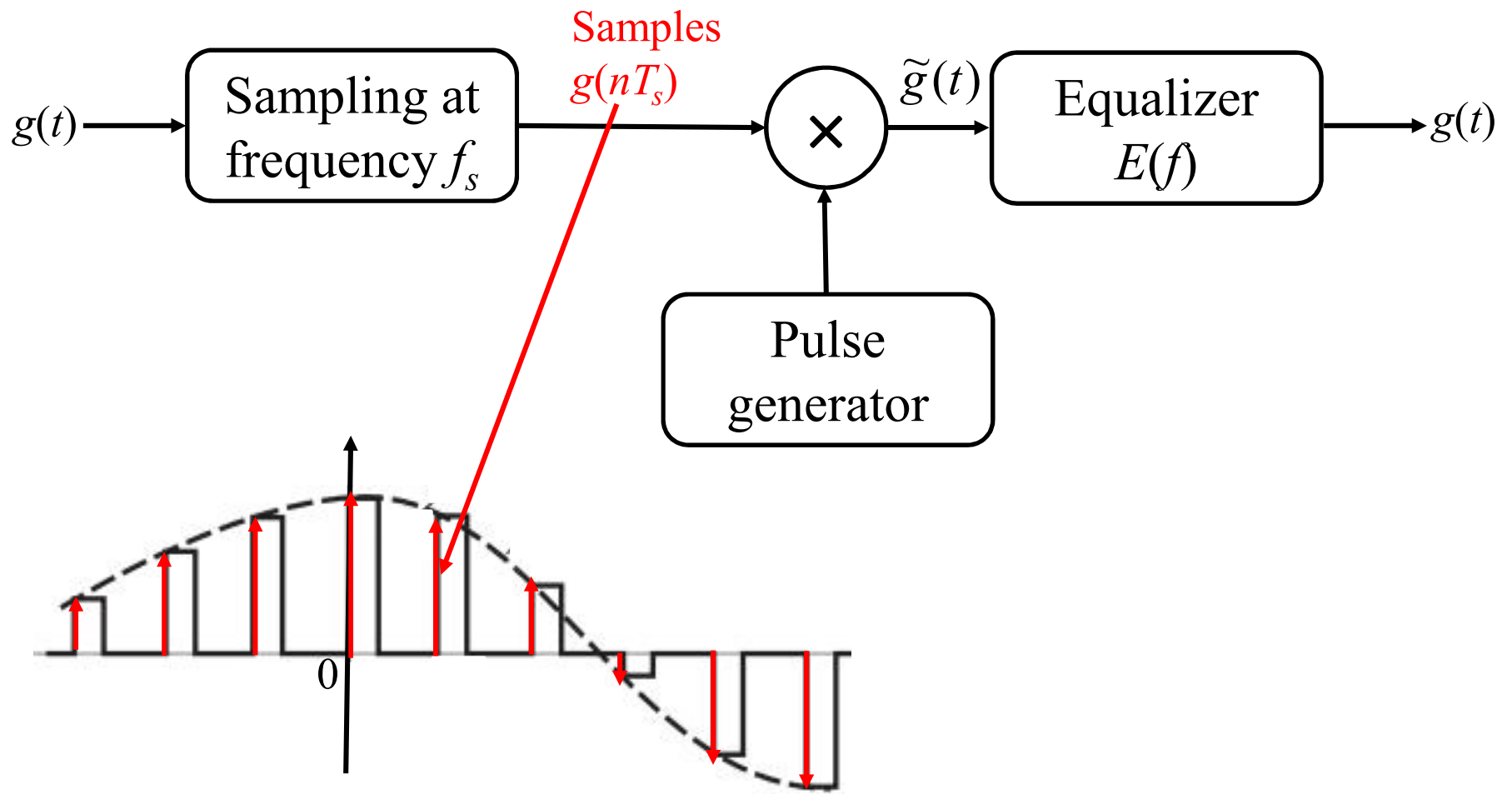
Practical Signal Reconstruction (Interpolation)



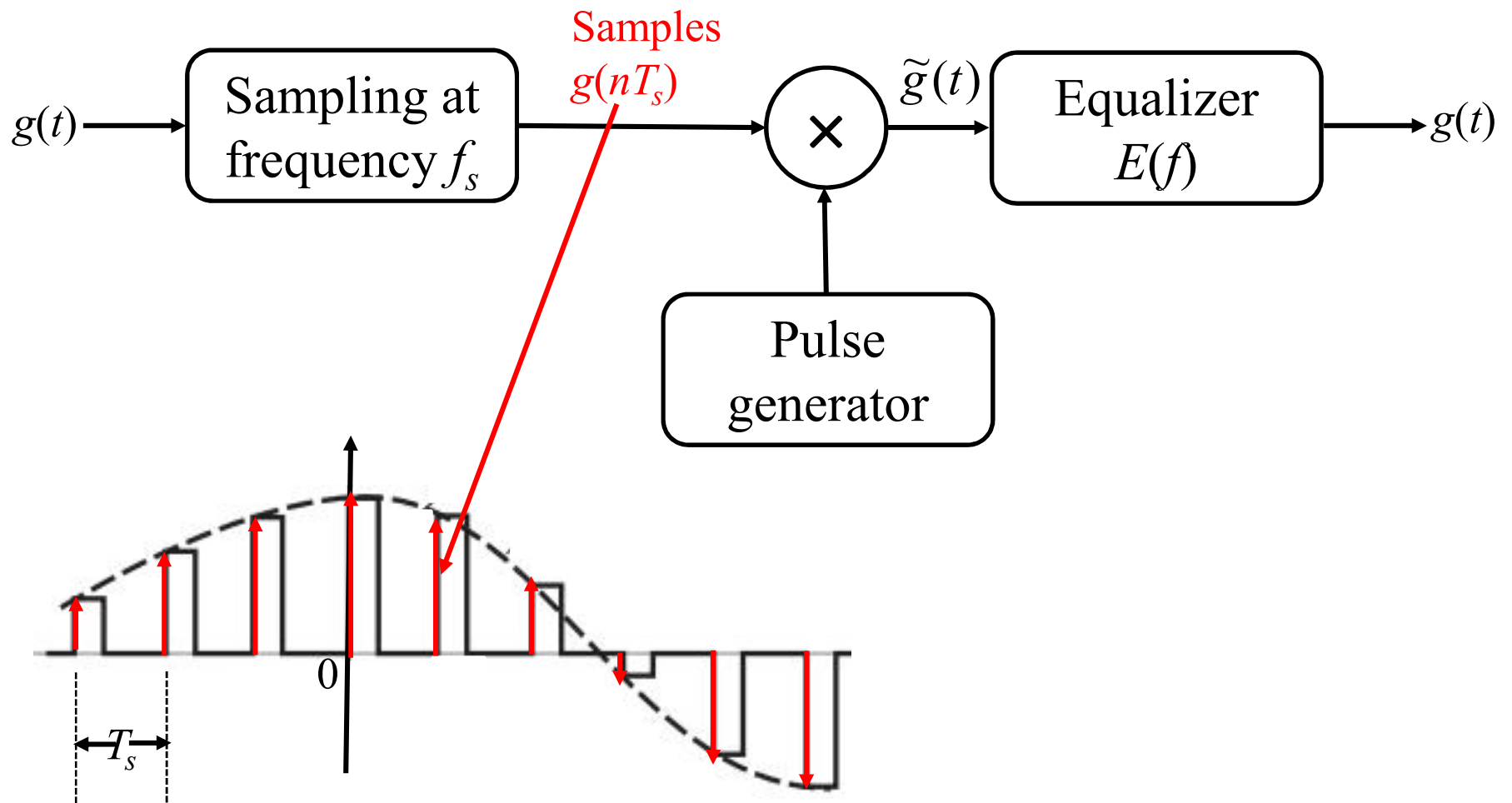
Practical Signal Reconstruction (Interpolation)



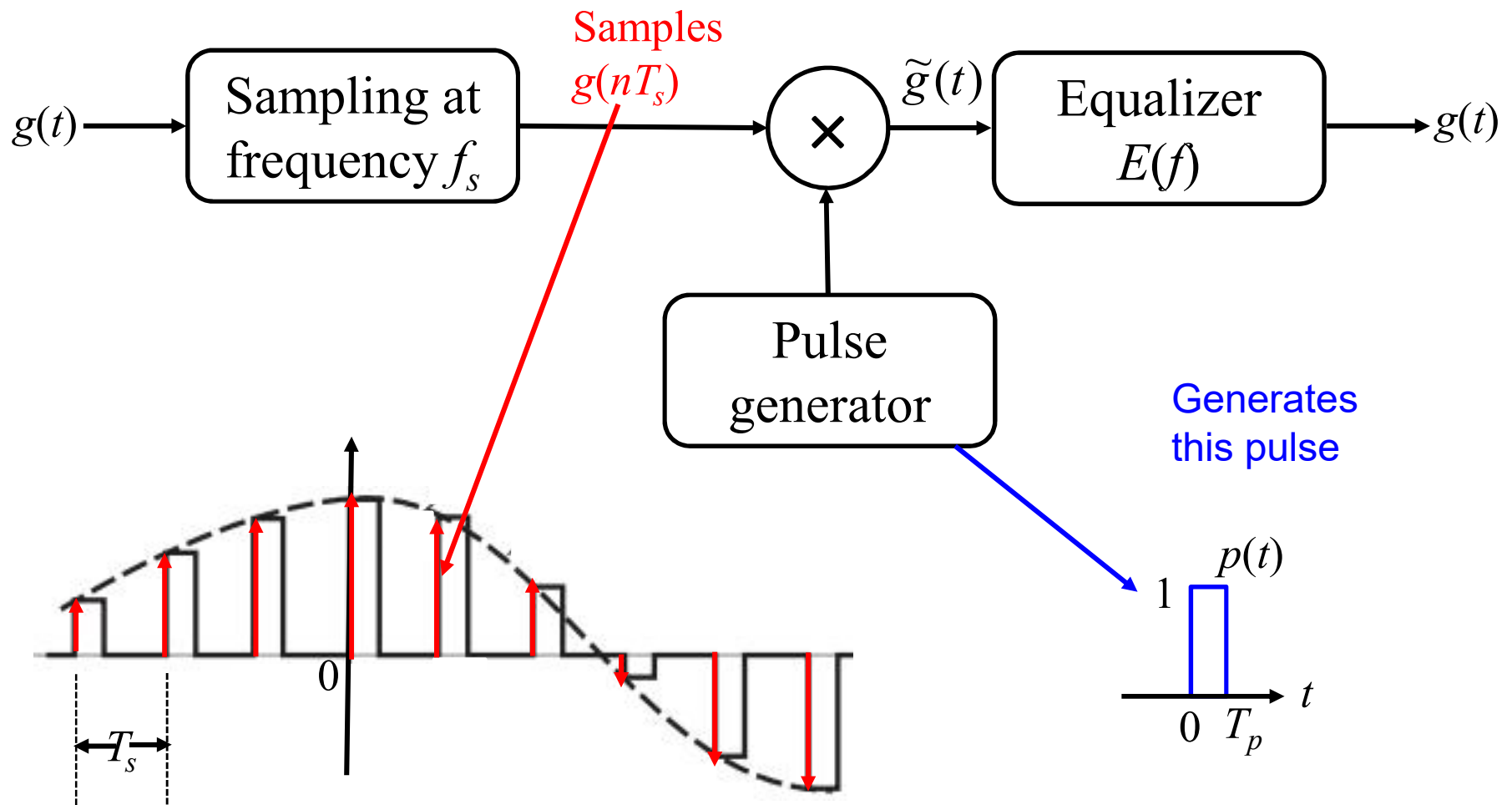
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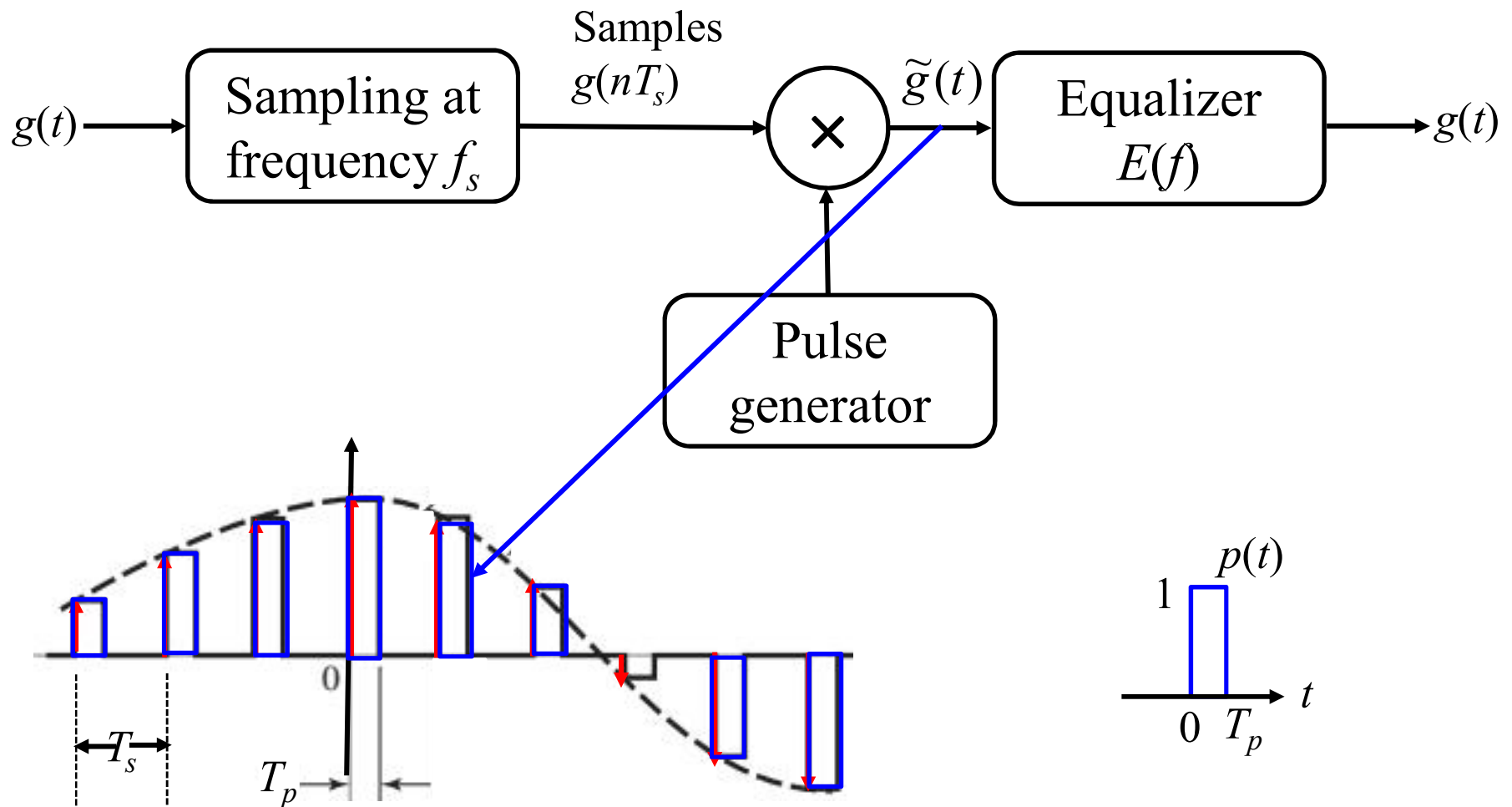
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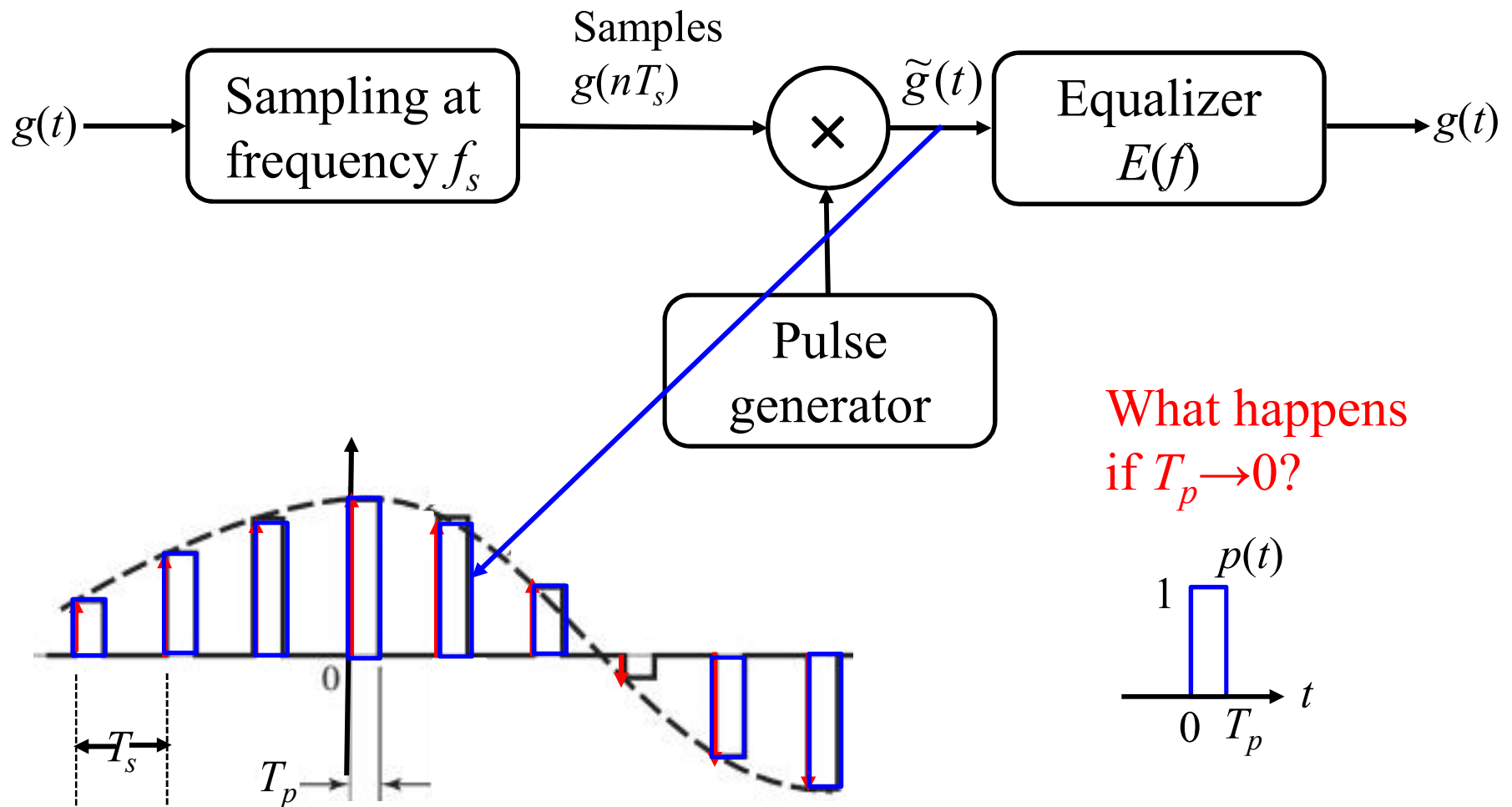
Practical Signal Reconstruction (Interpolation)



Practical Signal Reconstruction (Interpolation)

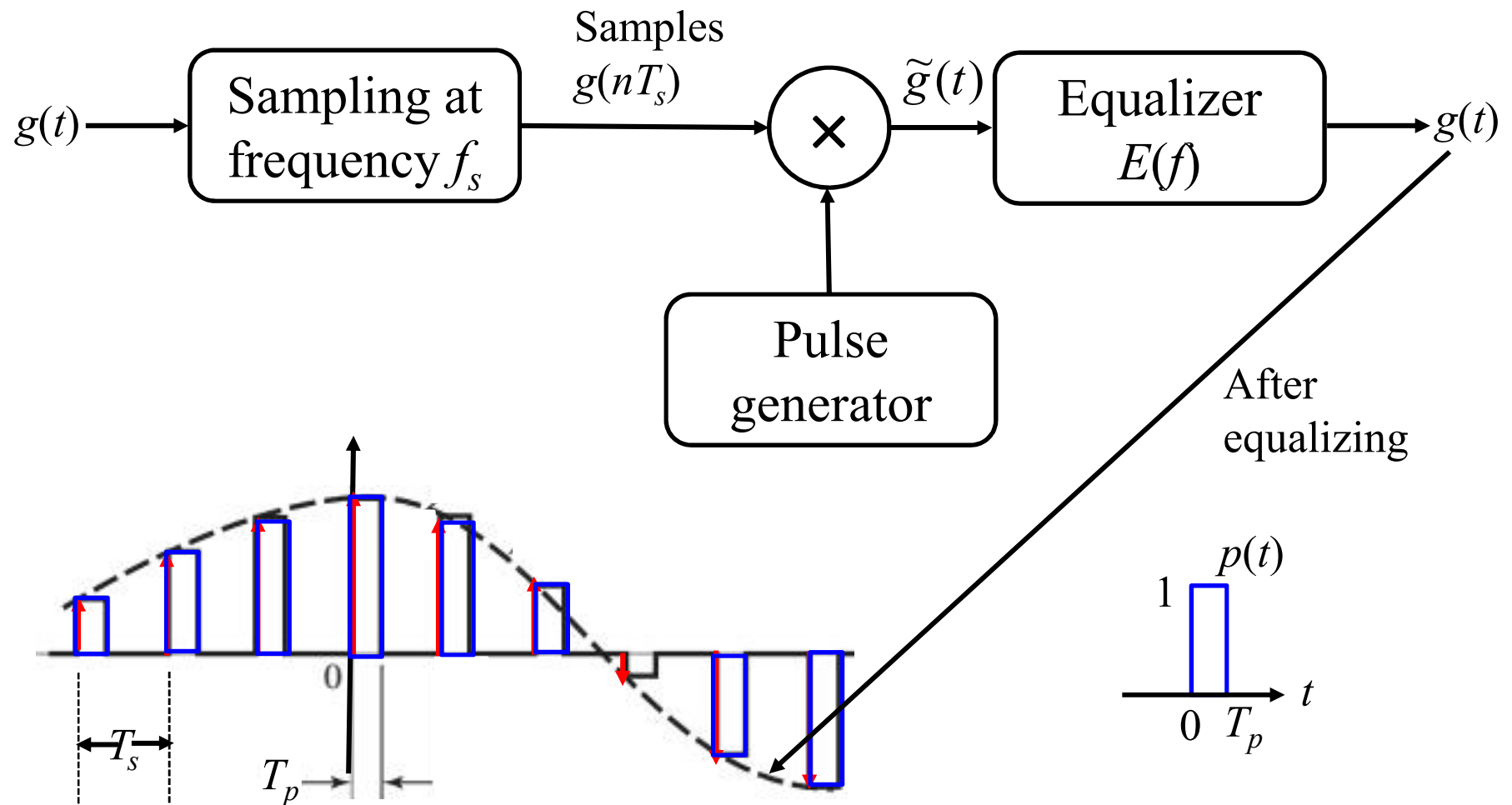


Practical Signal Reconstruction (Interpolation)

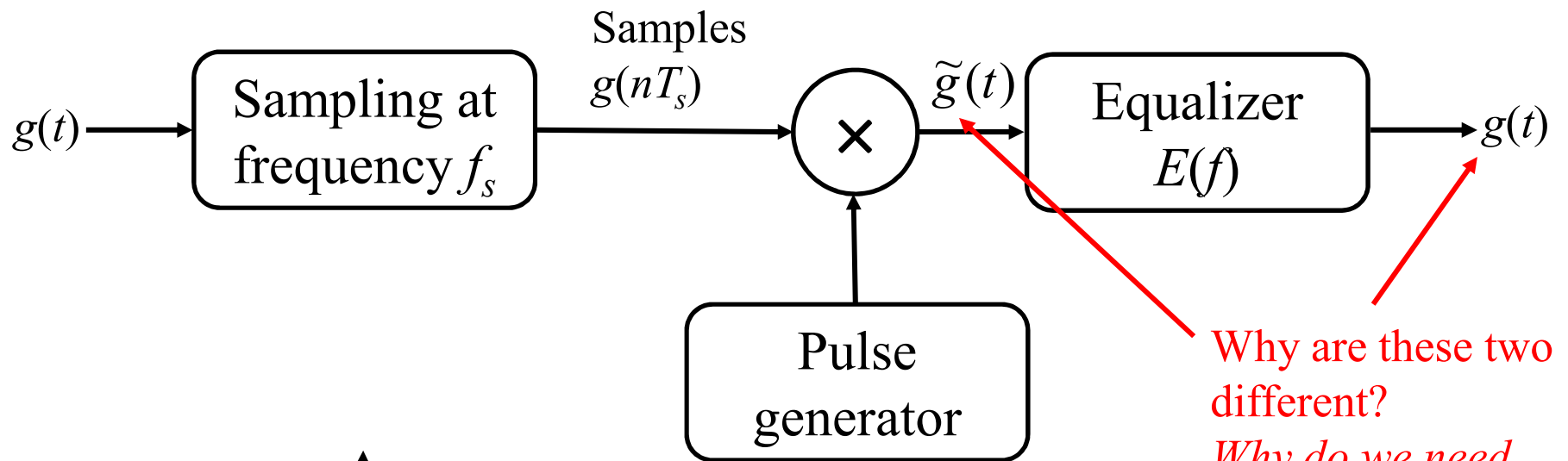


What happens
if $T_p \rightarrow 0$?

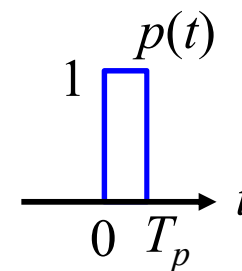
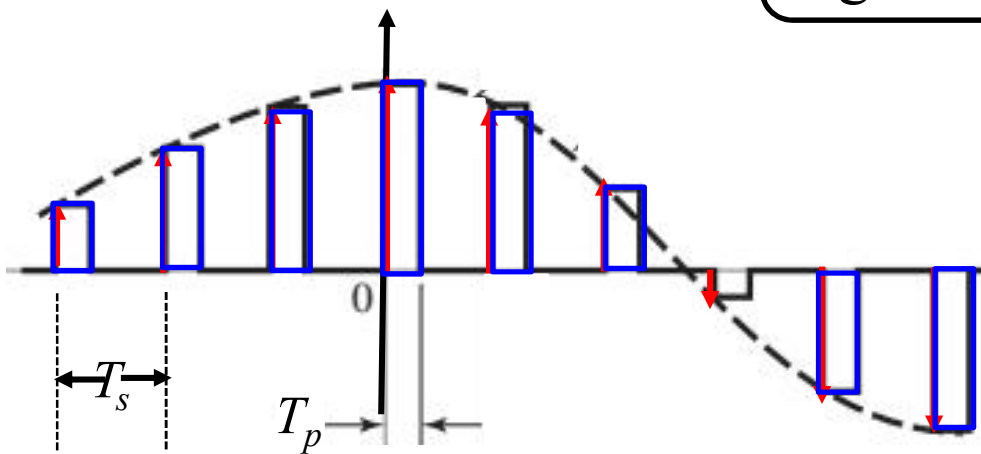
Practical Signal Reconstruction (Interpolation)



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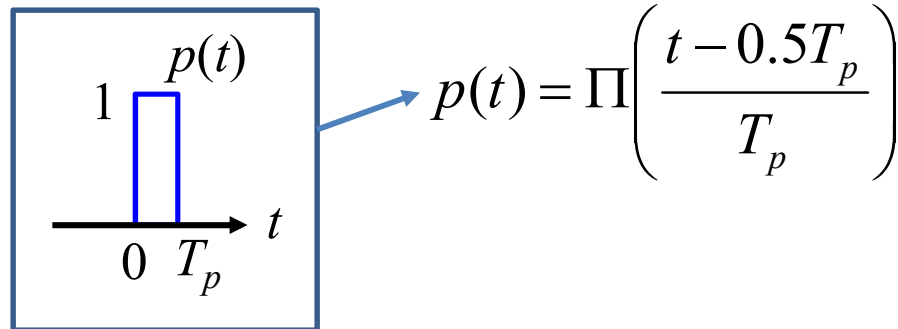


Why are these two different?
Why do we need equalizer?



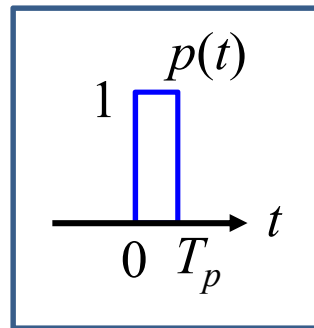
Practical Signal Reconstruction (Interpolation)

Let's find the FT
response of the
rectangular
reconstruction
pulse



Practical Signal Reconstruction (Interpolation)

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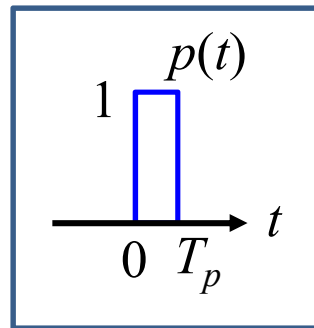


$$p(t) = \Pi\left(\frac{t - 0.5T_p}{T_p}\right)$$

$$P(f) = T_p \text{sinc}(\pi f T_p) e^{-j\pi f T_p}$$

Practical Signal Reconstruction (Interpolation)

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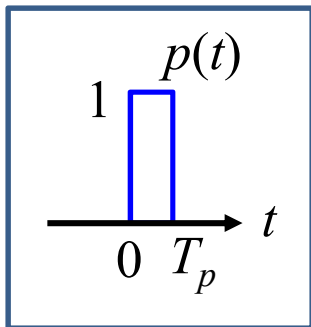
$$P(f) = T_p \text{sinc}(\pi f T_p) e^{-j\pi f T_p}$$

$$\Pi\left(\frac{t}{\tau}\right) \Leftrightarrow \tau \text{sinc}(\pi f \tau)$$

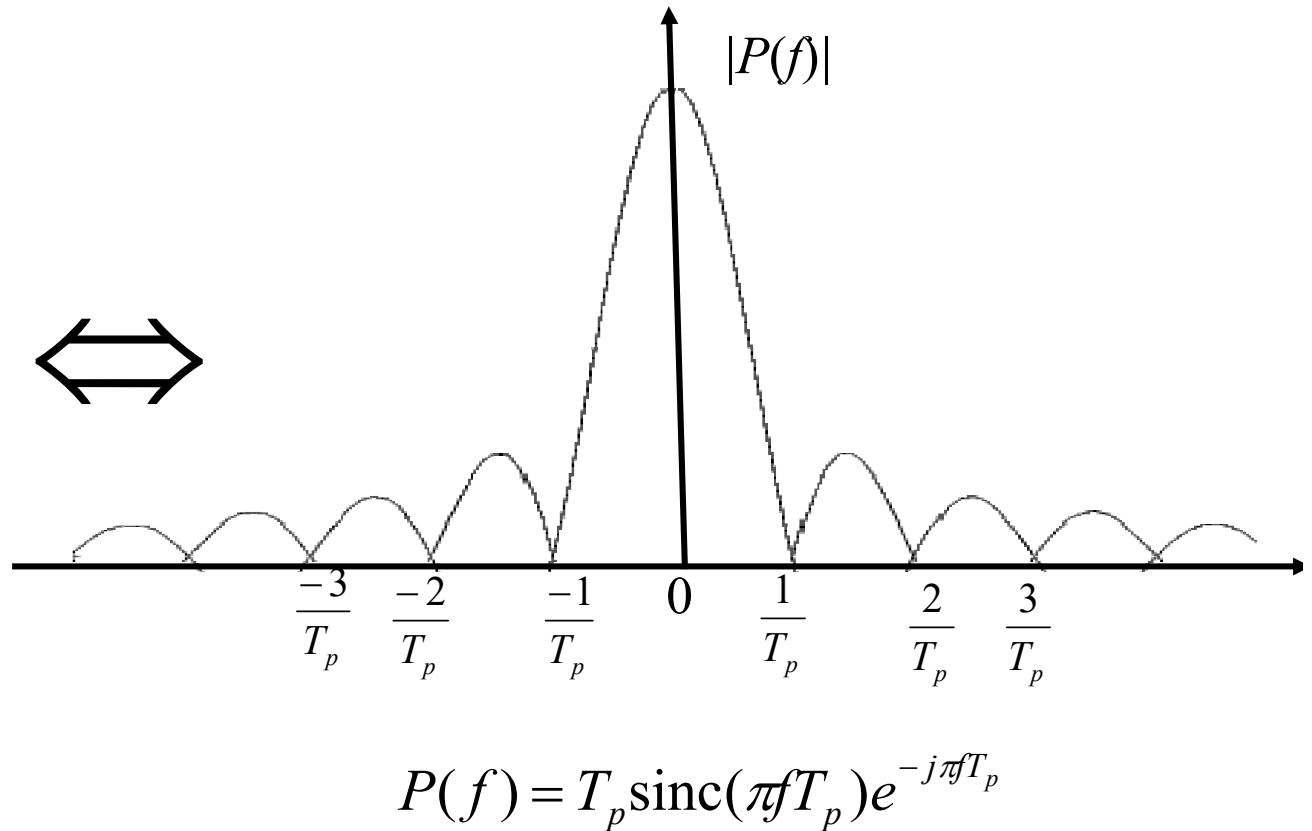
$$g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$$

Practical Signal Reconstruction (Interpolation)

rectangular
reconstruction pulse

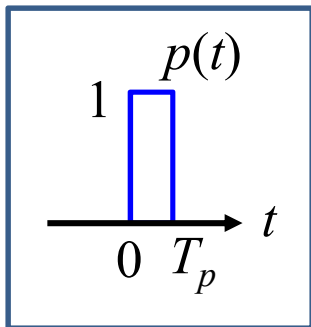


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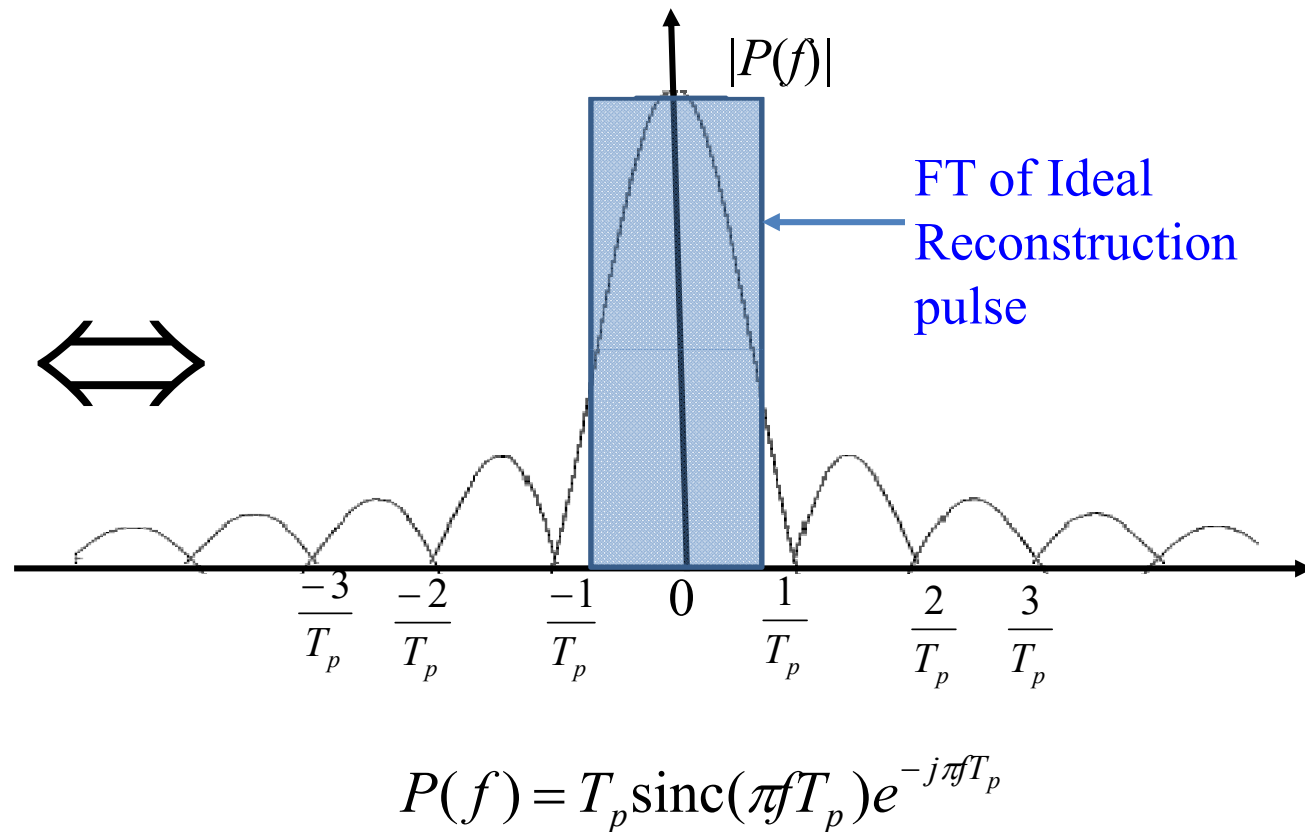


Practical Signal Reconstruction (Interpolation)

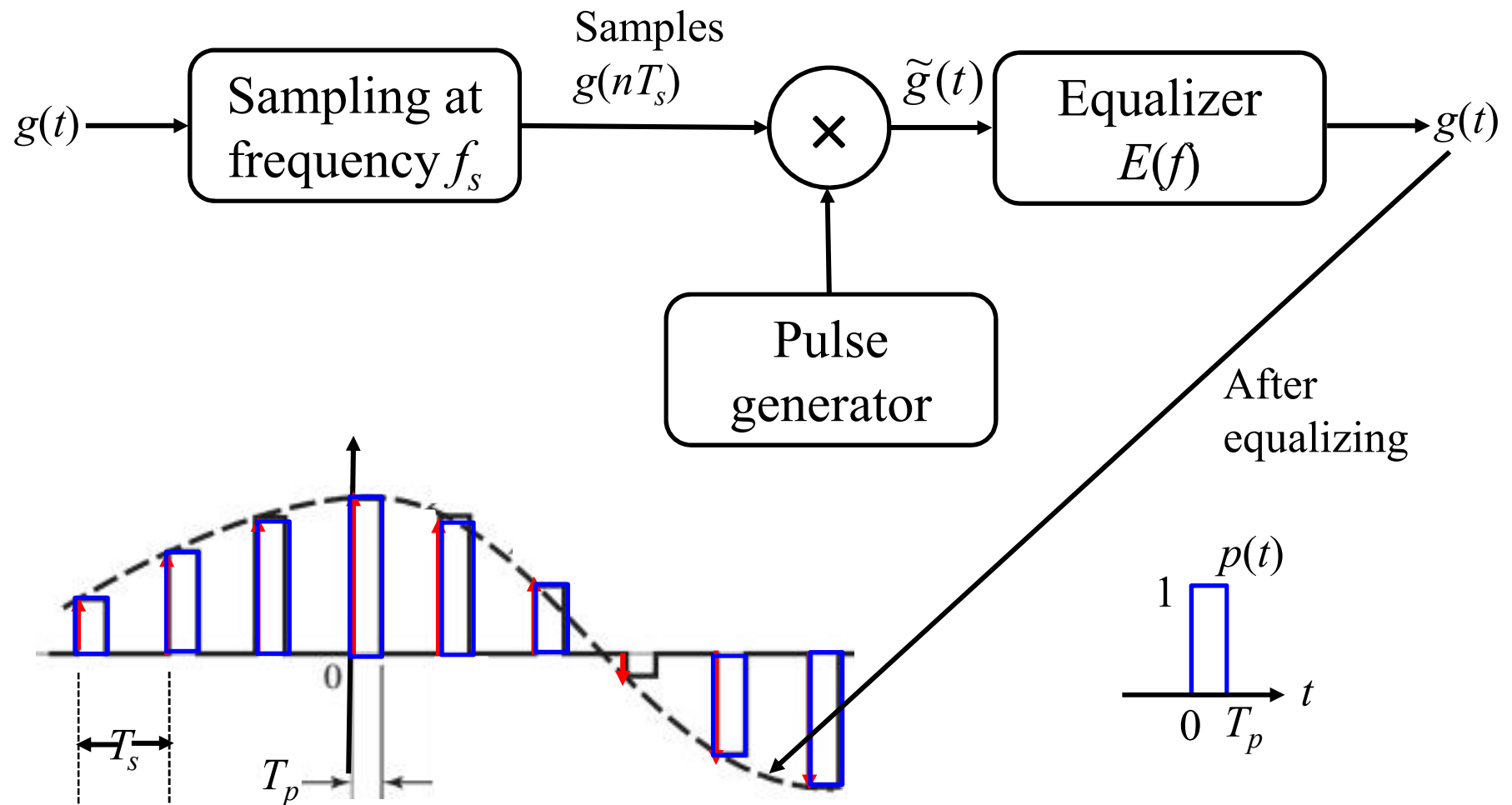
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Practical Signal Reconstruction (Interpolation)



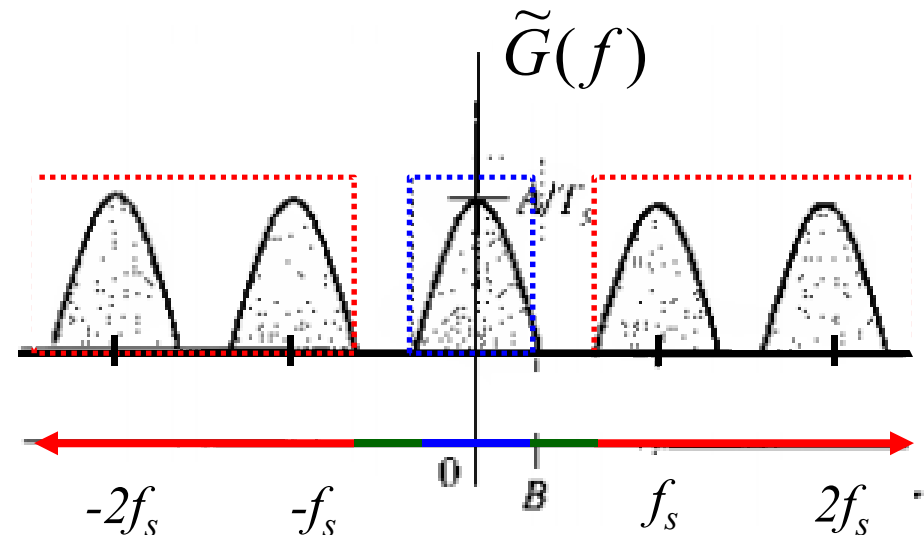
Practical Signal Reconstruction (Interpolation)

$$\tilde{G}(f) = P(f) \frac{1}{T_s} \sum_n G(f - nf_s)$$

$$\begin{aligned} G(f) &= E(f) \tilde{G}(f) \\ &= E(f) P(f) \frac{1}{T_s} \sum_n G(f - nf_s) \end{aligned}$$

Equalizer

$$E(f) = \begin{cases} T_s / P(f) & |f| \leq B \\ \text{Flexible} & B < |f| \leq f_s - B \\ 0 & |f| \geq f_s - B \end{cases}$$

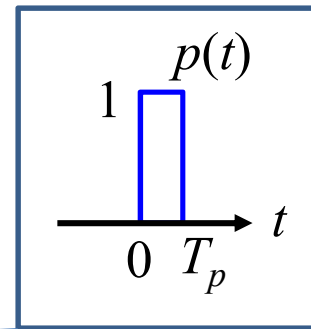


Practical Signal Reconstruction (Interpolation)

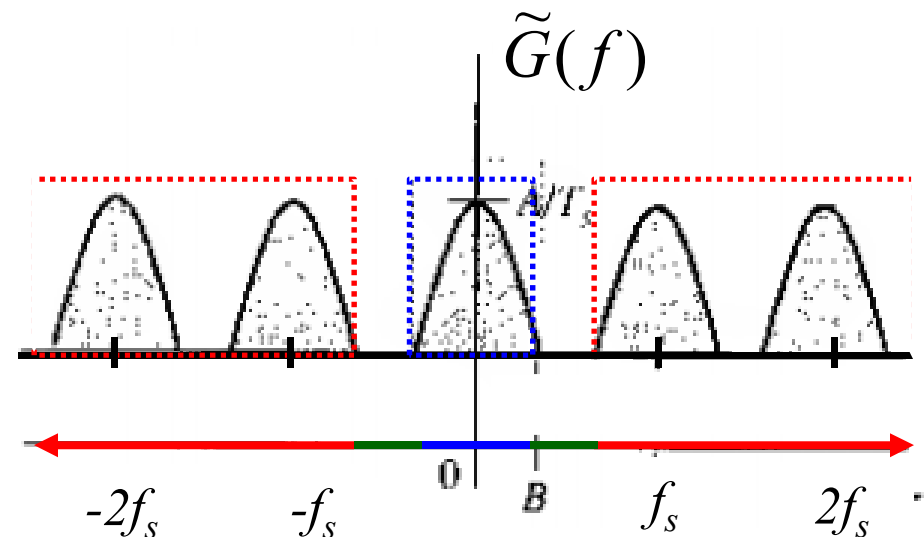
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$$G(f) = E(f) \tilde{G}(f)$$

$$= E(f) P(f) \frac{1}{T_s} \sum_n G(f - nf_s)$$



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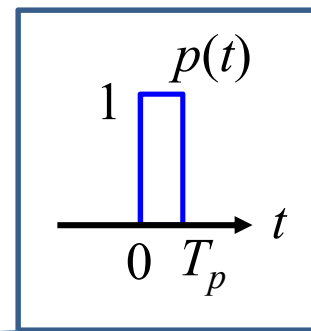


Practical Signal Reconstruction (Interpolation)

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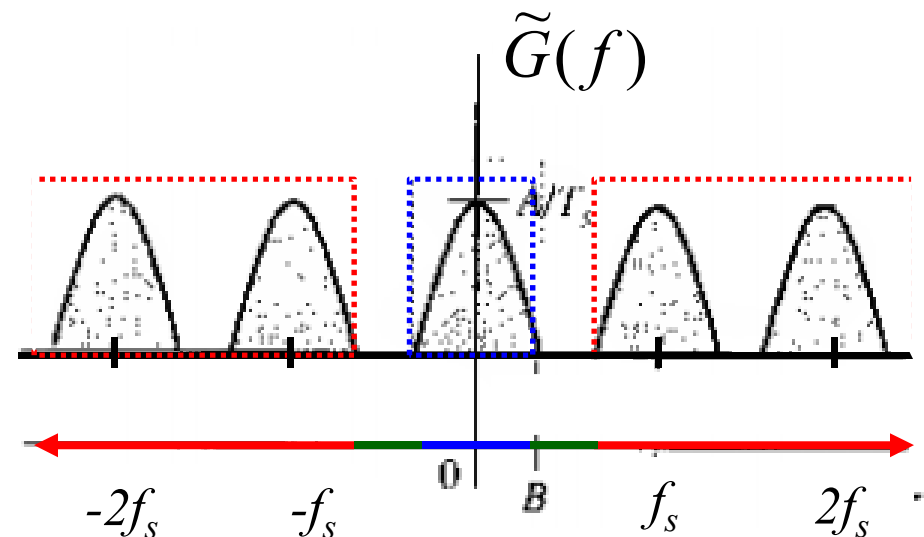
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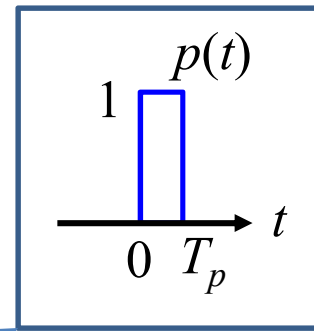


Practical Signal Reconstruction (Interpolation)

$$\tilde{G}(f) = P(f) \frac{1}{T_s} \sum_n G(f - nf_s)$$

$$G(f) = E(f) \tilde{G}(f)$$

$$= E(f) P(f) \frac{1}{T_s} \sum_n G(f - nf_s)$$



$$p(t) = \Pi\left(\frac{t - 0.5T_p}{T_p}\right)$$

$$P(f) = T_p \text{sinc}(\pi f T_p) e^{-j\pi f T_p}$$

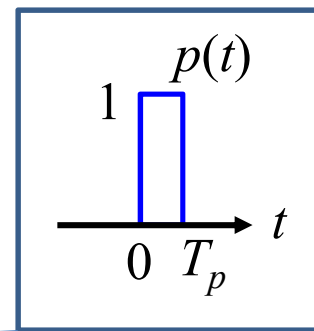
$$E(f) = \begin{cases} T_s / P(f) & |f| \leq B \\ \text{Flexible} & B < |f| \leq f_s - B \\ 0 & |f| \geq f_s - B \end{cases}$$

Practical Signal Reconstruction (Interpolation)

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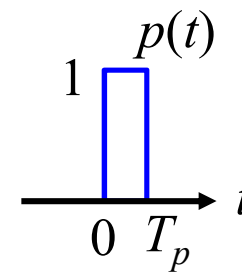
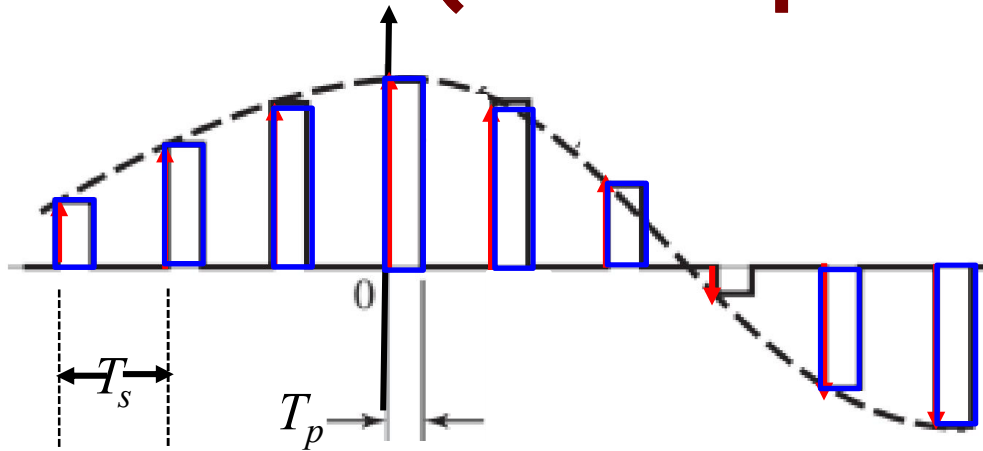
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$$\longrightarrow |E(f)| = T_s \cdot \left| \frac{\pi f}{\sin(\pi f T_p)} \right| \quad |f| \leq B$$

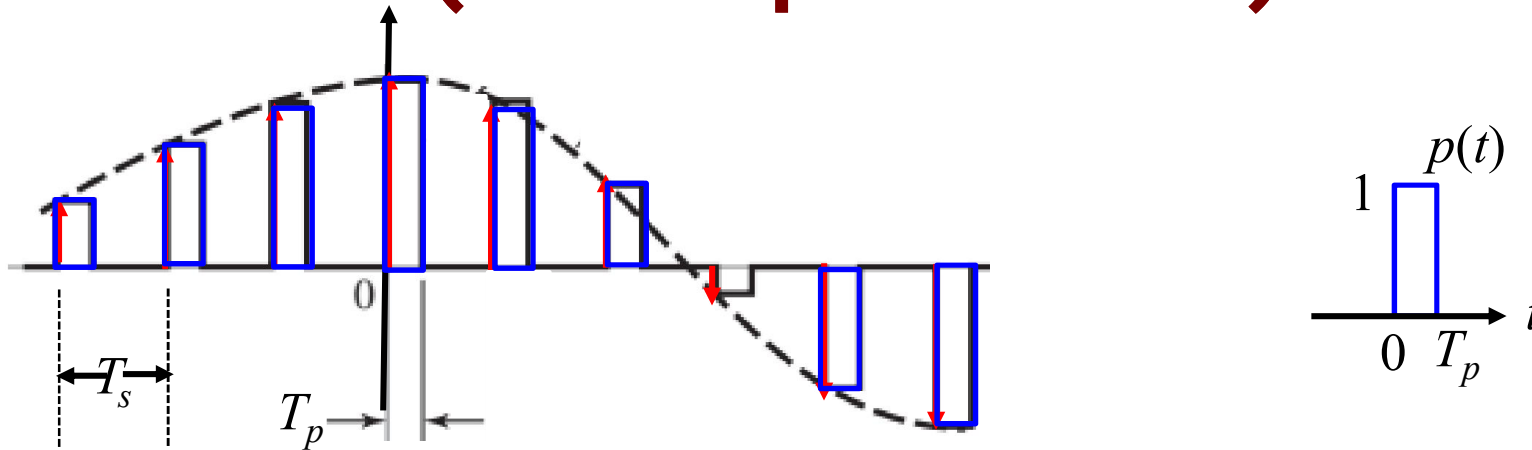
Practical Signal Reconstruction (Interpolation)



$$|E(f)| = T_s \cdot \left| \frac{\pi f}{\sin(\pi f T_p)} \right| \quad |f| \leq B$$

T_p must be $< 1/B$,
otherwise signal will be
distorted

Practical Signal Reconstruction (Interpolation)

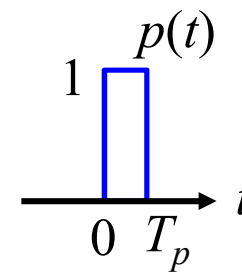
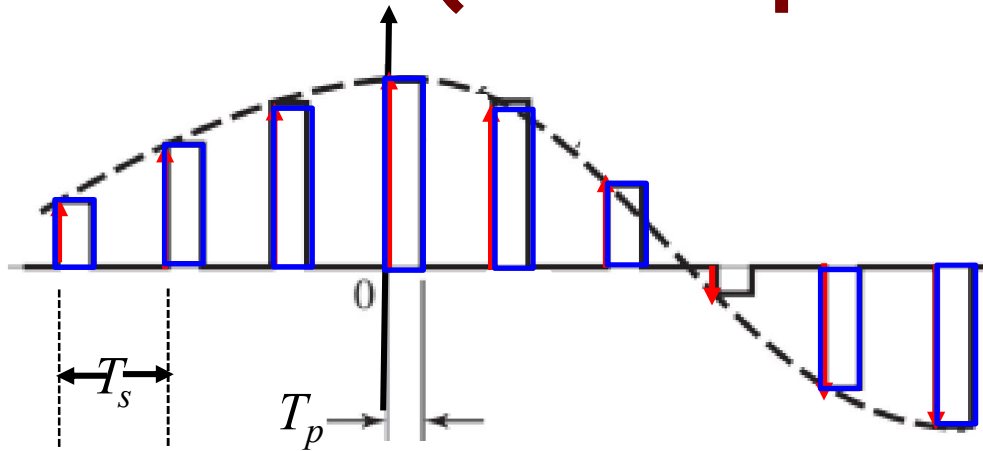


$$|E(f)| = T_s \cdot \left| \frac{\pi f}{\sin(\pi f T_p)} \right| \quad |f| \leq B$$

T_p must be $< 1/B$,
otherwise signal will be
distorted

$T_p > 0$, otherwise $E(f)$ will
have infinite gain

Practical Signal Reconstruction (Interpolation)

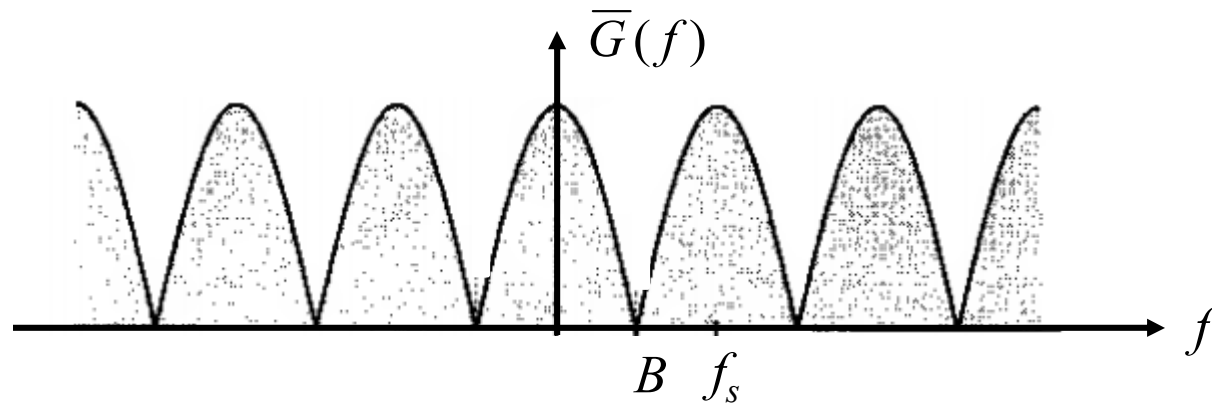


in special case when T_p is very small,

$$|E(f)| = T_s \cdot \left| \frac{\pi f}{\sin(\pi f T_p)} \right| \approx \frac{T_s}{T_p} \quad |f| \leq B$$

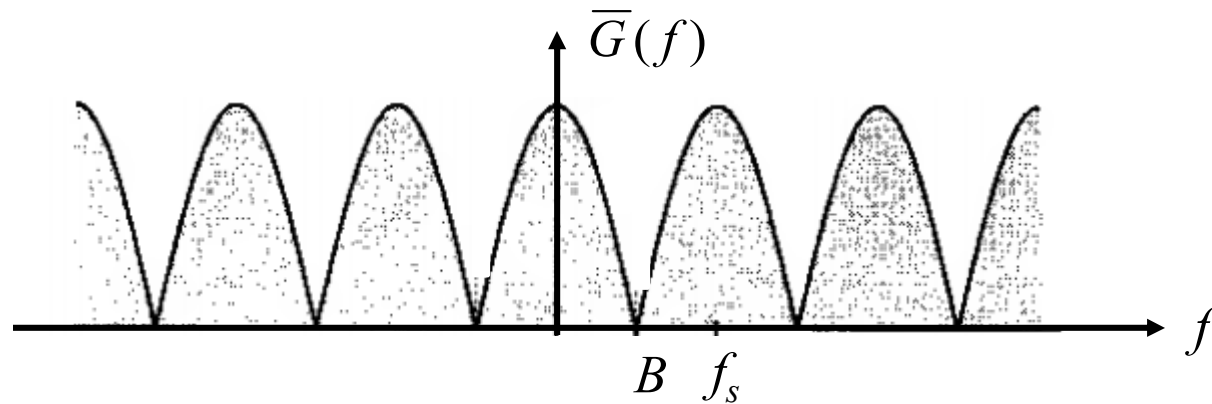
Realizability of Reconstruction Filters

Spectrum $\bar{G}(f)$ when signal $g(t)$ is
sampled at the Nyquist rate $f_s = 2B$



Realizability of Reconstruction Filters

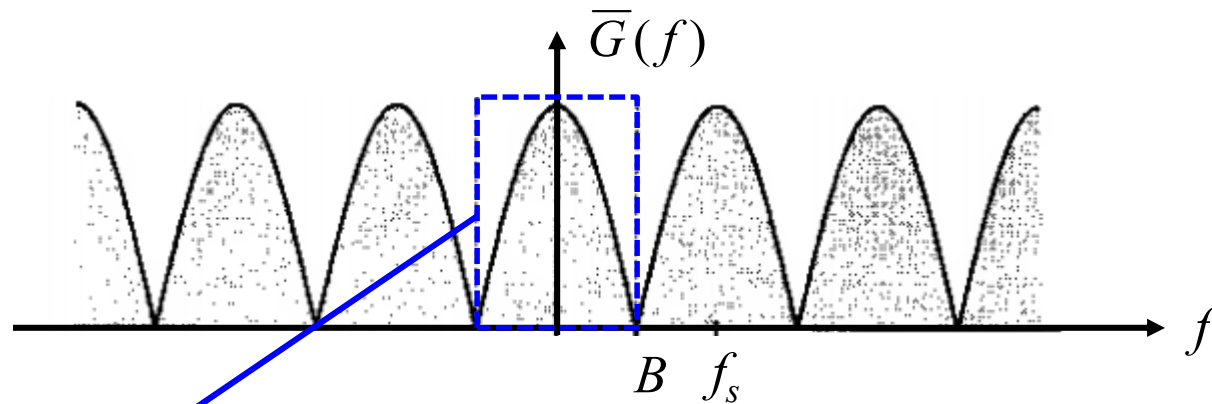
Spectrum $\overline{G}(f)$ when signal $g(t)$ is sampled at the Nyquist rate $f_s = 2B$



Repetitions of $G(f)$ **without any gap** between successive cycles

Realizability of Reconstruction Filters

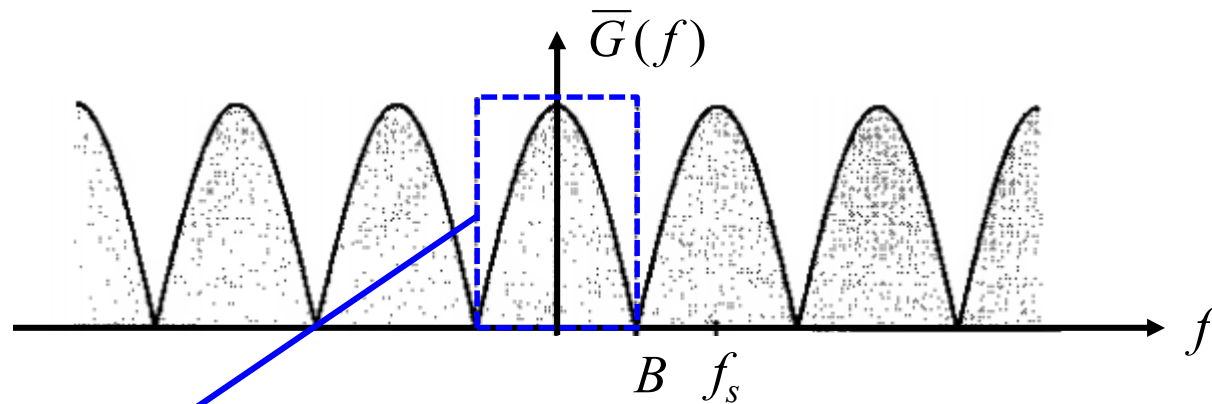
Spectrum $\bar{G}(f)$ when signal $g(t)$ is sampled at the Nyquist rate $f_s = 2B$



Ideal reconstruction
filter (*Box filter*)

Realizability of Reconstruction Filters

Spectrum $\bar{G}(f)$ when signal $g(t)$ is sampled at the Nyquist rate $f_s = 2B$

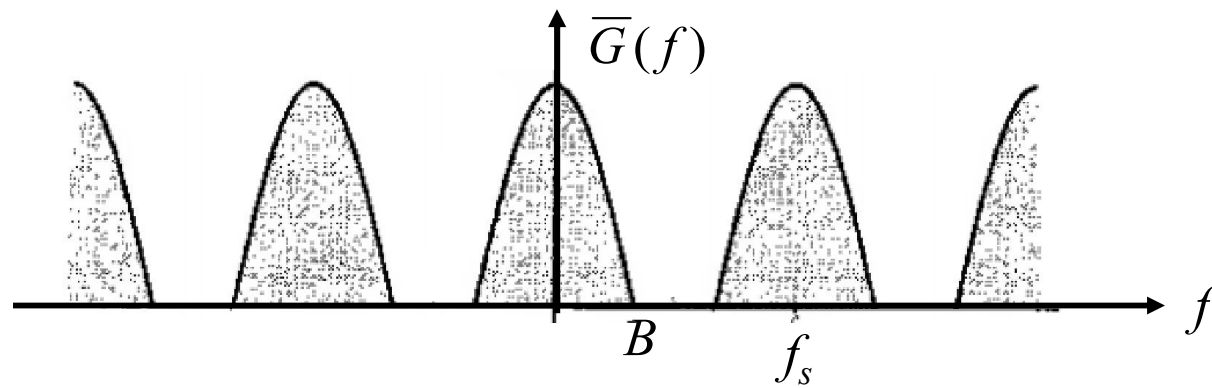


Ideal reconstruction
filter (*Box filter*)

But box filter is unrealizable; can be
approximated only with infinite time delay

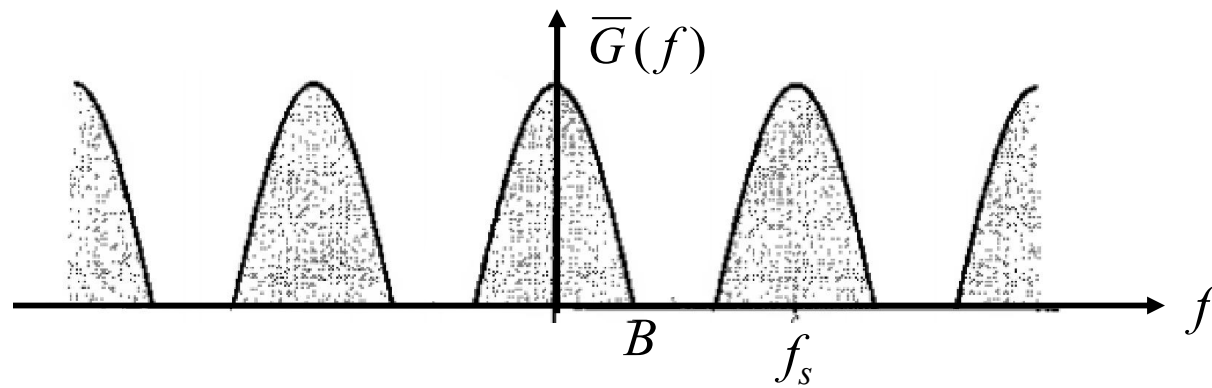
Realizability of Reconstruction Filters

Alternate: Sample at a rate higher than
Nyquist rate $f_s > 2B$



Realizability of Reconstruction Filters

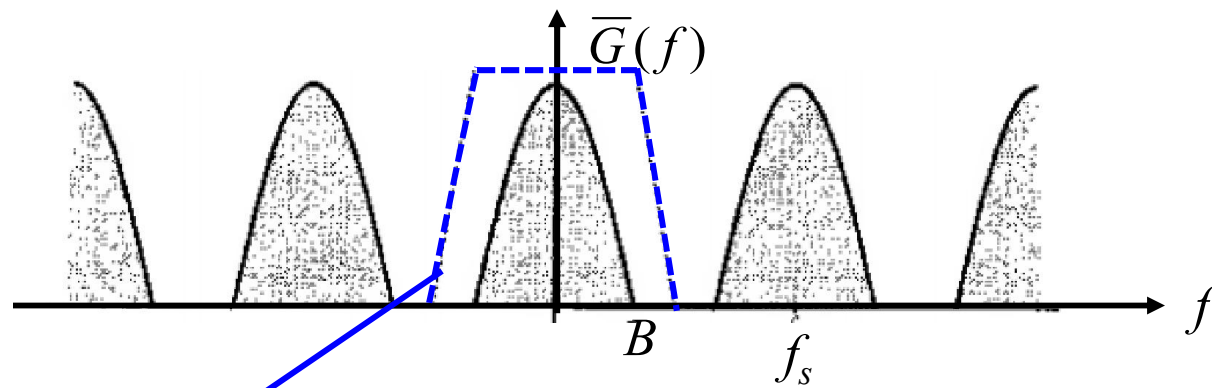
Alternate: Sample at a rate higher than
Nyquist rate $f_s > 2B$



Repetitions of $G(f)$ with finite gap
between successive cycles

Realizability of Reconstruction Filters

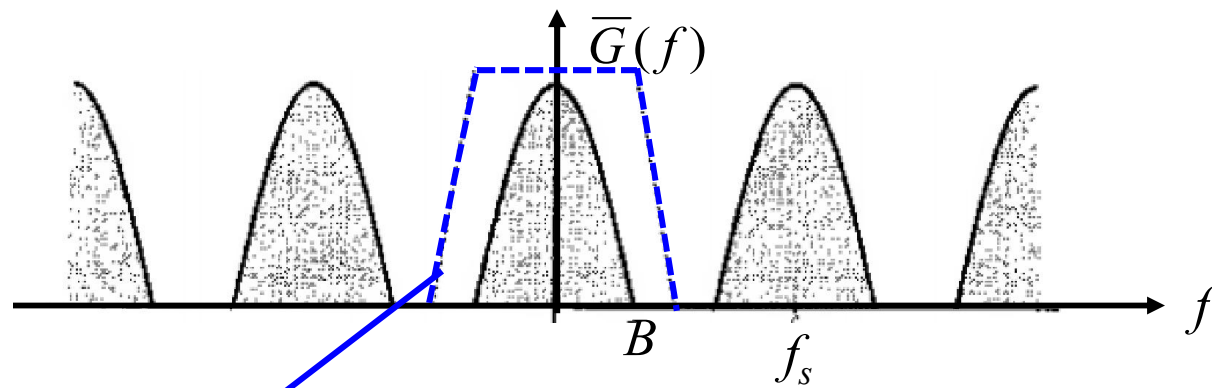
Alternate: Sample at a rate higher than Nyquist rate $f_s > 2B$



Low pass filter with gradual cutoff characteristics

Realizability of Reconstruction Filters

Alternate: Sample at a rate higher than Nyquist rate $f_s > 2B$

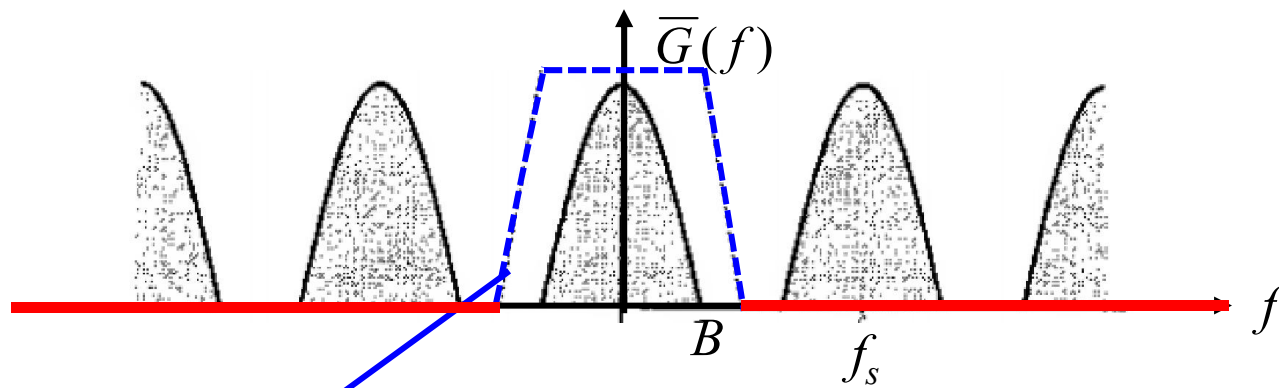


Low pass filter with gradual cutoff characteristics

Advantage: requires smaller time delay

Realizability of Reconstruction Filters

Alternate: Sample at a rate higher than Nyquist rate $f_s > 2B$



Low pass filter with gradual cutoff characteristics

Disadvantage: filter gain is zero beyond first cycle of $G(f)$ $\rightarrow$ fails to satisfy Paley-Wiener criteria

Realizability of Reconstruction Filters

Findings:

- Exact recovery of $g(t)$ from samples is impossible, even $f_s > 2B$
- As f_s increases, recovered signal approaches $g(t)$

Wrong Assumption on Bandwidth: Treachery of Aliasing

Definition of aliasing

- A **false** or **assumed** identity: same person, different names
- an **alternative name** or **label** that refers to a file, command, address, or other item, and can be used to locate or access it.

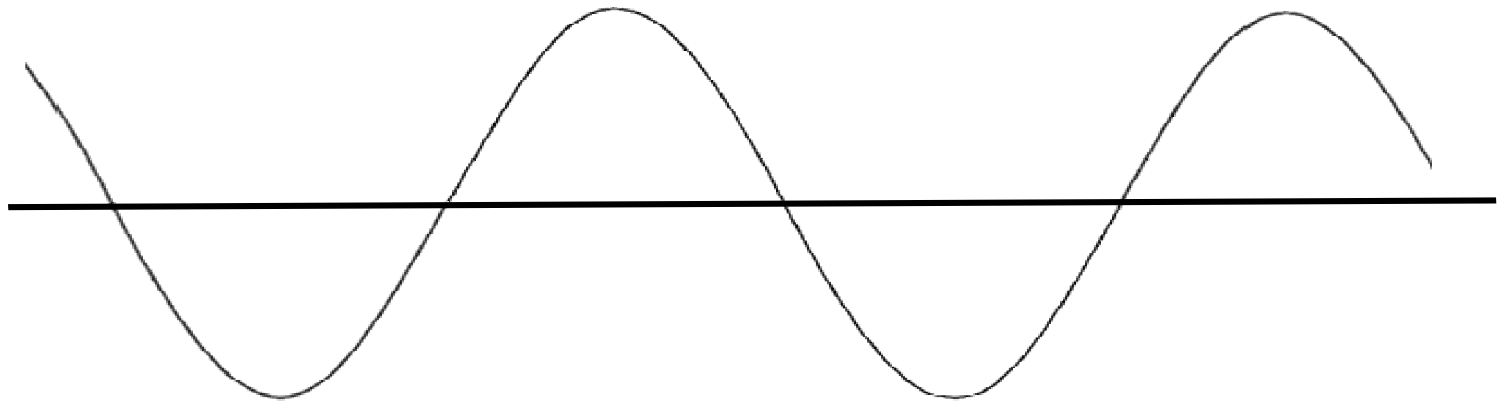
Wrong Assumption on Bandwidth: Treachery of Aliasing

Definition of aliasing in signal processing

- each of a group of signal frequencies that, when sampled at a given uniform rate, would give the same set of sampled values, and thus might be incorrectly substituted for one another when reconstructing the original signal

Wrong Assumption on Bandwidth: Treachery of Aliasing

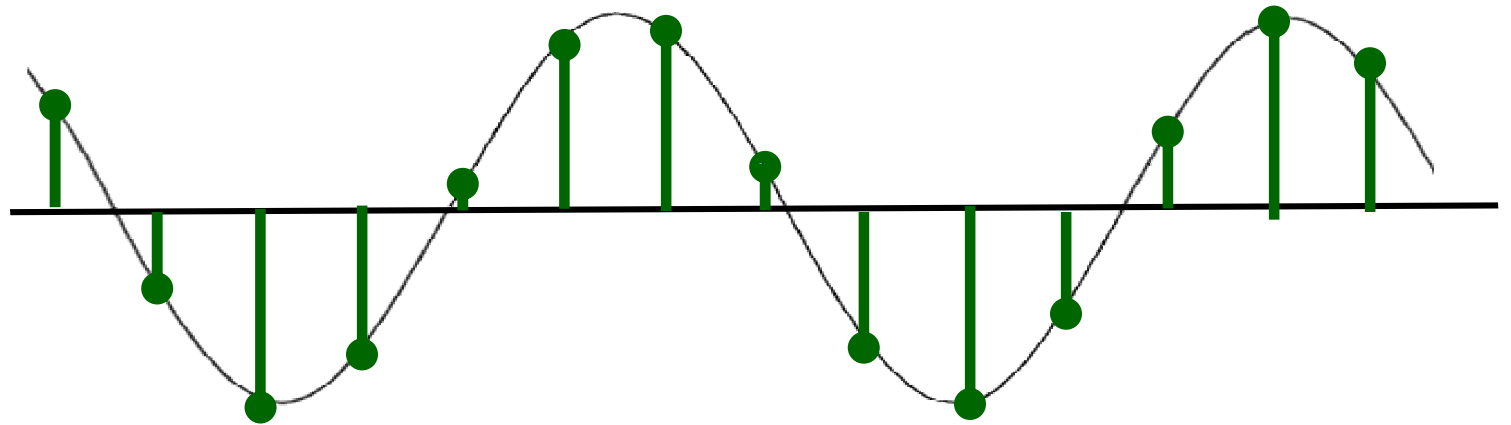
Definition of aliasing



A signal of 60 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

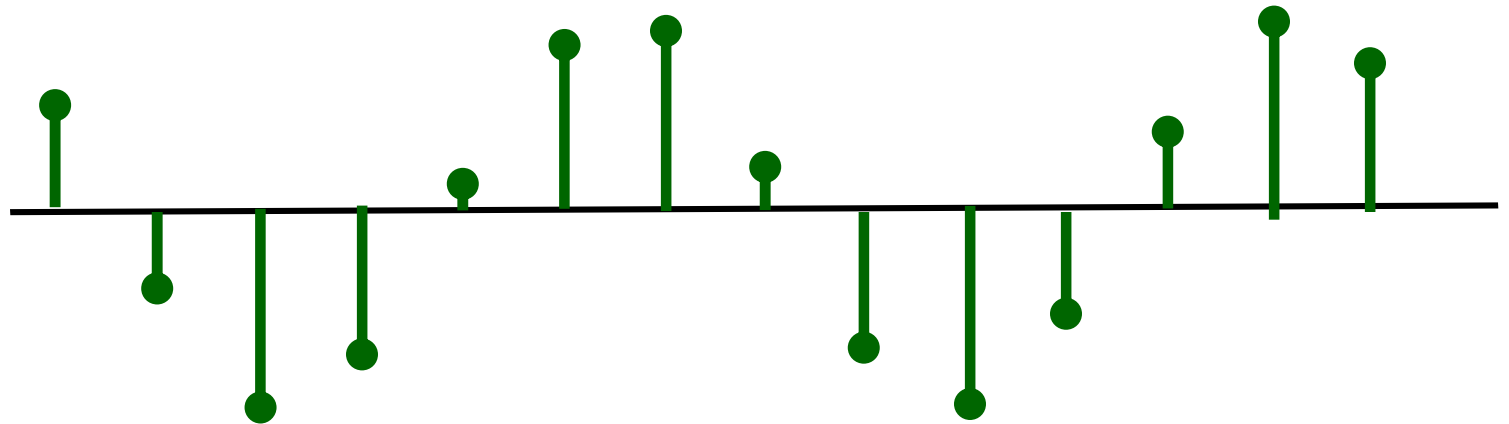
Definition of aliasing



Sampled at 400 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

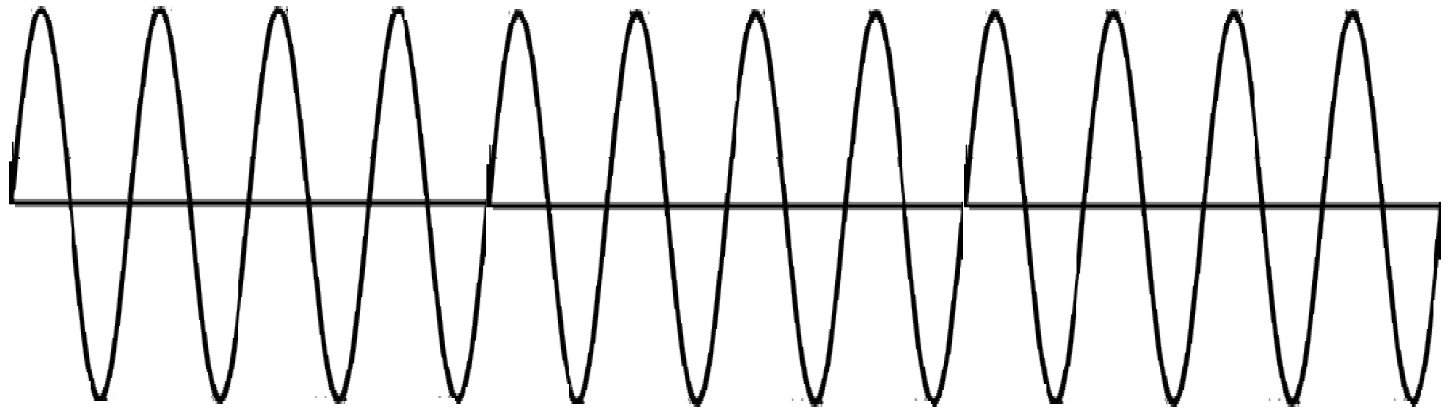
Definition of aliasing



Sampled Signal

Wrong Assumption on Bandwidth: Treachery of Aliasing

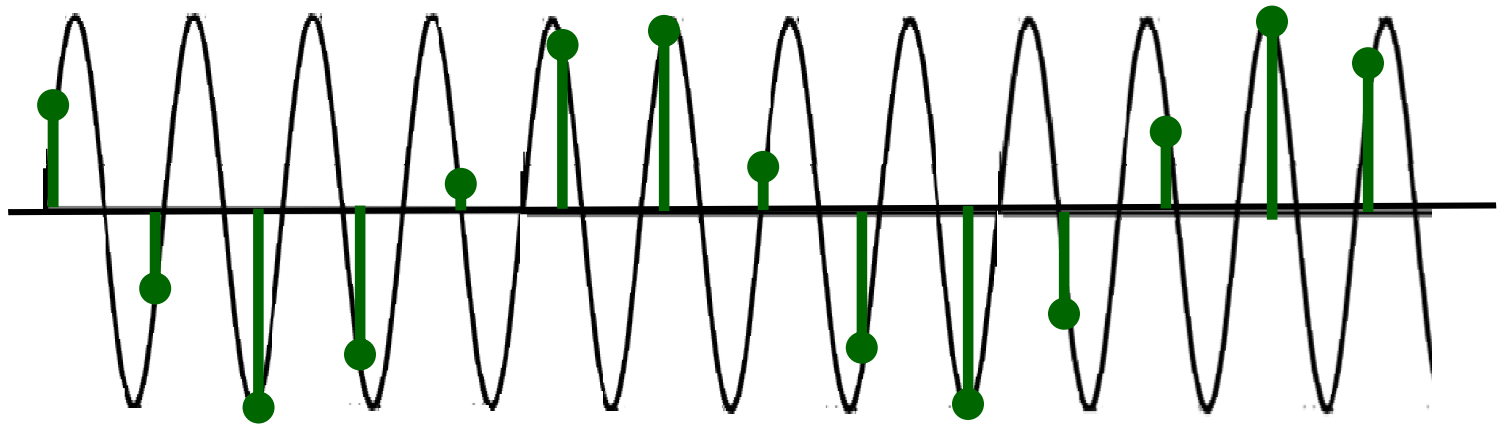
Definition of aliasing



Another signal with 340 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

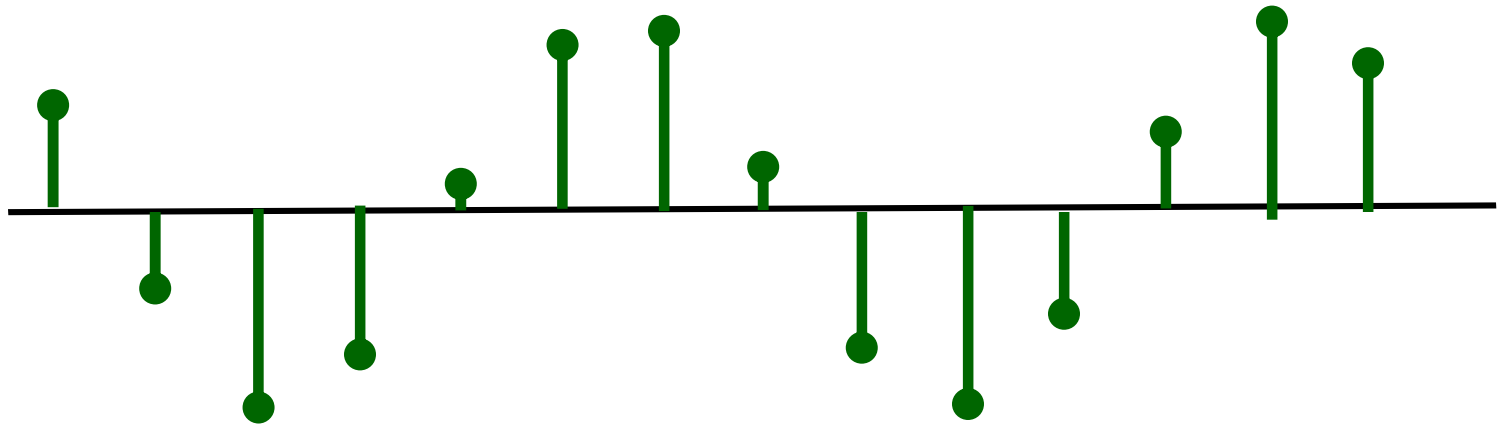
Definition of aliasing



Sampled at 400 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

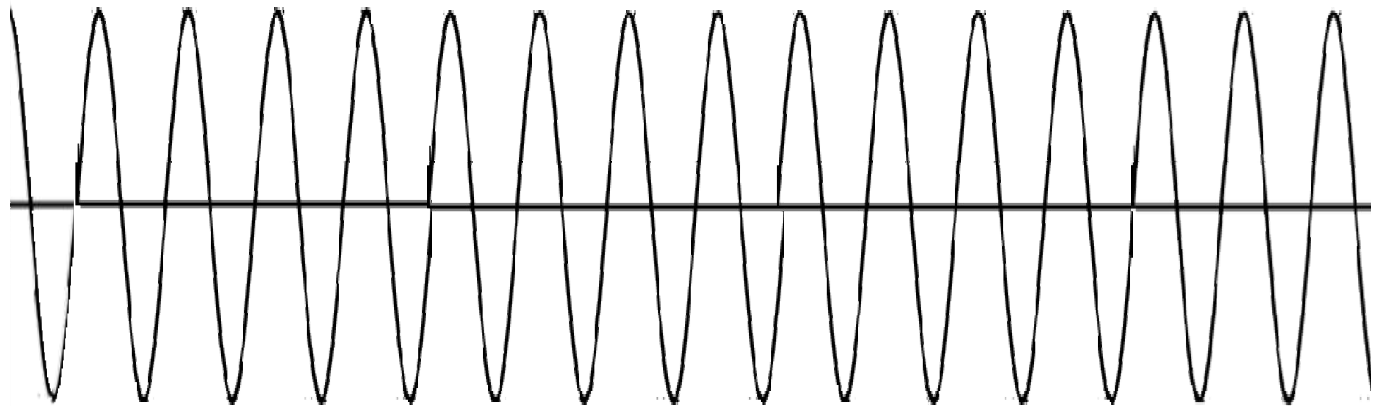
Definition of aliasing



We got the same sampled signal

Wrong Assumption on Bandwidth: Treachery of Aliasing

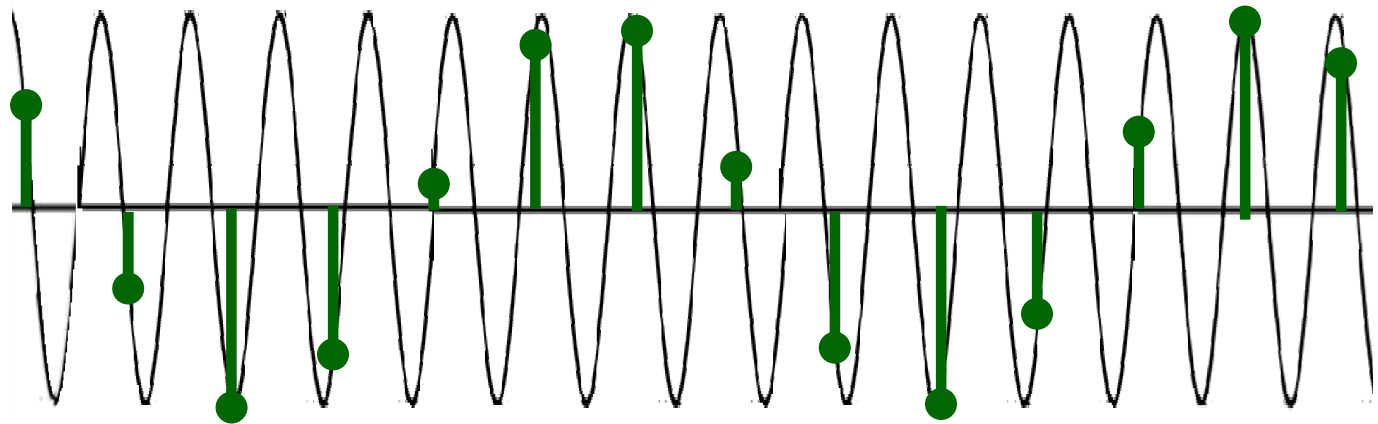
Definition of aliasing



The third signal: 460 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

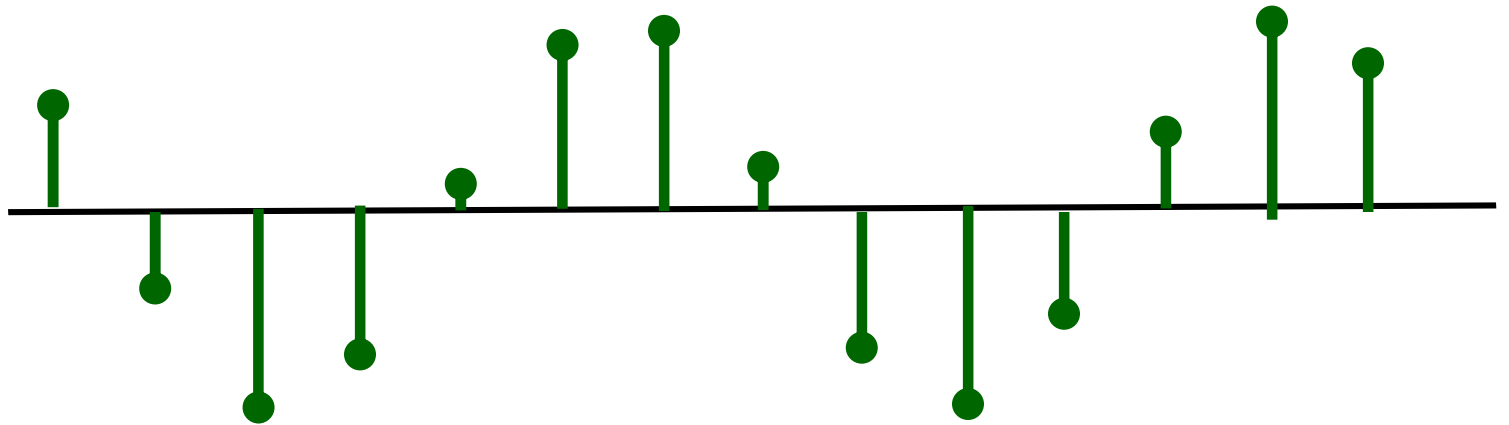
Definition of aliasing



Sampled at 400 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

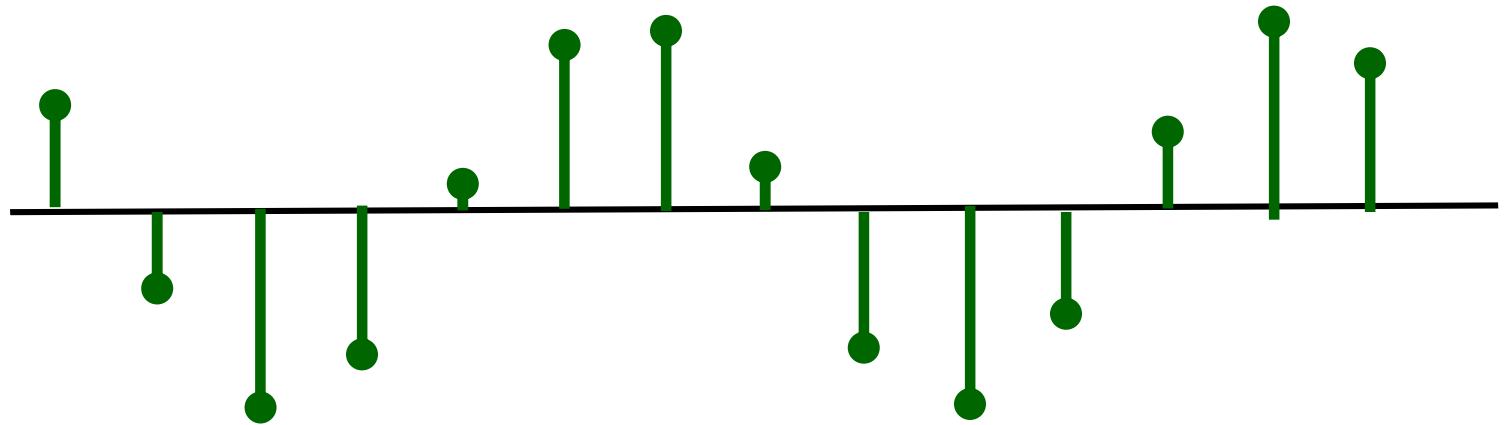
Definition of aliasing



No difference: the same sampled signal is obtained

Wrong Assumption on Bandwidth: Treachery of Aliasing

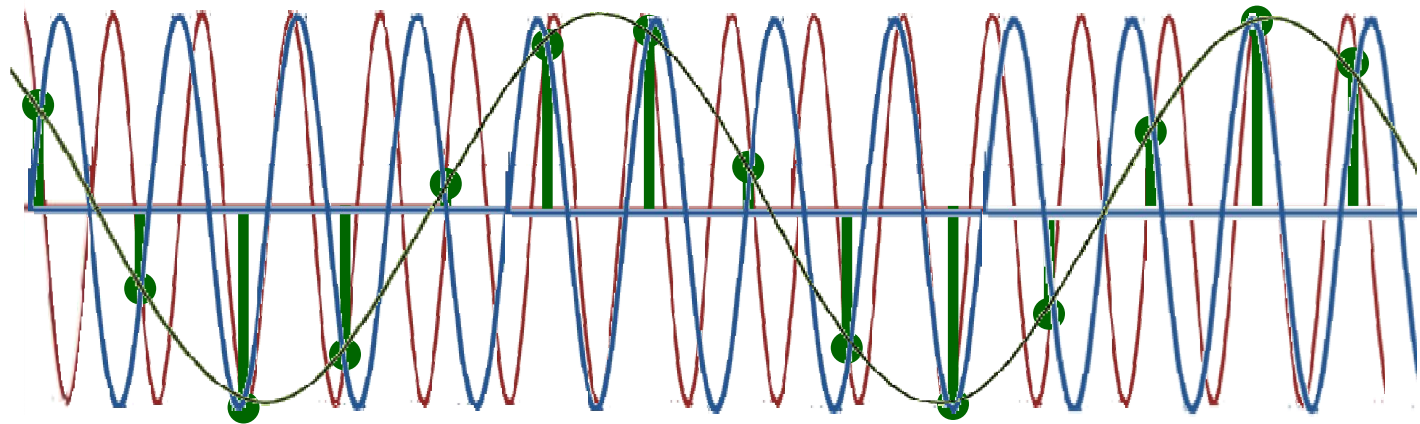
Definition of aliasing



While reconstructing, these samples may generate all three signals

Wrong Assumption on Bandwidth: Treachery of Aliasing

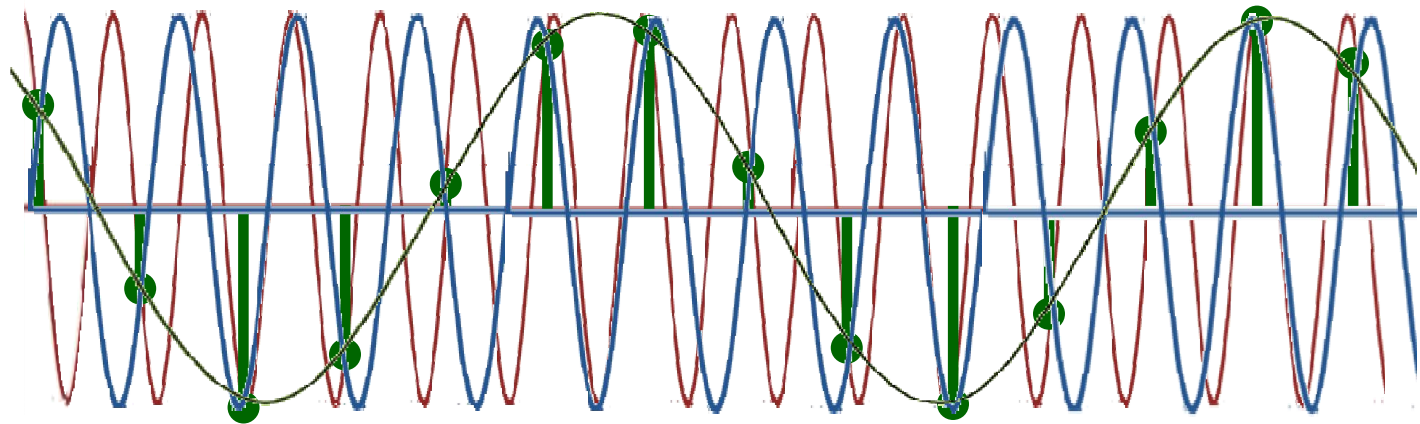
Definition of aliasing



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Wrong Assumption on Bandwidth: Treachery of Aliasing

Definition of aliasing



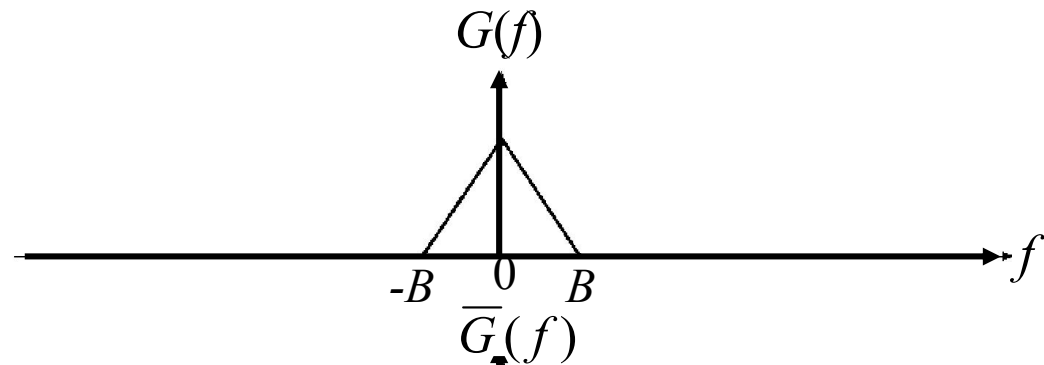
60, 340, and 460, are aliased frequencies for 400 Hz
sampling rate

Wrong Assumption on Bandwidth: Treachery of Aliasing

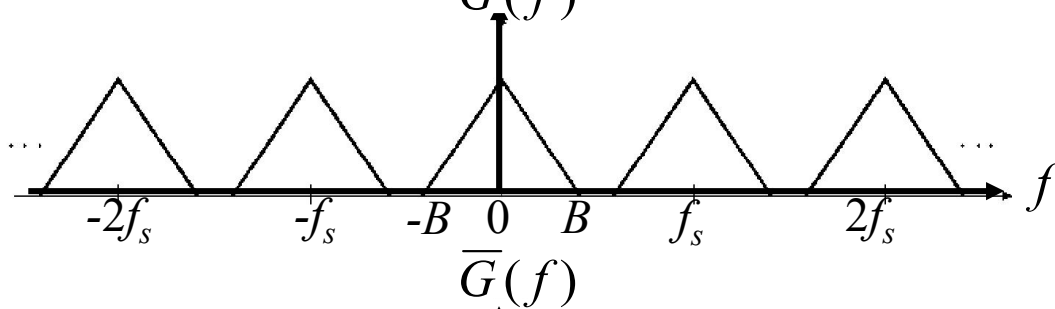
Definition of aliasing

- Aliasing occurs when sampling rate is $<$ Nyquist rate
- Spectral cycles overlaps

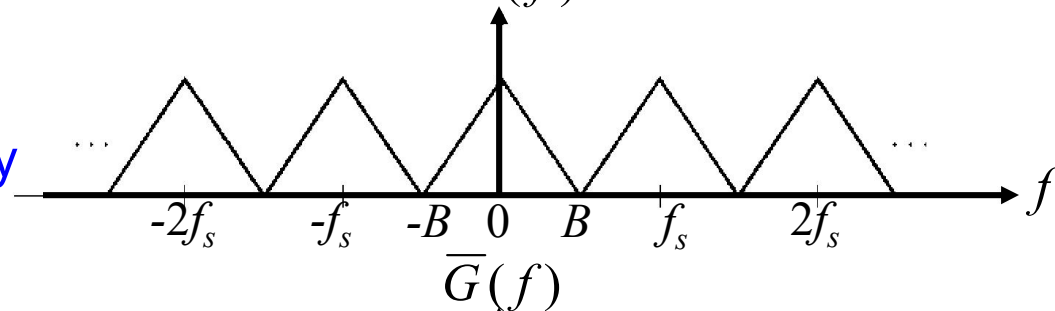
Spectrum of
unsampled $g(t)$



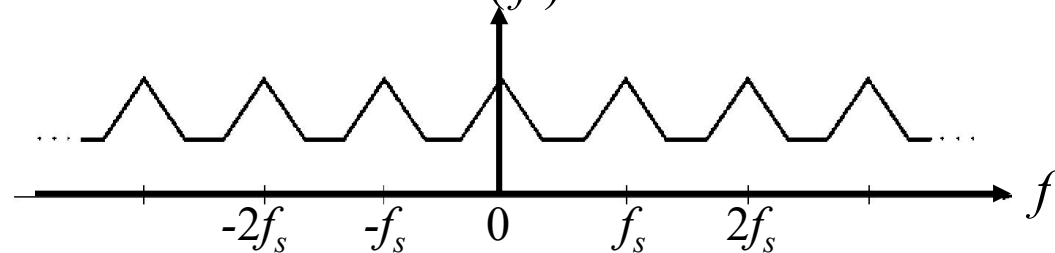
Spectrum of over
sampled $g(t)$



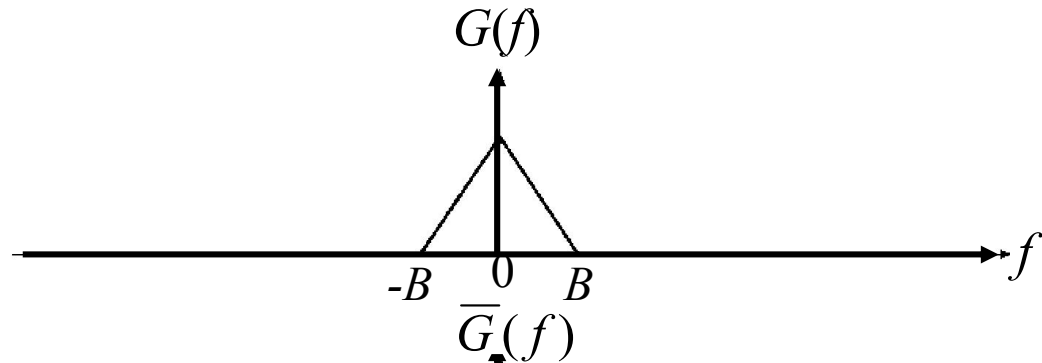
Spectrum of critically
sampled $g(t)$



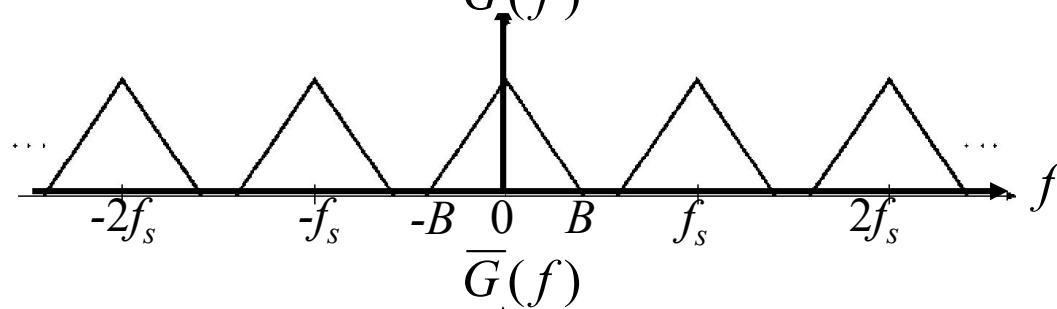
Spectrum of under
sampled $g(t)$



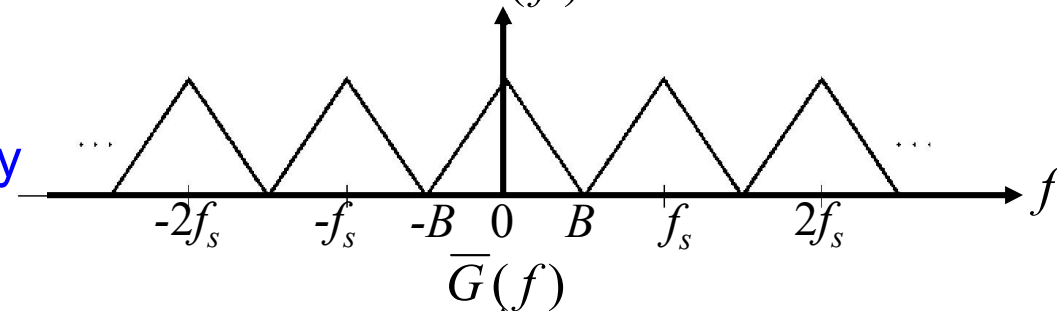
Spectrum of
unsampled $g(t)$



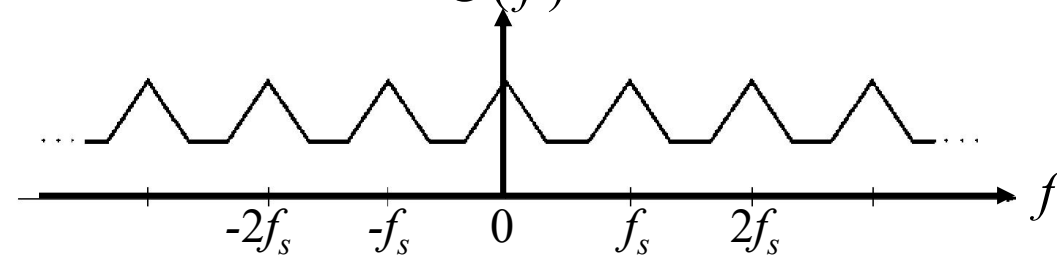
Spectrum of over
sampled $g(t)$



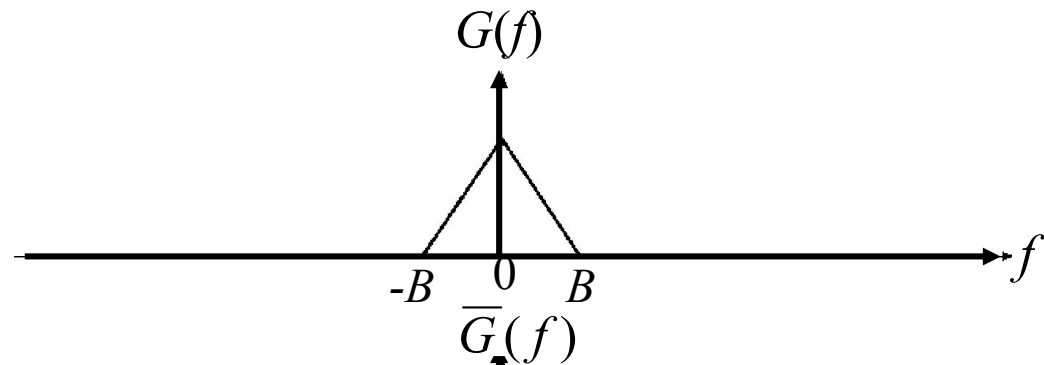
Spectrum of critically
sampled $g(t)$



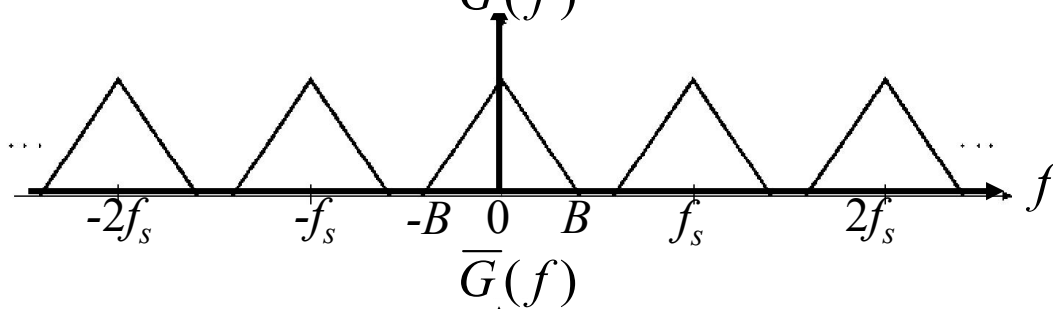
Overlapping cycles
of $G(f)$



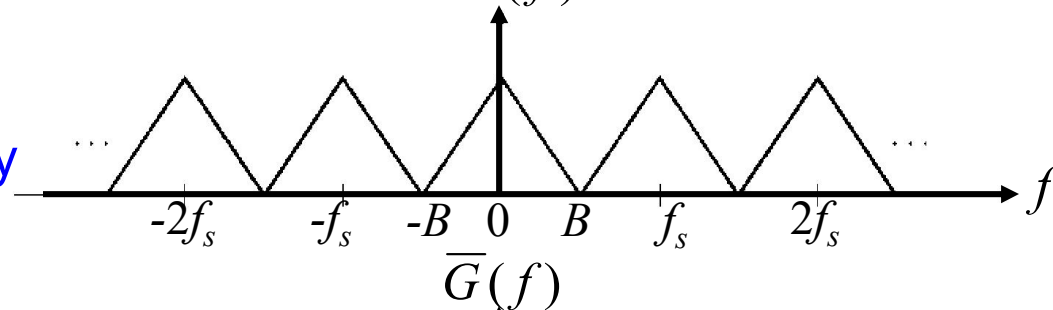
Spectrum of
unsampled $g(t)$



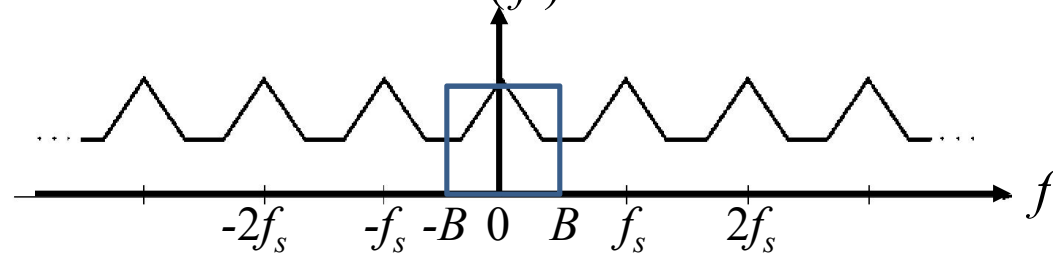
Spectrum of over
sampled $g(t)$



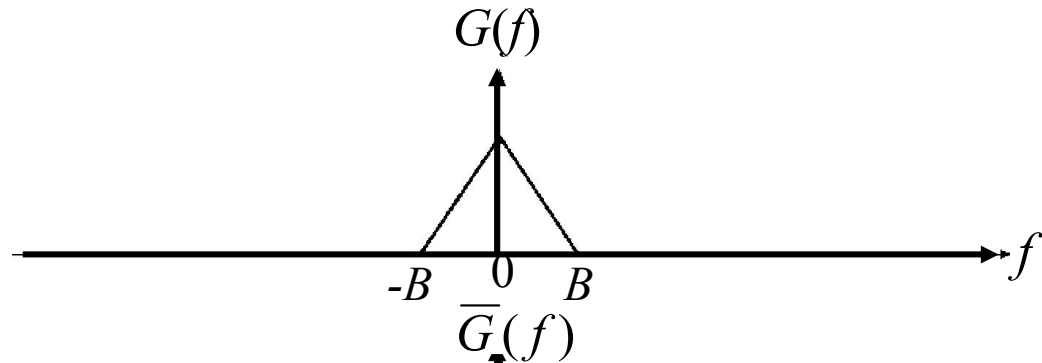
Spectrum of critically
sampled $g(t)$



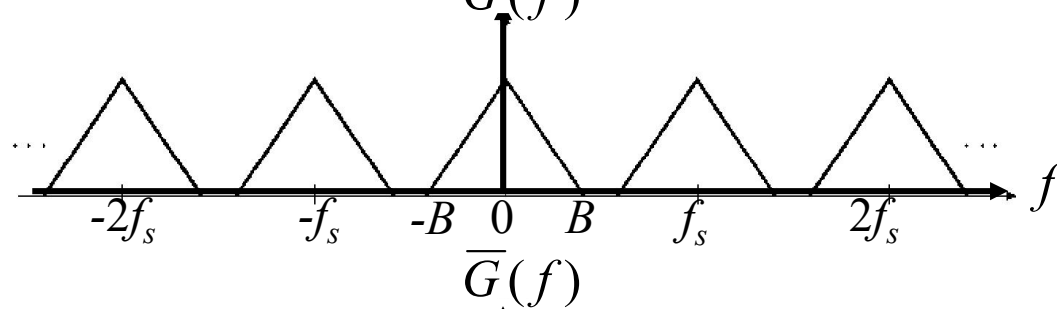
a complete cycle of $G(f)$
cannot be recovered



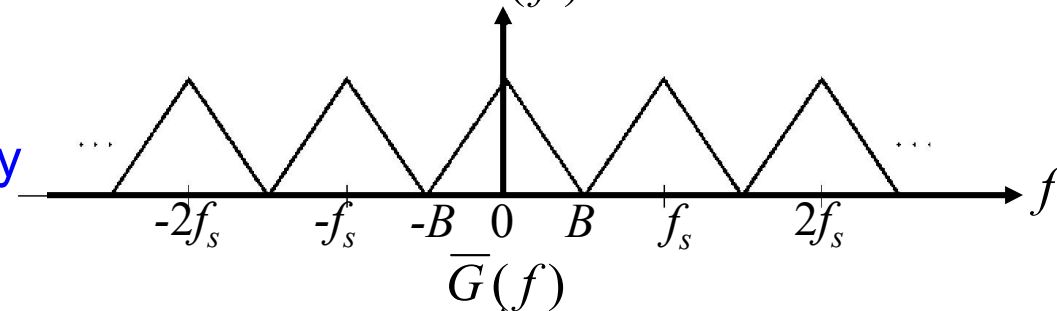
Spectrum of
unsampled $g(t)$



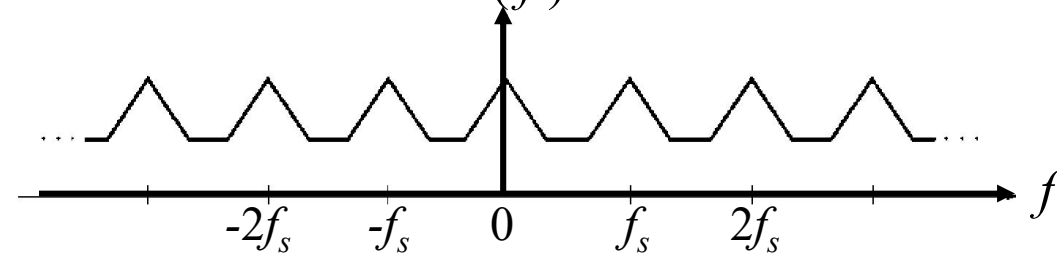
Spectrum of over
sampled $g(t)$



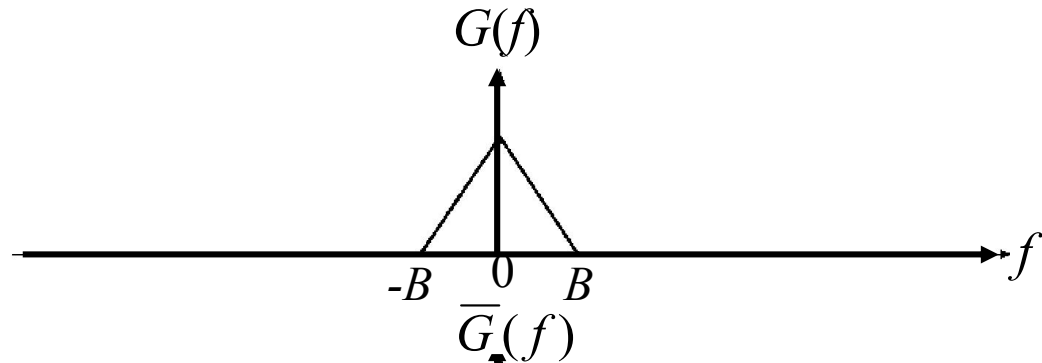
Spectrum of critically
sampled $g(t)$



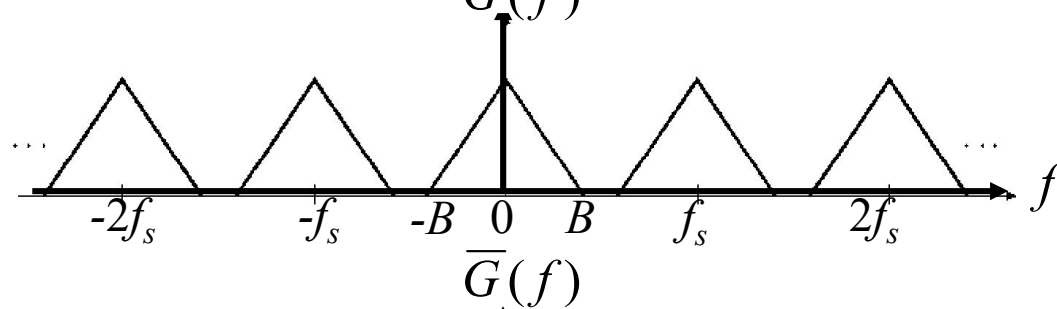
Original $g(t)$ may not be
recovered, too



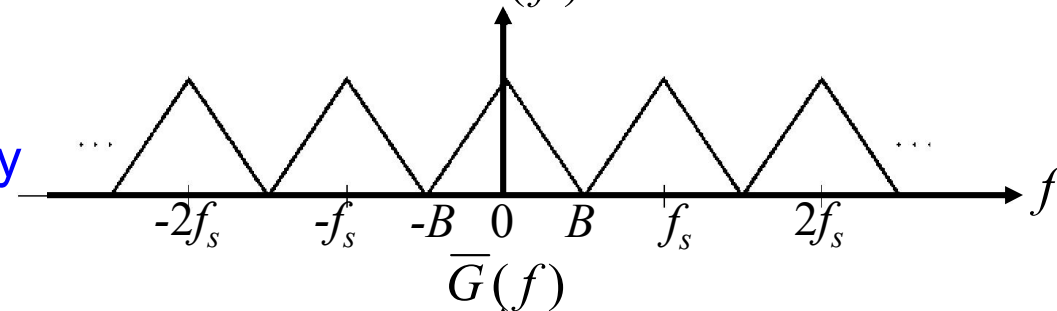
Spectrum of
unsampled $g(t)$



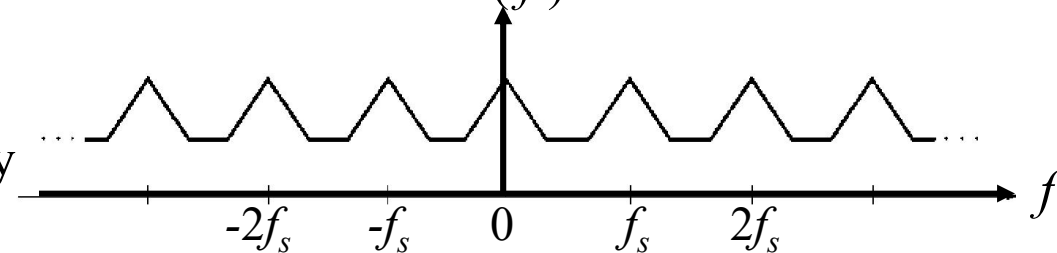
Spectrum of over
sampled $g(t)$



Spectrum of critically
sampled $g(t)$



Rather aliased signals may
be reconstructed



Wrong Assumption on Bandwidth: Treachery of Aliasing

While sampling and reconstructing,

- Signal $g(t)$ is assumed to be band-limited to B

Wrong Assumption on Bandwidth: Treachery of Aliasing

While sampling and reconstructing,

- Signal $g(t)$ is assumed to be band-limited to B

However,

- all practical signals are time-limited

Wrong Assumption on Bandwidth: Treachery of Aliasing

While sampling and reconstructing,

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However,

- all practical signals are time-limited
- time limited signal CANNOT be band-limited

Wrong Assumption on Bandwidth: Treachery of Aliasing

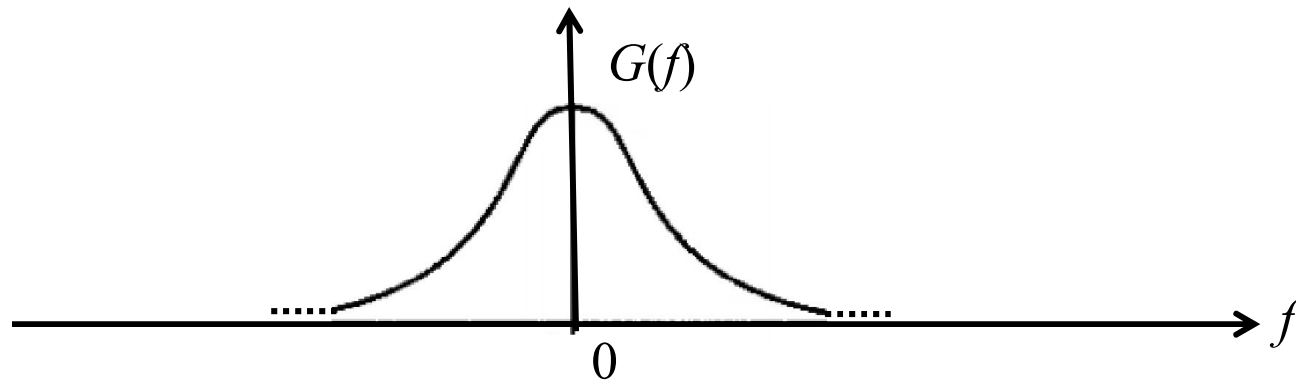
While sampling and reconstructing,

- Signal $g(t)$ is assumed to be band-limited to B

However,

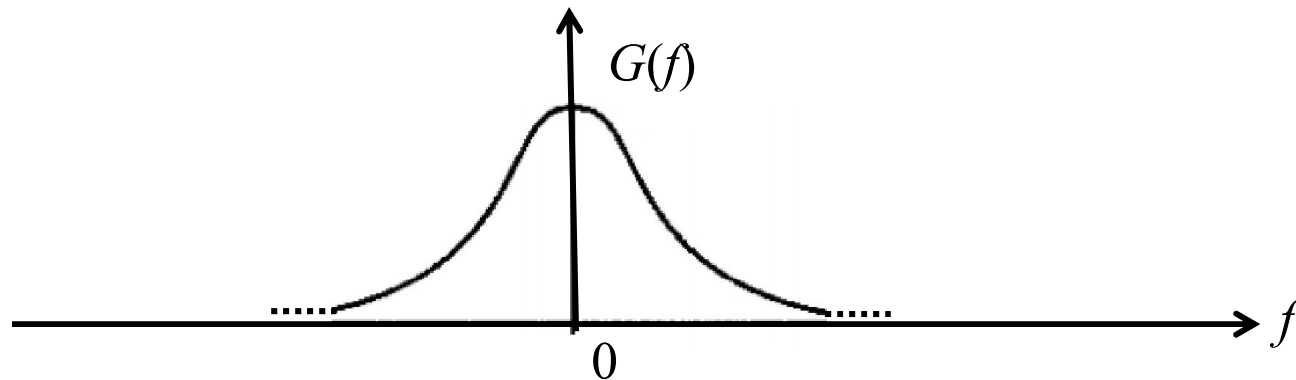
- all practical signals are time-limited
- time limited signal CANNOT be band-limited
- practical signals are therefore NON-band-limited

Wrong Assumption on Bandwidth: Treachery of Aliasing



A practical signal with
infinite bandwidth (B)

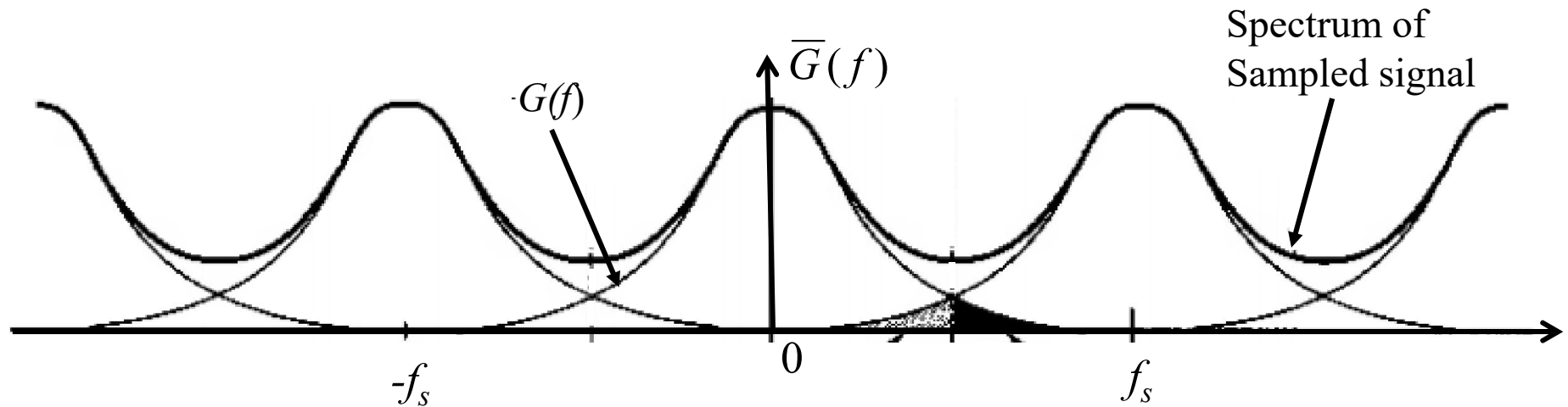
Wrong Assumption on Bandwidth: Treachery of Aliasing



A practical signal with
infinite bandwidth (B)

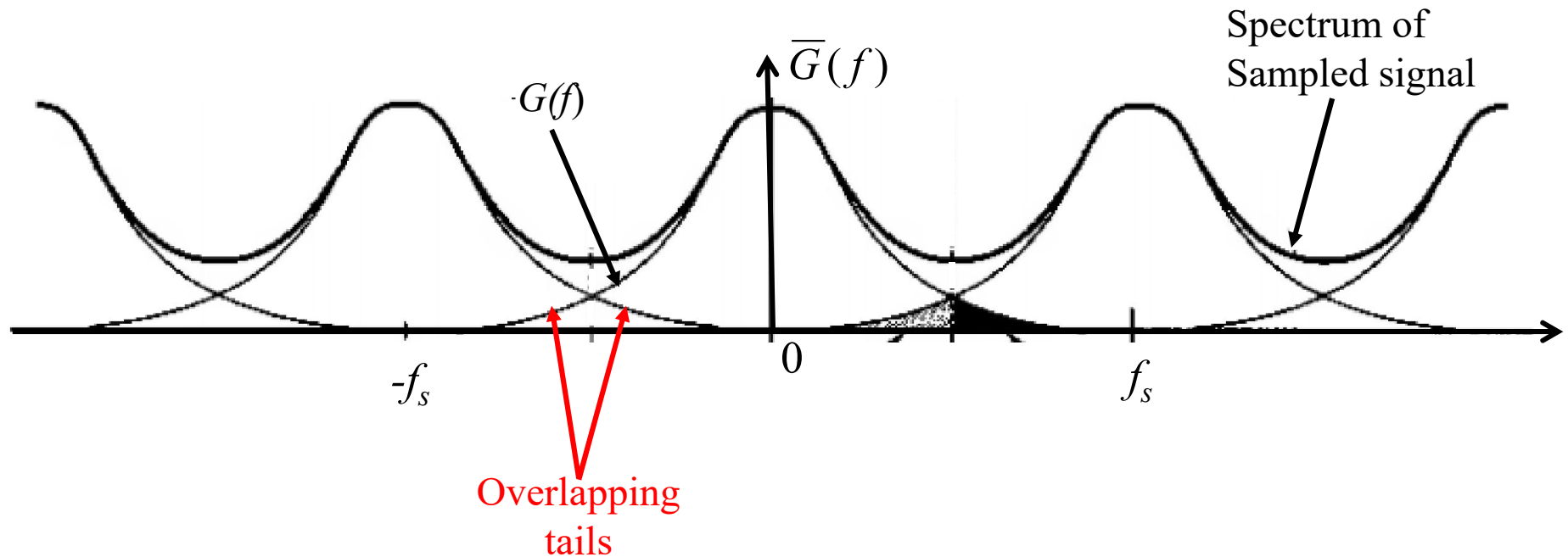
Sampling will result $\bar{G}(f)$ consisting of overlapping $G(f)$ at every f_s

Wrong Assumption on Bandwidth: Treachery of Aliasing



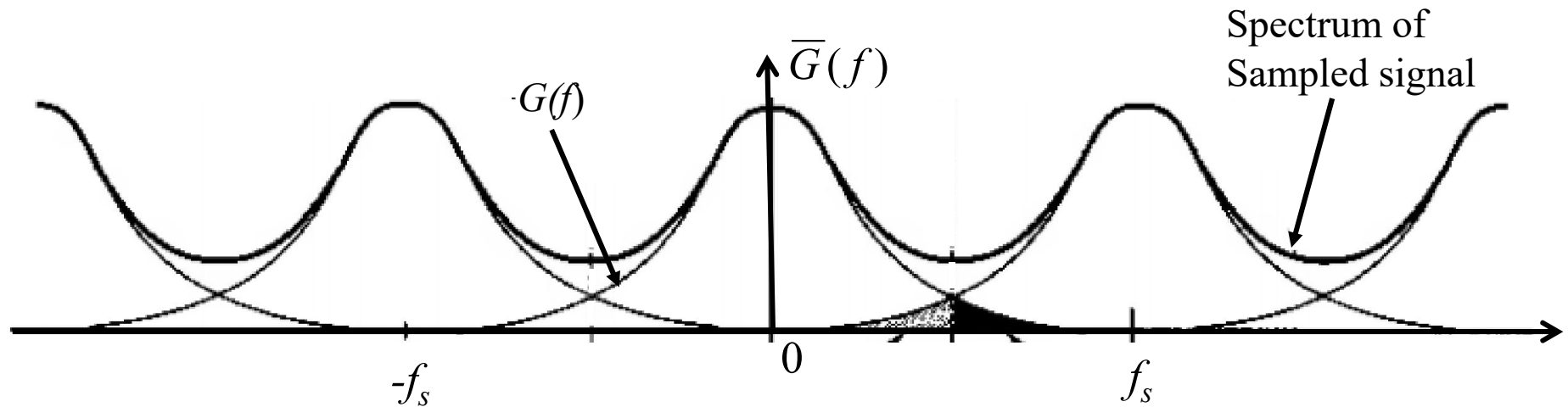
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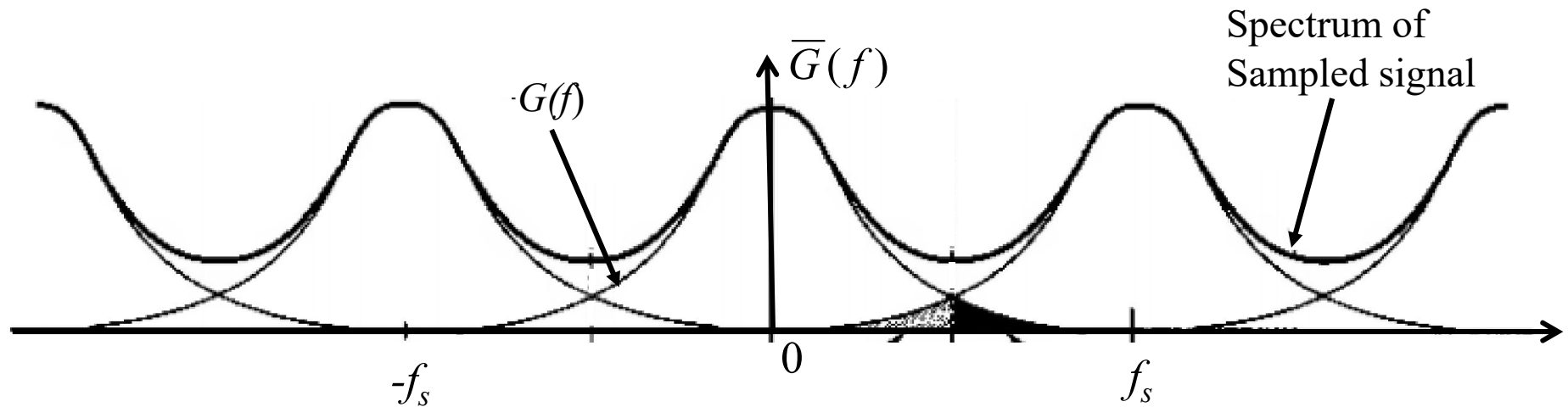
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Wrong Assumption on Bandwidth: Treachery of Aliasing



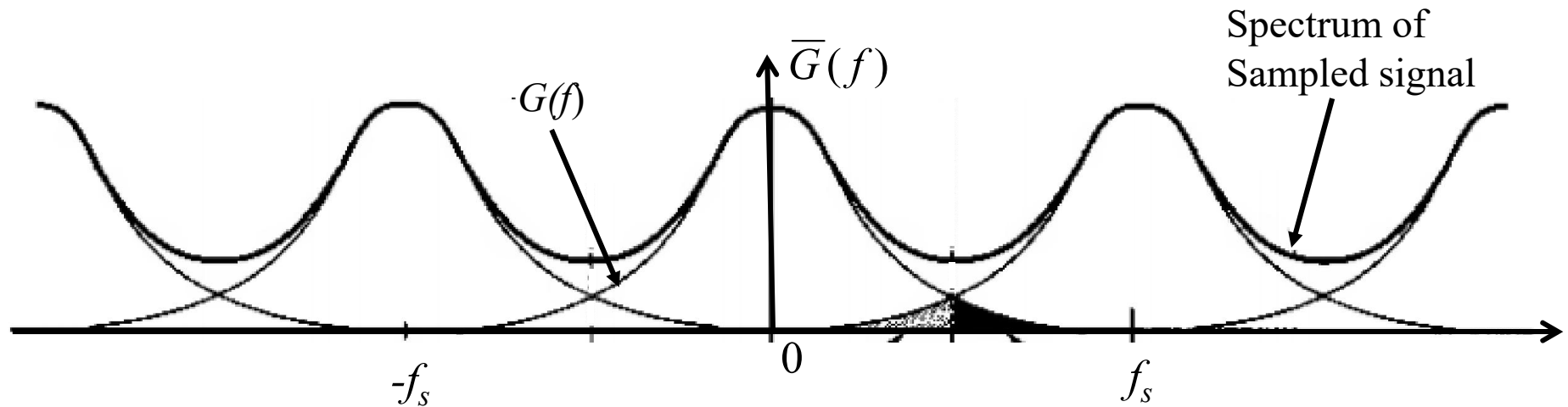
Spectral **overlap is unavoidable** as bandwidth is infinite

Wrong Assumption on Bandwidth: Treachery of Aliasing



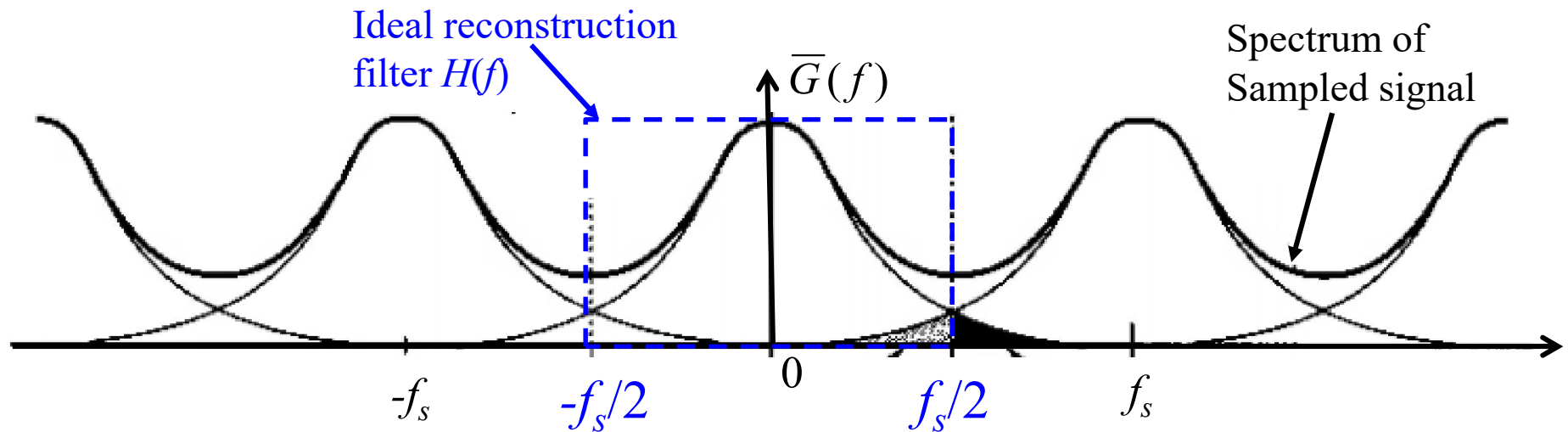
Higher sampling rate **ONLY** reduces overlap; but **cannot eliminate**

Wrong Assumption on Bandwidth: Treachery of Aliasing



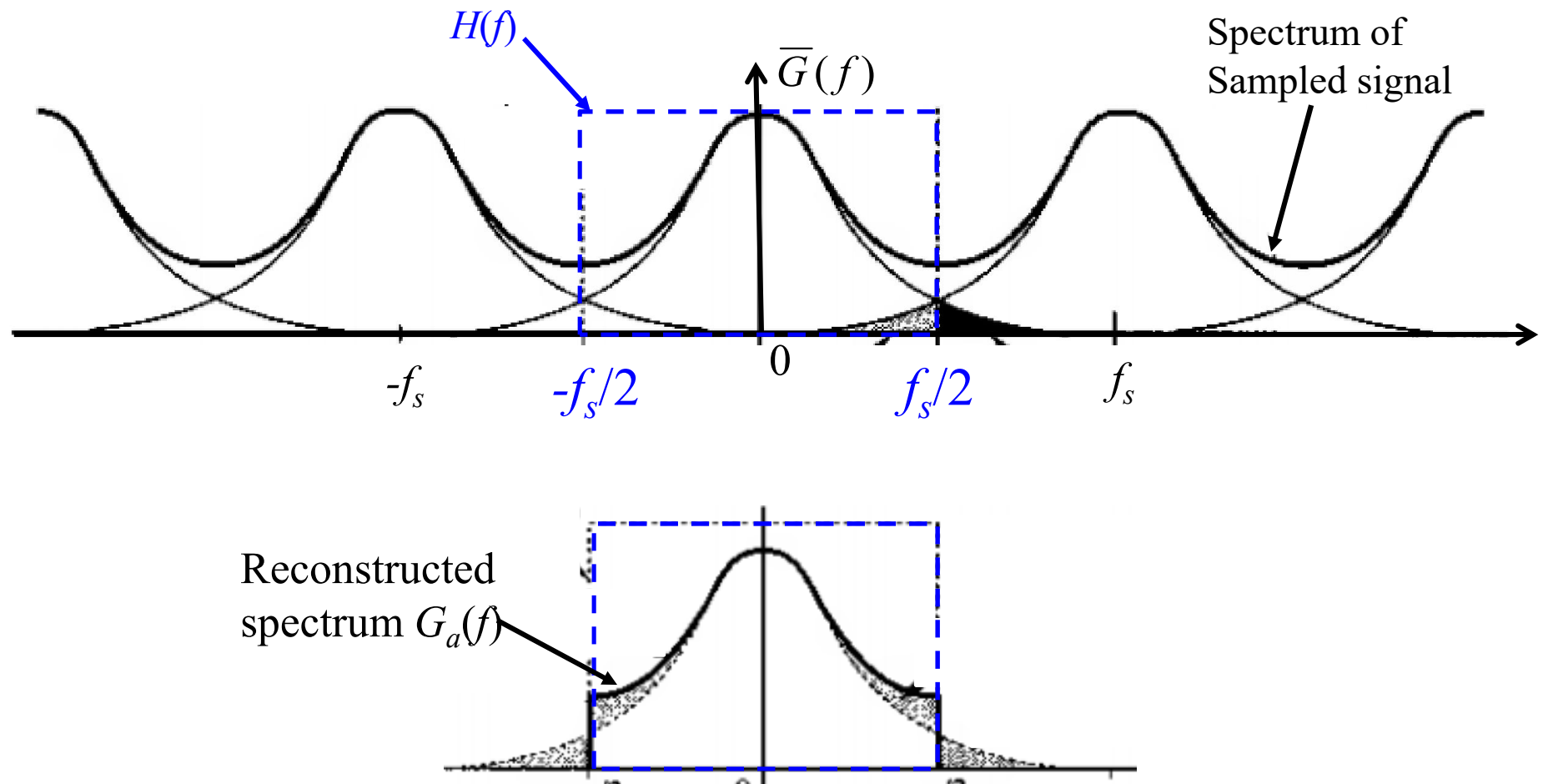
$\bar{G}(f)$ has NO complete information of $G(f)$
 $g(t)$ cannot be fully recovered from $\bar{g}(t)$

Wrong Assumption on Bandwidth: Treachery of Aliasing

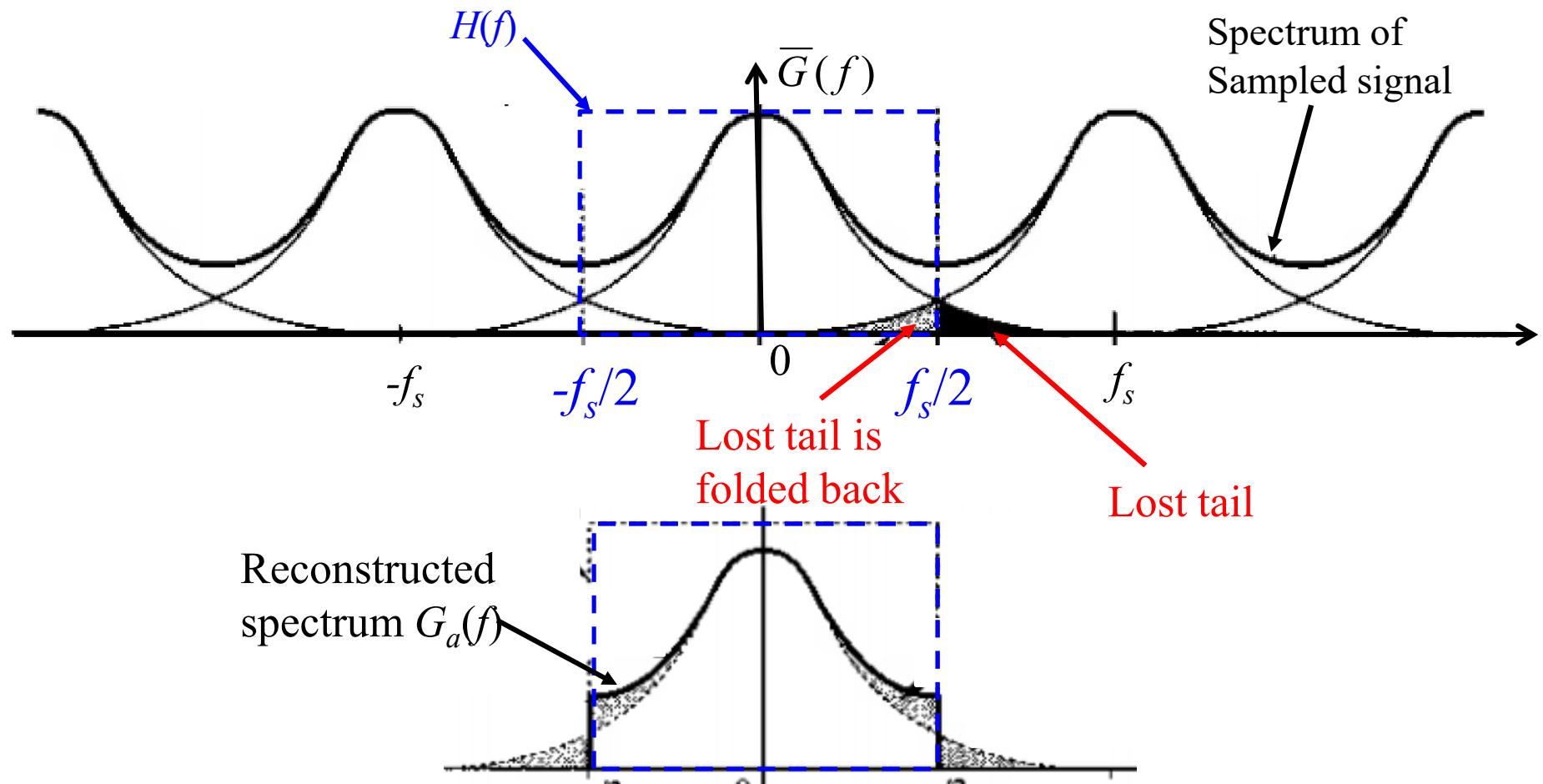


Ideal reconstruction filter CANNOT separate $G(f)$, rather generates $G_a(f)$ which spectrum of **aliased signal** $g_a(t)$

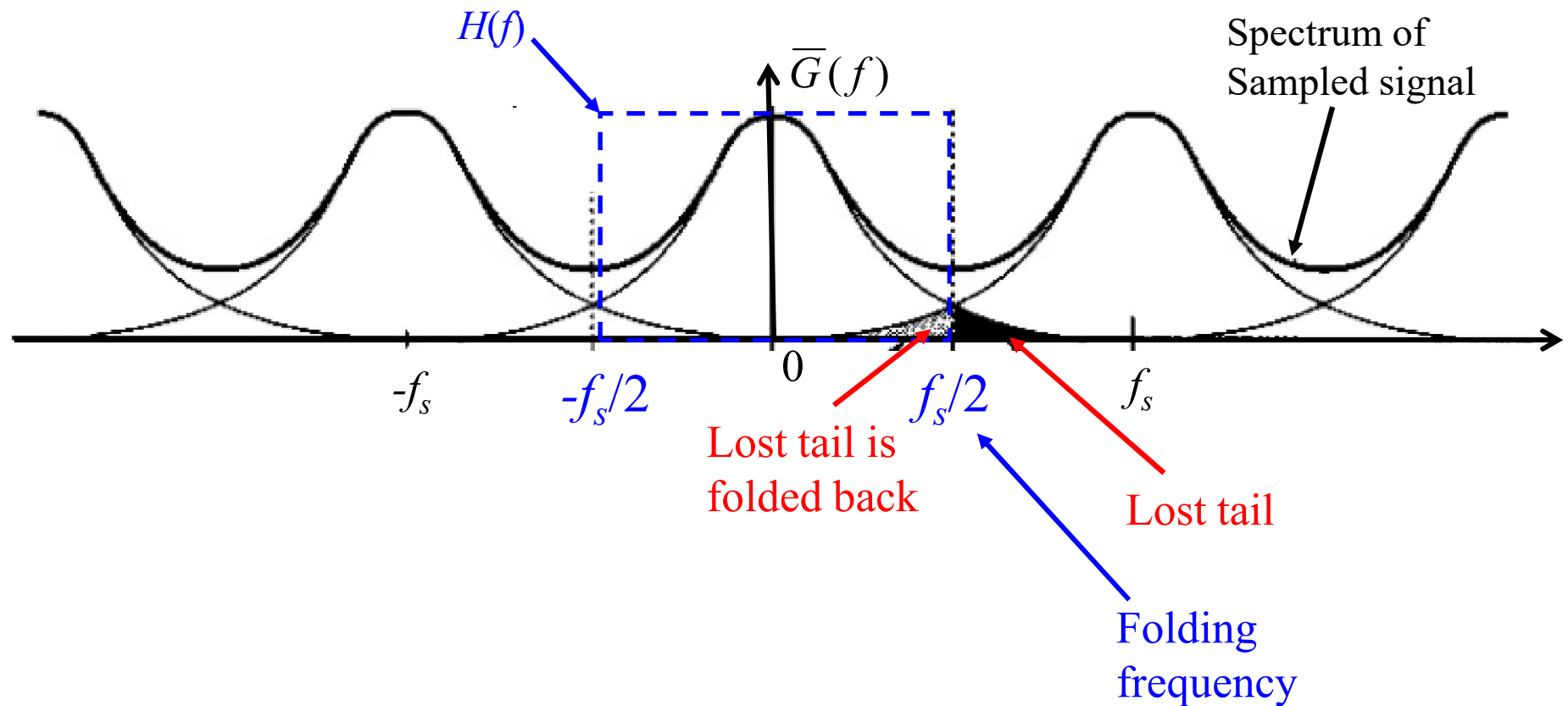
Wrong Assumption on Bandwidth: Treachery of Aliasing



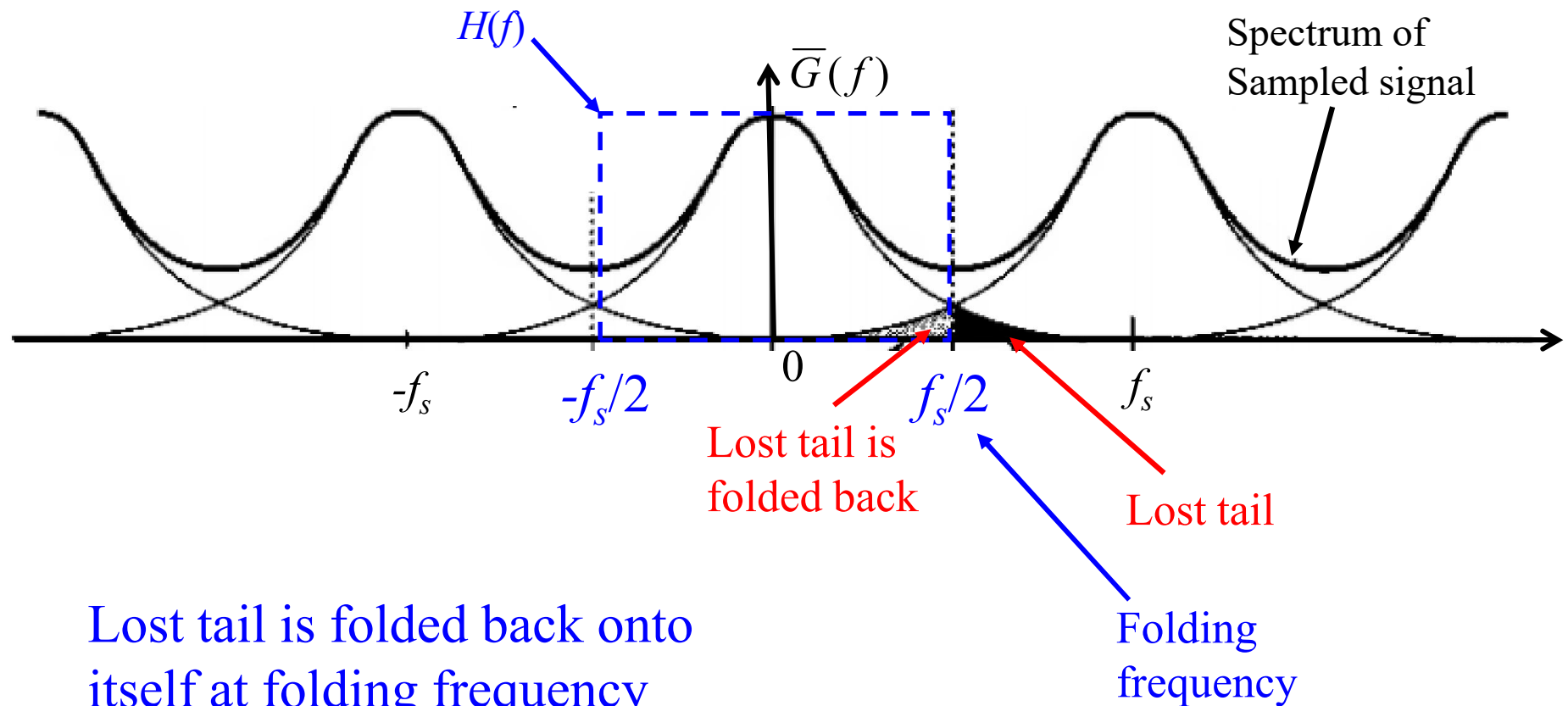
Wrong Assumption on Bandwidth: Treachery of Aliasing



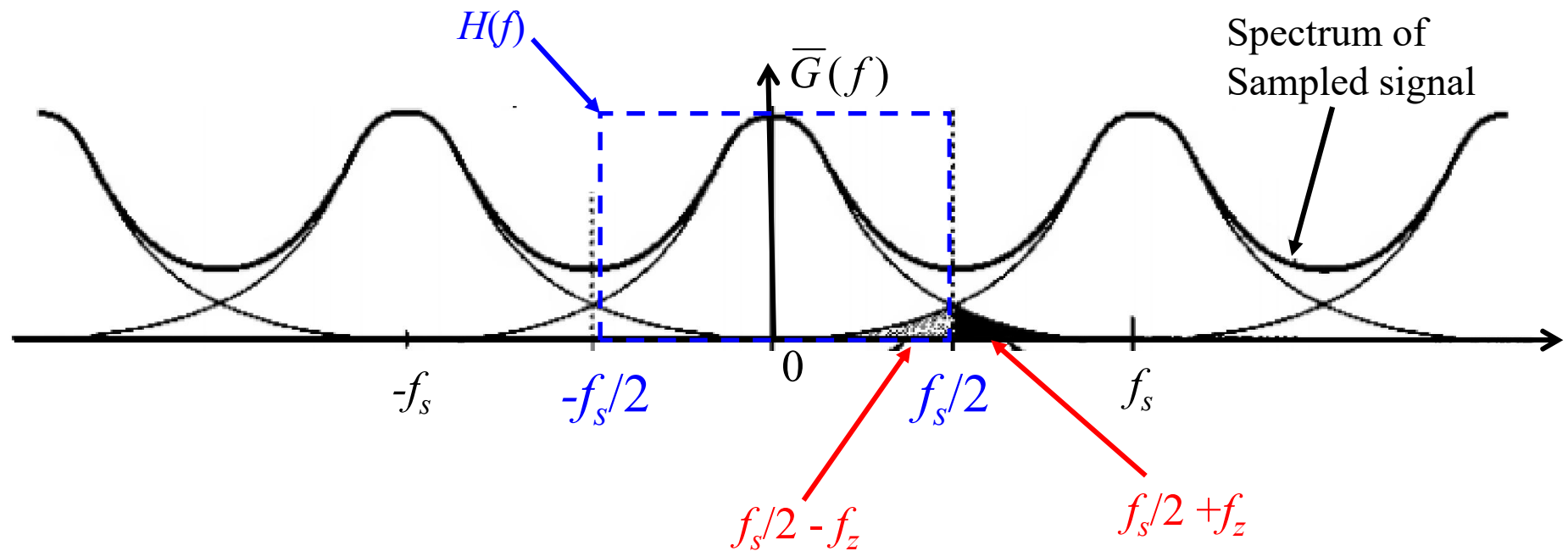
Wrong Assumption on Bandwidth: Treachery of Aliasing



Wrong Assumption on Bandwidth: Treachery of Aliasing

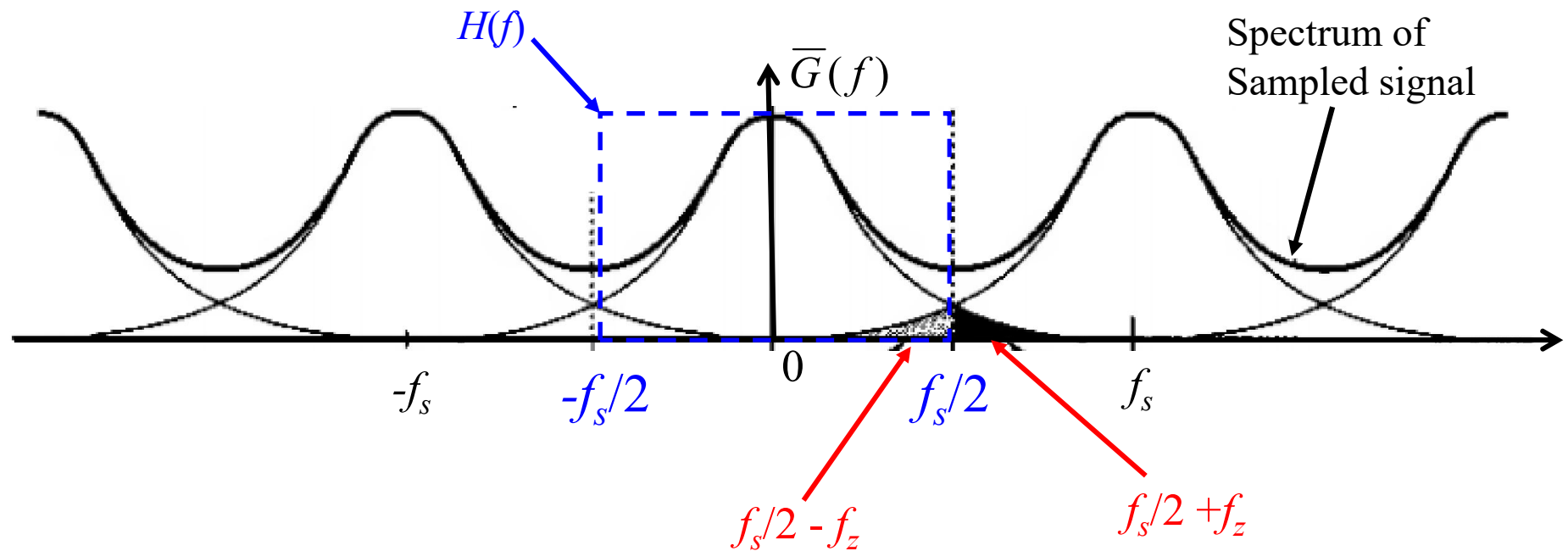


Wrong Assumption on Bandwidth: Treachery of Aliasing



Component at $f_s/2 + f_z$ is folded
back to $f_s/2 - f_z$

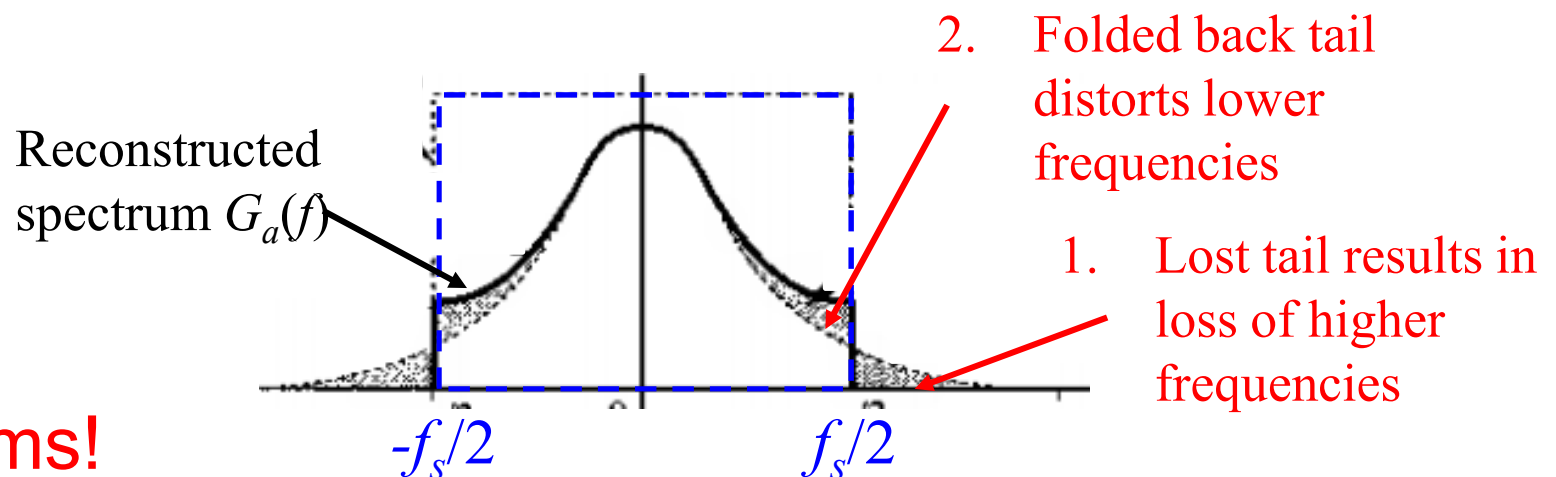
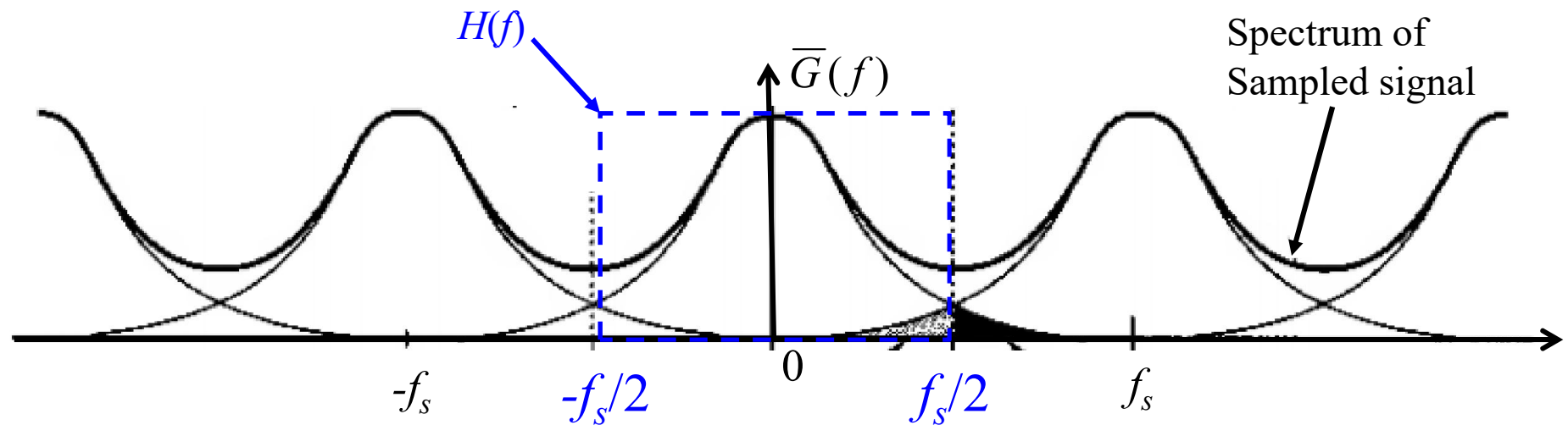
Wrong Assumption on Bandwidth: Treachery of Aliasing



Component at $f_s/2 + f_z$ is folded
back to $f_s/2 - f_z$

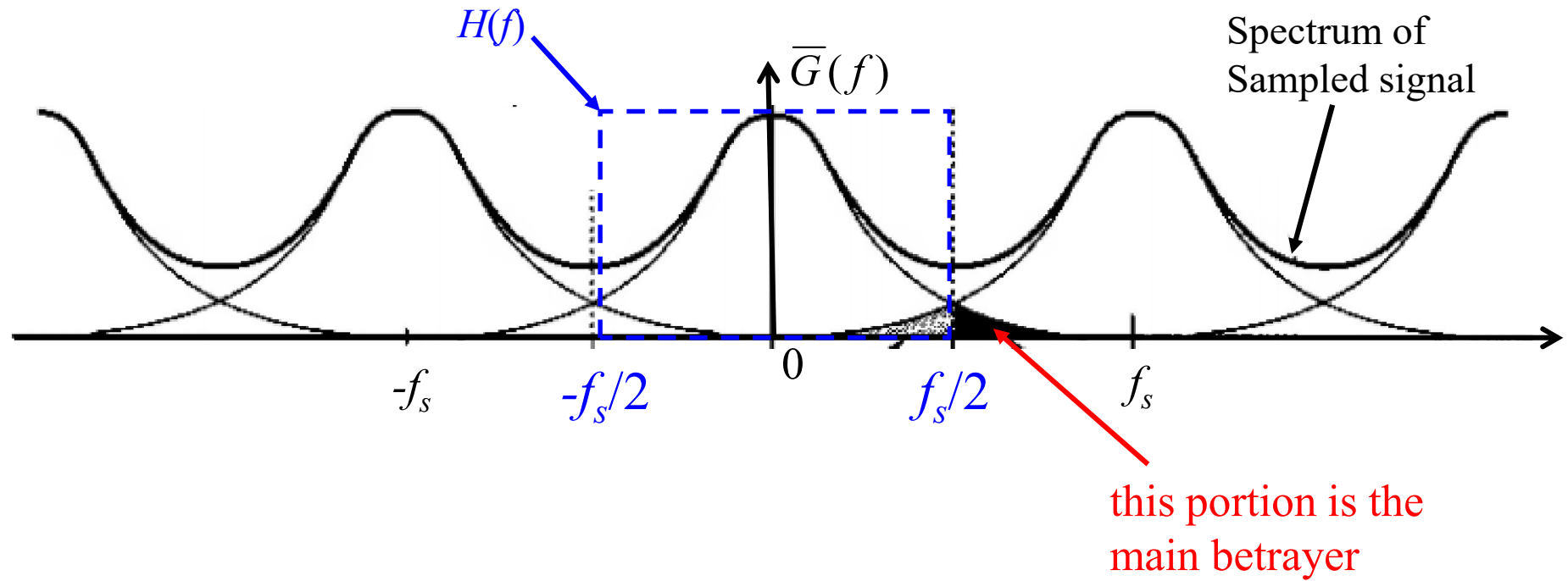
Tail folding is known as
spectral folding or aliasing

Wrong Assumption on Bandwidth: Treachery of Aliasing

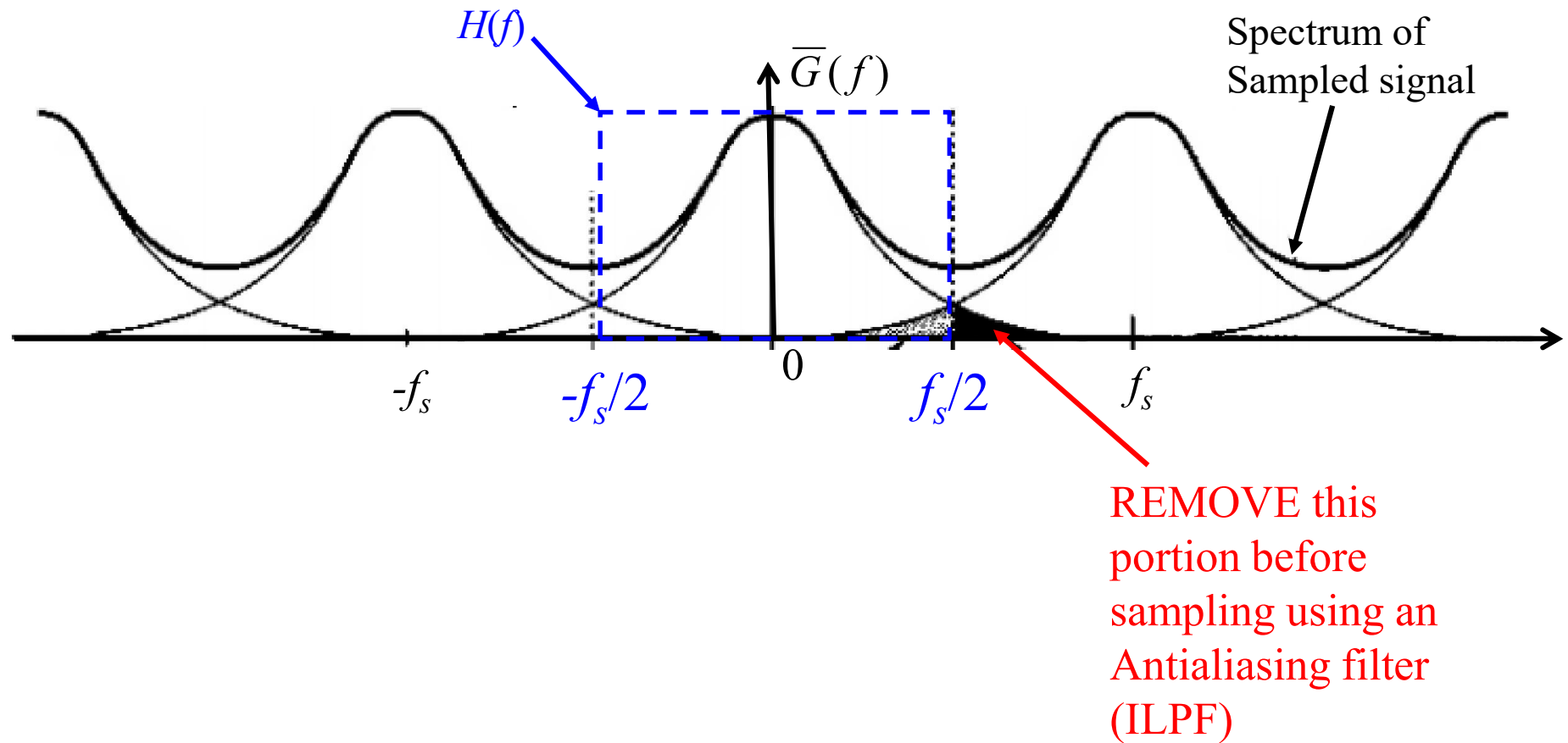


**TWO
Problems!**

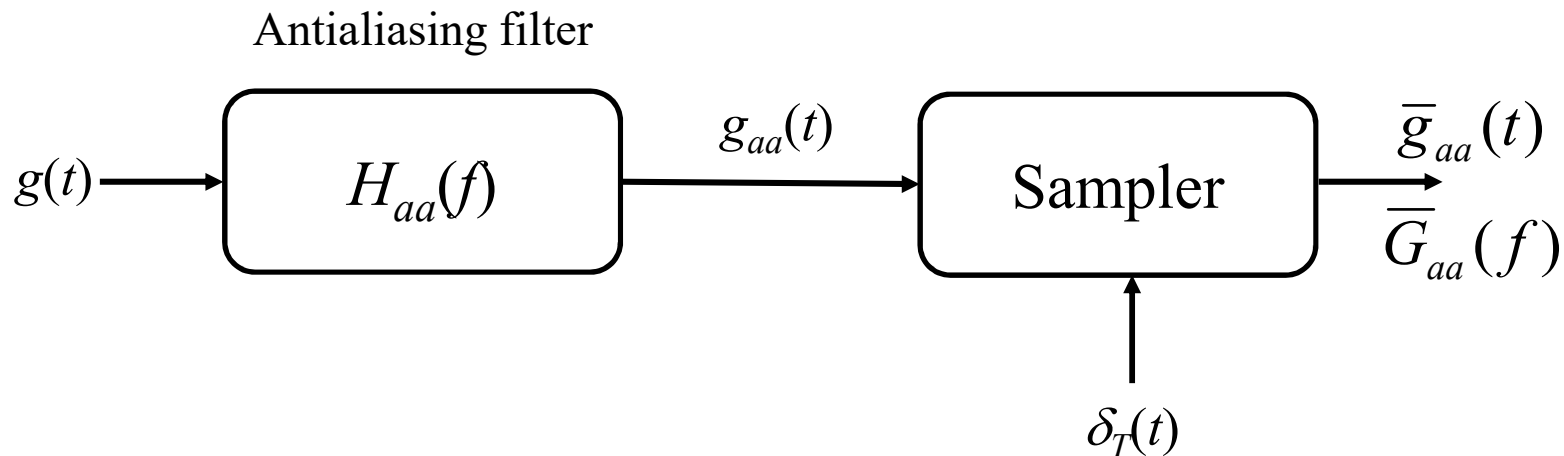
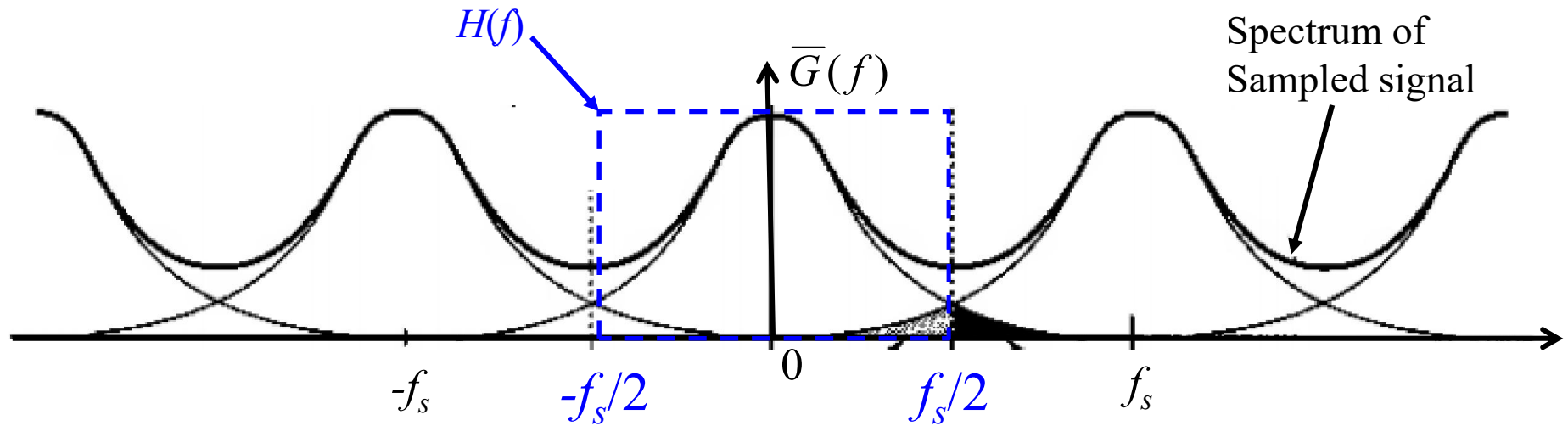
Combating Aliasing



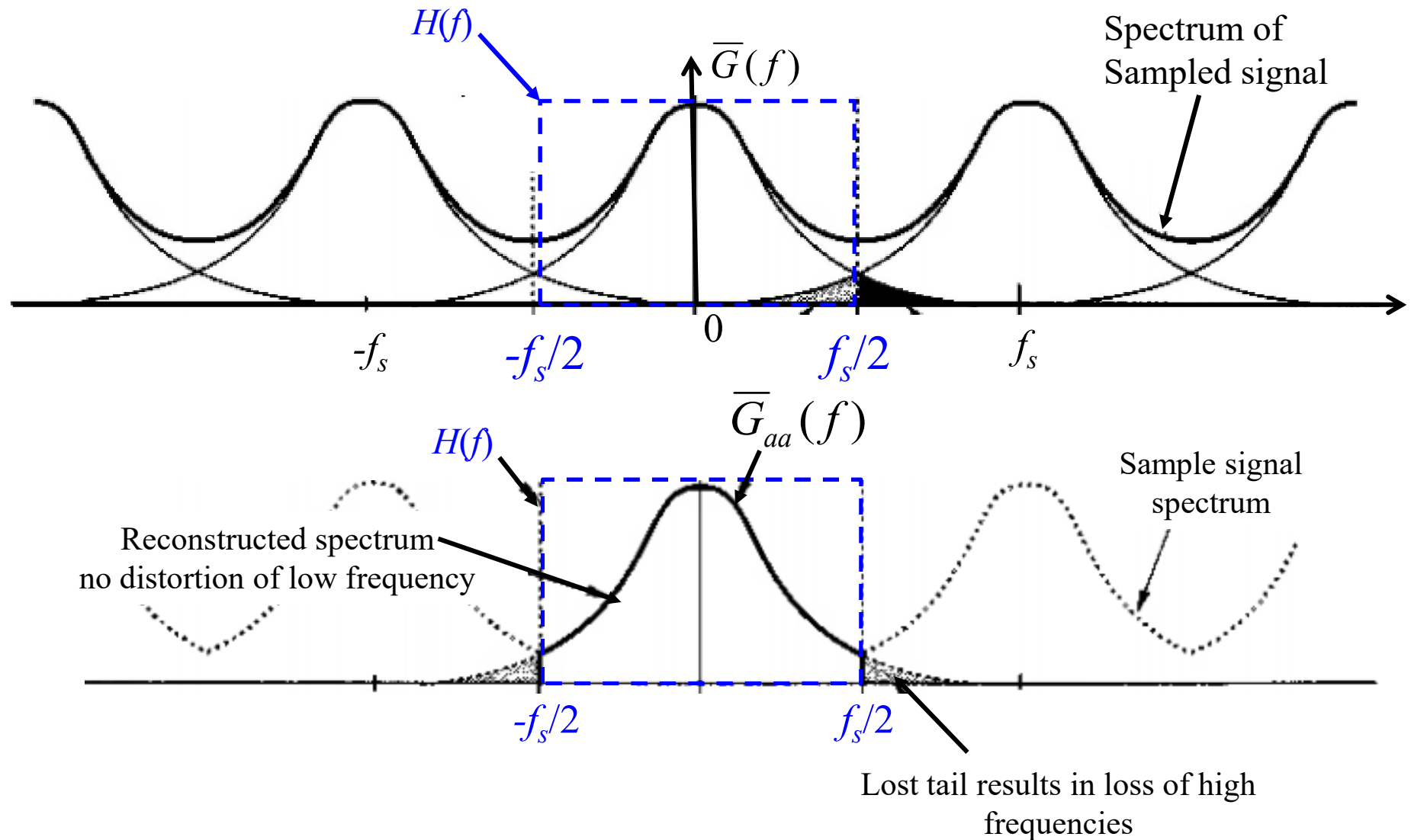
Combating Aliasing



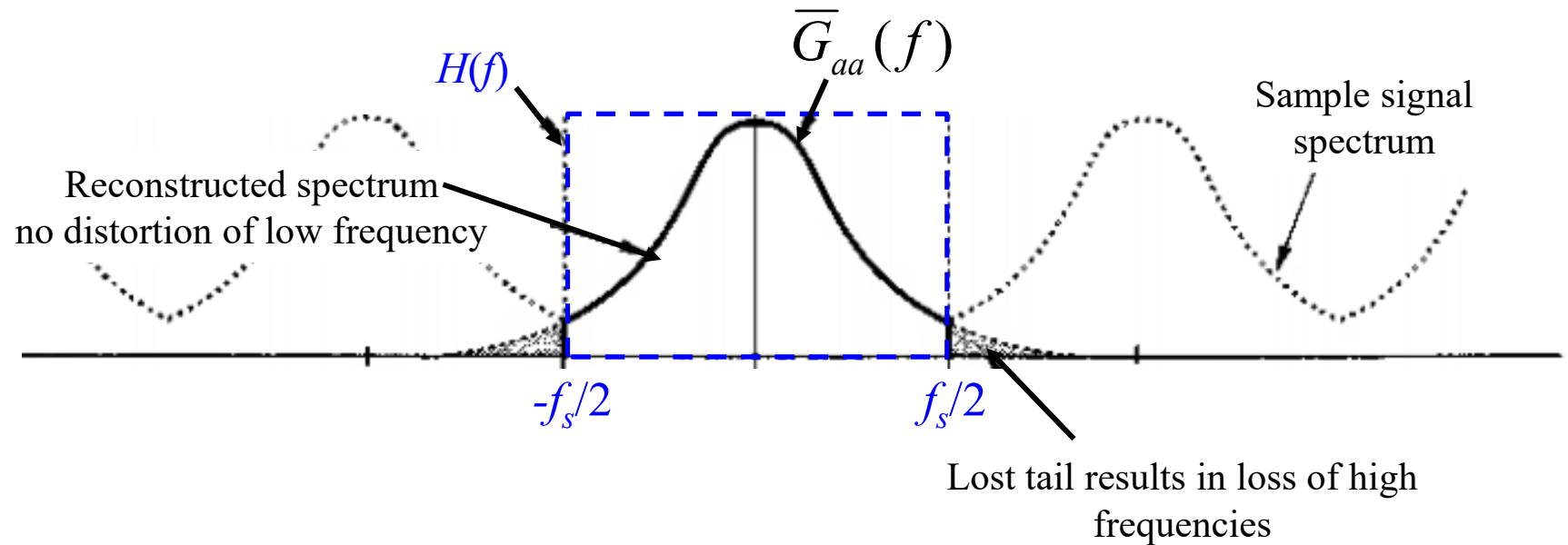
Combating Aliasing



Combating Aliasing



Combating Aliasing



$$\overline{G}_{aa}(f) = G(f) \quad |f| < f_s / 2$$

Advantage: suppressed
component cannot reappear
in low frequency
components